Harrington's Insights: Best Results

This assembly puts the best recorded result of every tool the author built or modified, originals (HCM, hcmx, the Harrington Exchange), modified encoders read by stock decoders (xz-hi, lzma-hi, 7z-hi, zip-hi, bzip2-hi, zstd-hi, Brotli-hi, flac-hi, the PDF row filter, RAR with detector delta) and mutants (record delta front ends, lpaq1-h, the Exchange->Blosc2 hybrid), beside every standard tool measured on the same benchmark.

Sizes are shown as 1,000,000 bits down to X bits. 'Harrington best-of' is the smallest of our own tools picked file by file, the same way a production system would pick per file; 'Best standard per file' does the same for the stock tools, so the comparison is fair both ways.

No new measurements were made for this assembly. Results come from the book's recorded tables (Chapters 4, 5, 6, 13, 15-18 and Appendix A) and from the 29 October 2026 same-machine run of Goodwin's three sets. Entries marked 'decode not recorded' appear in the size rankings only.

Order: a scoreboard, then for each benchmark a full ranking, a size-versus-decode-speed chart and a stock-versus-ours chart for each tool we modified; then spotlights on kennedy.xls, the x-ray image and JPEG XL; then the detailed Goodwin-set charts.

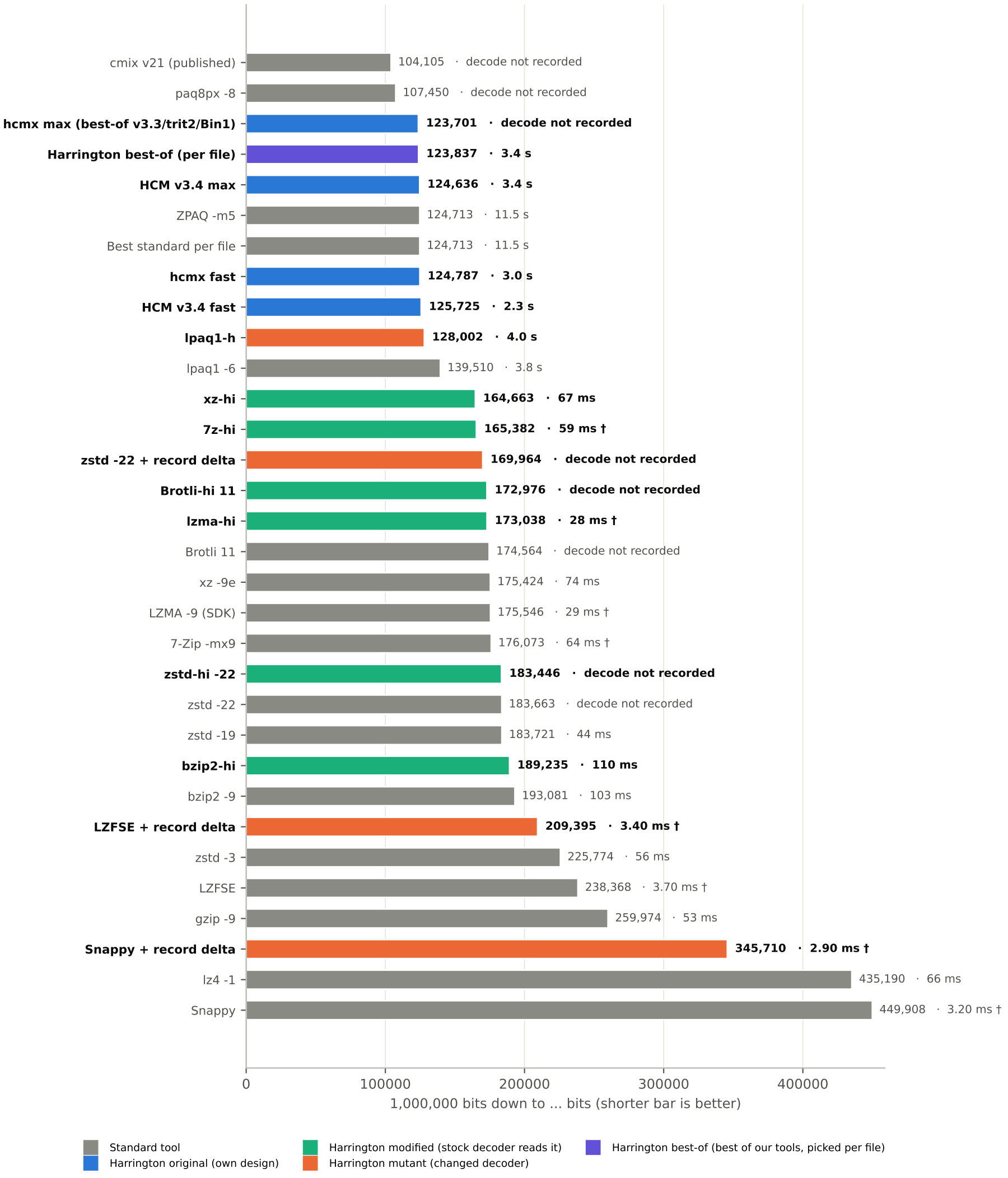
Results that are not exceptional are listed here for completeness: Kodak photos, WebP lossless: 11,204,040 -> 11,202,034 bytes (0.018%), decode 262 -> 258 ms. enwik8: HCM max 21,701,018 bytes vs ZPAQ -m5 19,625,015 and BSC max 21,480,530 (5.98 s decode vs 115 s) - HCM loses here.

Scoreboard: our best against the best standard tool, every benchmark

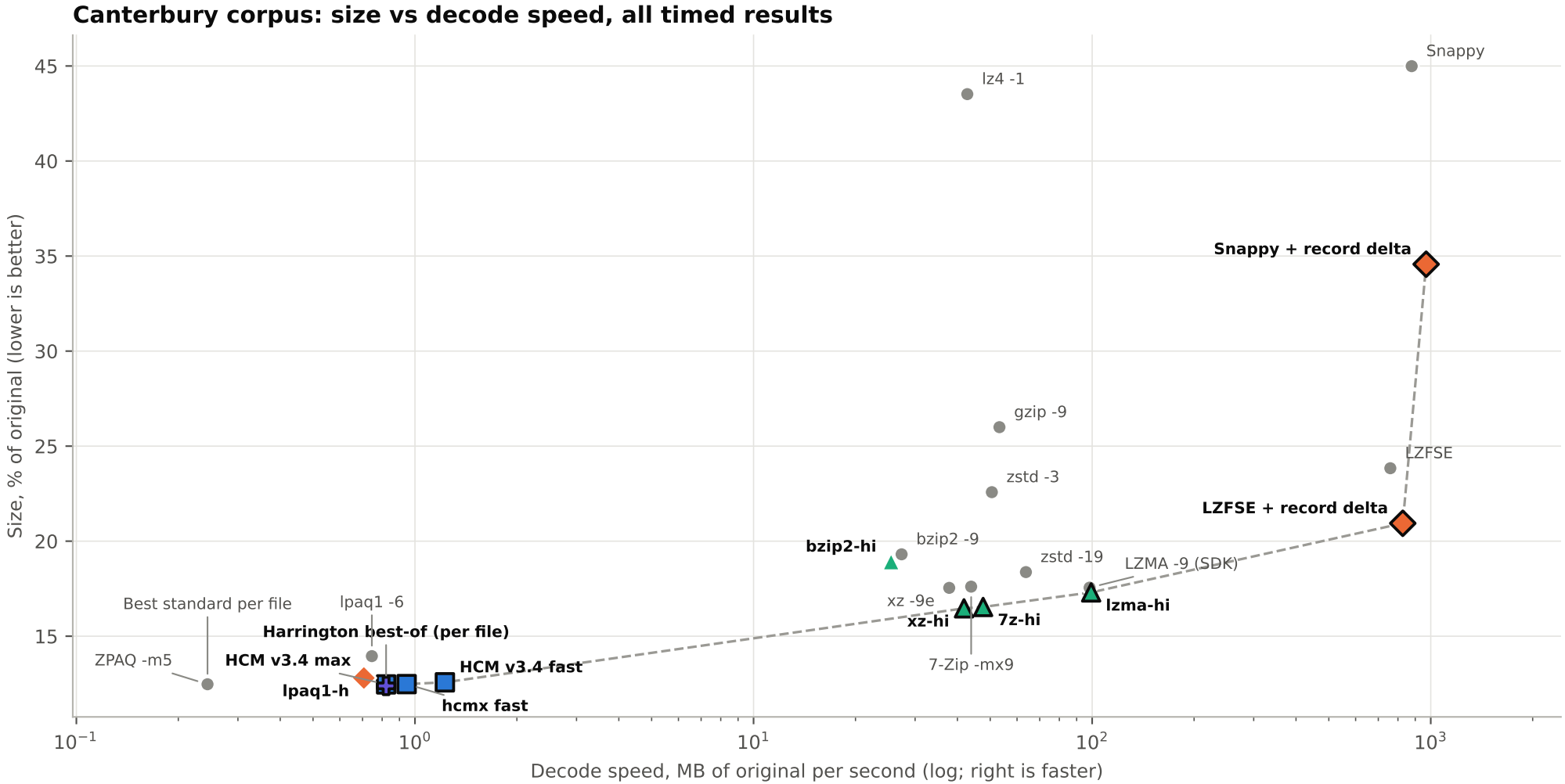
Sizes are 1,000,000 bits down to X bits. "Smaller AND faster" is our smallest entry that beats a standard tool on both size and decode time.

Benchmark	Our best (size, decode)	Best standard tool (size, decode)	Size verdict	Smaller AND faster
Canterbury corpus	hcmx max (best-of v3.3/trit2/Bin1) 123,701 (decode not recorded)	cmix v21 (published) 104,105 (decode not recorded)	18.8% behind cmix v21 (published)	Harrington best-of (per file) 123,837 in 3.4 s beats ZPAQ -m5 124,713 in 11.5 s
Silesia corpus (full)	Harrington best-of (per file) 197,371 (258.9 s)	ZPAQ -m5 184,547 (554.2 s)	6.9% behind ZPAQ -m5	zip-hi (7-Zip zip/PPMd, per-file) 207,119 in 13.6 s beats bzip3 221,432 in 23.1 s
Silesia 1 MB sample (Goodwin)	Harrington best-of (per file) 279,108 (14.6 s)	ZPAQ -m5 271,711 (54.7 s)	2.7% behind ZPAQ -m5	Harrington best-of (per file) 279,108 in 14.6 s beats lpaq1 -6 289,591 in 15.8 s
Synthetic set (Goodwin)	Harrington best-of (per file) 445,817 (22.2 s)	ZPAQ -m5 455,660 (83.1 s)	**OURS SMALLEST OF ALL**	Harrington best-of (per file) 445,817 in 22.2 s beats ZPAQ -m5 455,660 in 83.1 s
Biased packed data	Harrington best-of (per file) 465,170 (15 ms)	order-0 rANS 447,217 (28 ms)	4.0% behind order-0 rANS	Harrington best-of (per file) 465,170 in 15 ms beats ZPAQ -m5 465,419 in 10.6 s
Radio SDR captures	xz-hi 612,815 (171 ms)	ZPAQ -m5 (decode estimated) 577,465 (15.0 s)	6.1% behind ZPAQ -m5 (decode estimate	-
Modern logs and exports	xz-hi (smallest) 60,702 (390 ms)	xz -9e 61,305 (384 ms)	**OURS SMALLEST OF ALL**	xz-hi (smallest) 60,702 in 390 ms beats bzip2 -9 111,264 in 1.9 s
PDF documents (139 files)	Ours, smallest always 837,340 (11.6 s)	qpdf optimisation 851,978 (11.3 s)	**OURS SMALLEST OF ALL**	Ours, No-Pain pick 846,683 in 11.2 s beats qpdf optimisation 851,978 in 11.3 s
Speech audio (FLAC, 6 speakers)	flac-hi (No-Pain) 520,279 (123 ms)	flac -8 531,634 (136 ms)	**OURS SMALLEST OF ALL**	flac-hi (No-Pain) 520,279 in 123 ms beats flac -8 531,634 in 136 ms

Canterbury corpus: every recorded result, smallest first (2,810,784 bytes)

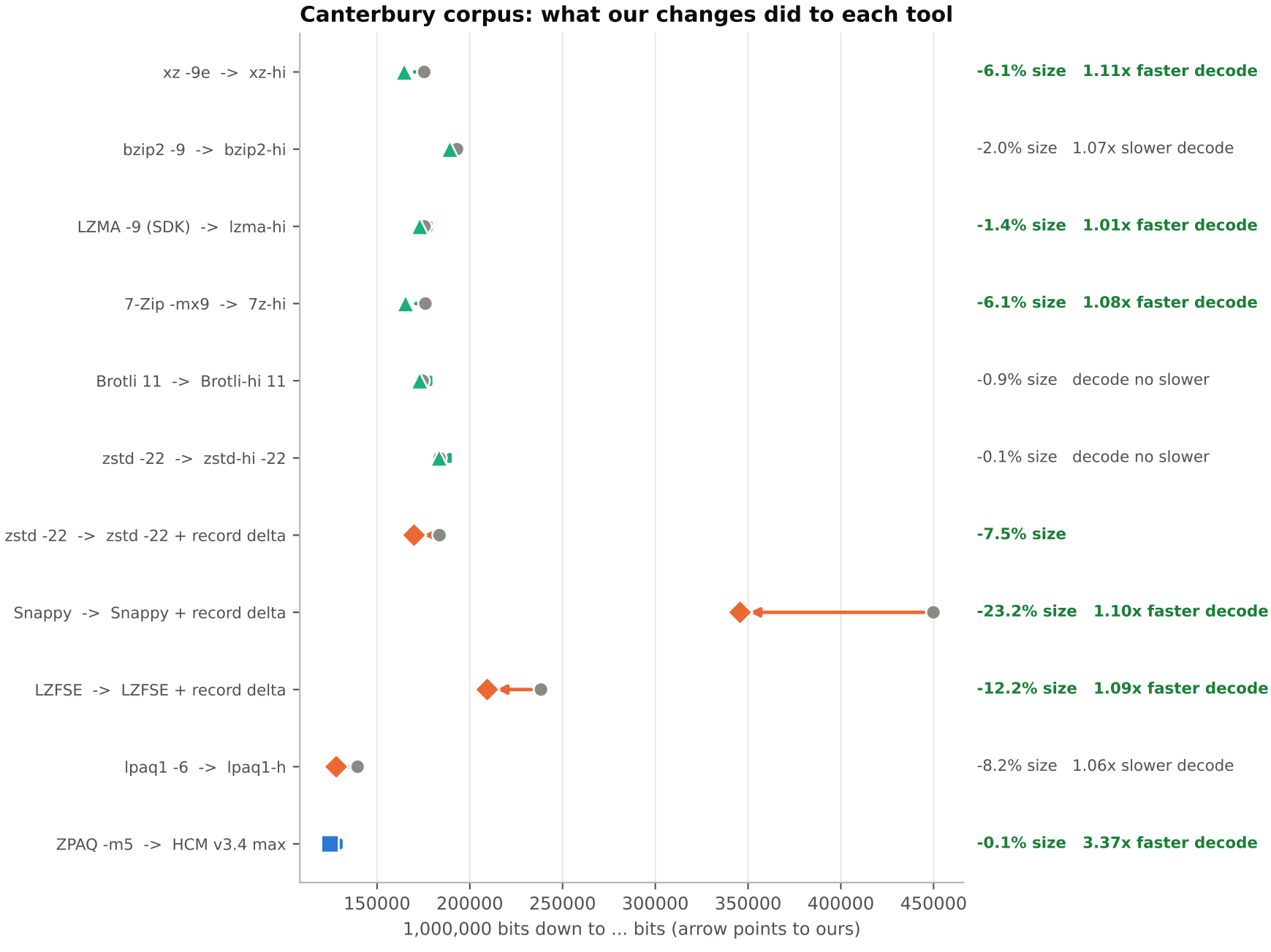


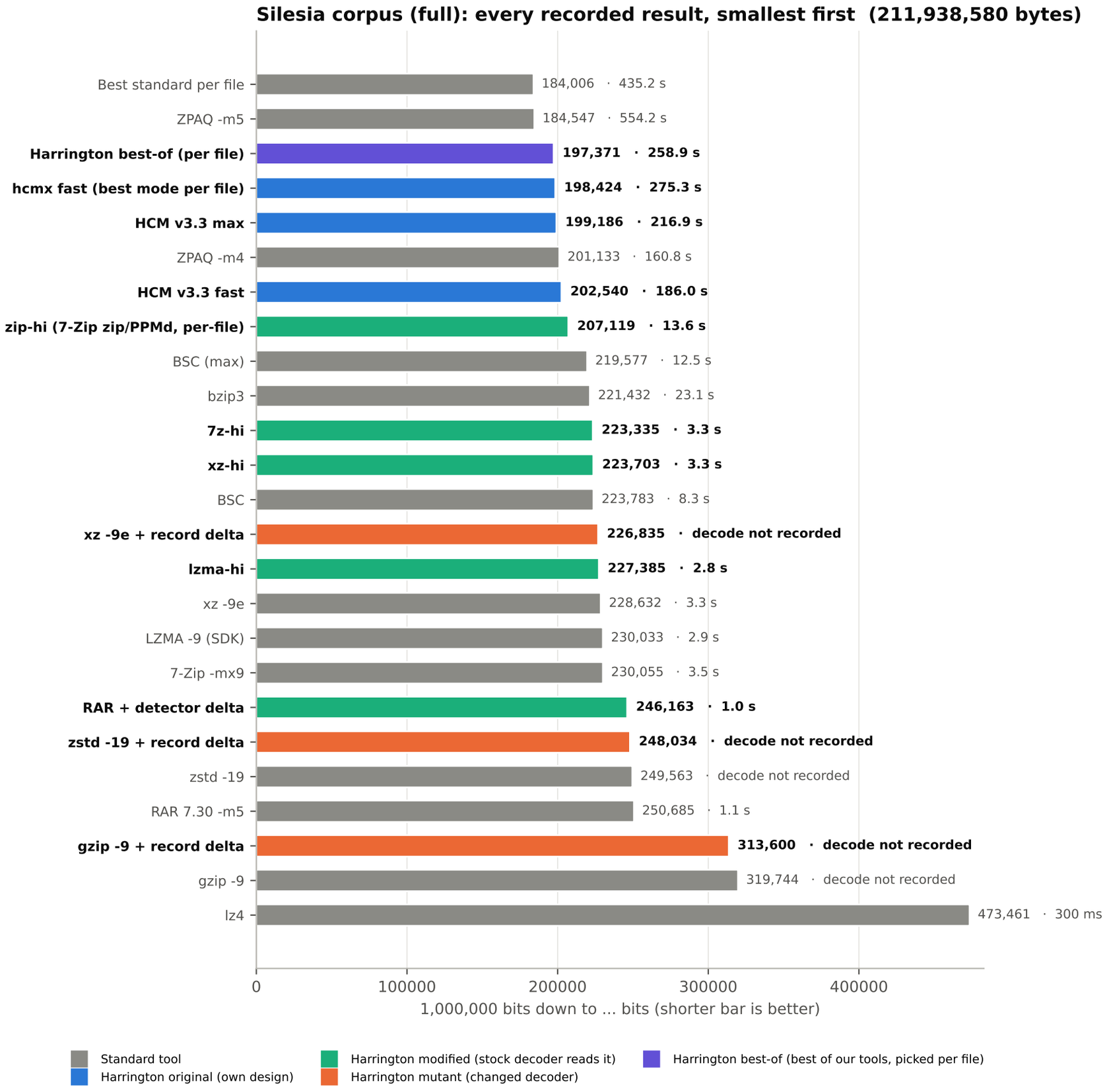
Decode times: † = recorded in a book chapter run (in-process); unmarked = 29 Oct 2026 same-machine run (wall clock).

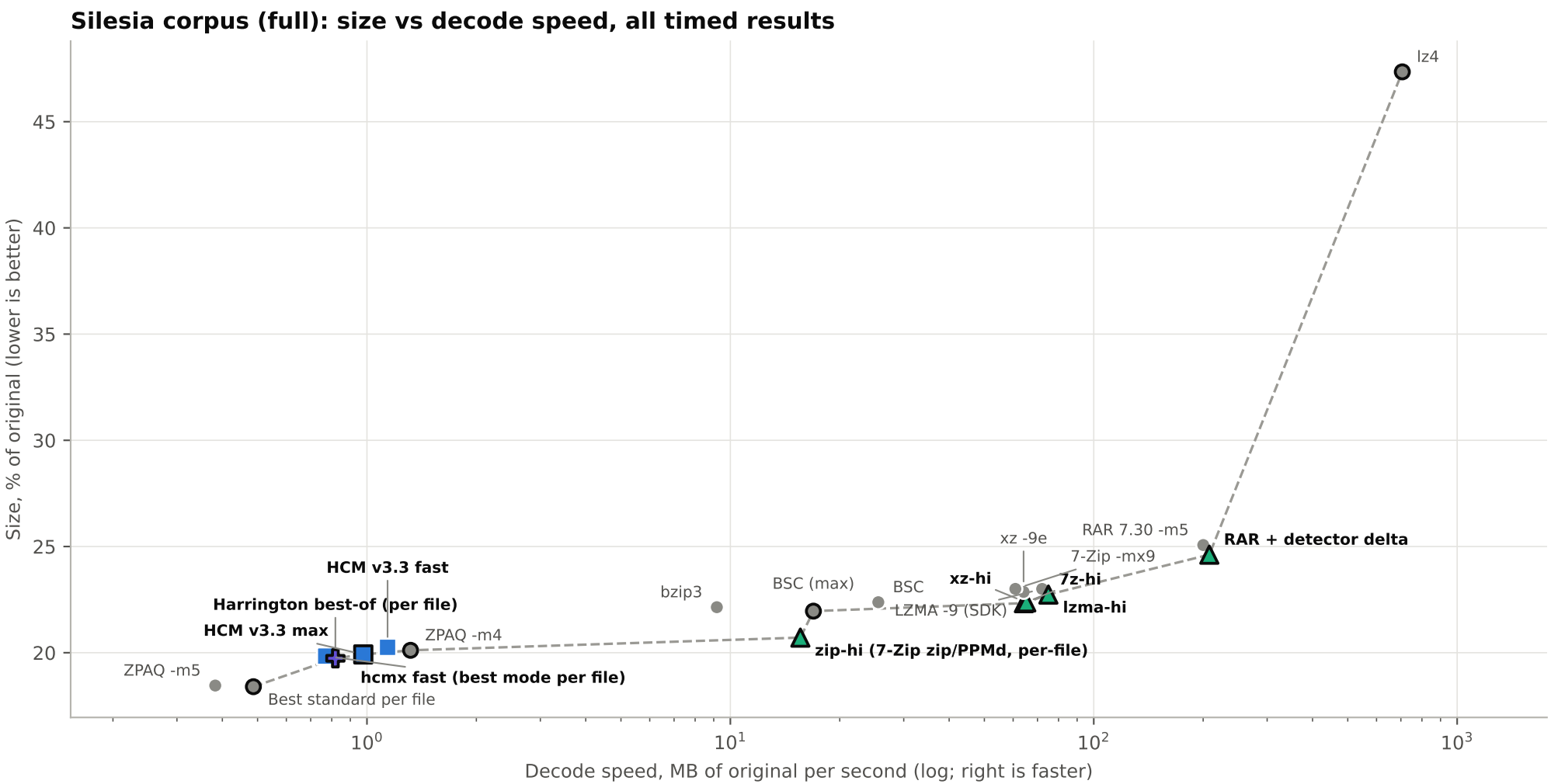


Timing sources: ch4, ch5, run. Times from different book chapters were measured in different runs; each stock-vs-ours pair comes from the same run.

- Standard tool
- Harrington original (own design)
- Harrington modified (stock decoder reads it)
- Harrington mutant (changed decoder)
- Harrington best-of (best of our tools, picked per file)
- Best-trade-off line (nothing beats these on both)
- On the line

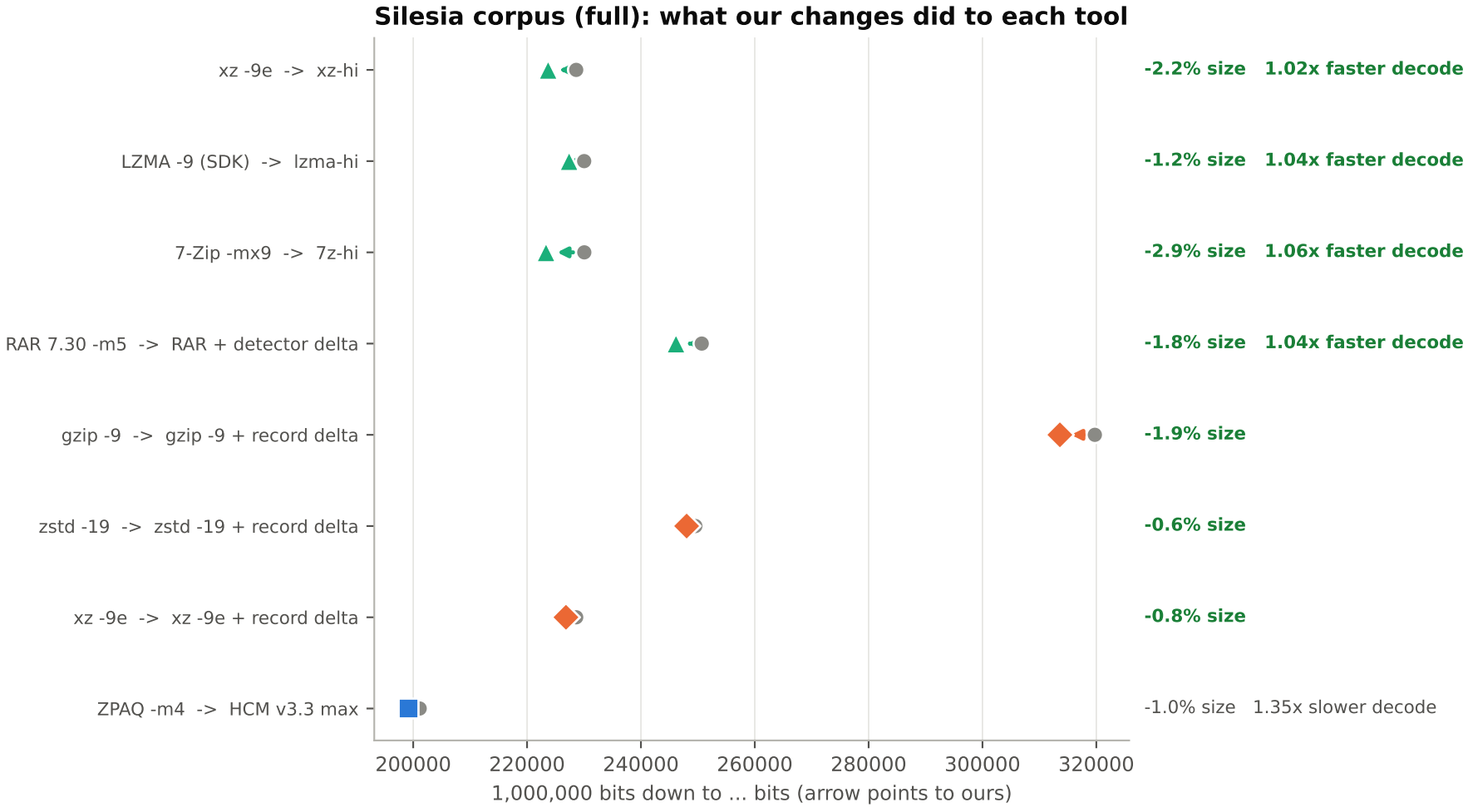

Canterbury corpus: what our changes did to each tool
xz -9e -> xz-hi
bzip2 -9 -> bzip2-hi
LZMA -9 (SDK) -> lzma-hi
7-Zip -mx9 -> 7z-hi
Brotli 11 -> Brotli-hi 11
zstd -22 -> zstd-hi -22
zstd -22 -> zstd -22 + record delta
Snappy -> Snappy + record delta
LZFSE -> LZFSE + record delta
lpaq1 -6 -> lpaq1-h
ZPAQ -m5 -> HCM v3.4 max
-6.1% size 1.11x faster decode
-2.0% size 1.07x slower decode
-1.4% size 1.01x faster decode
-6.1% size 1.08x faster decode
-0.9% size decode no slower
-0.1% size decode no slower
-7.5% size
-23.2% size 1.10x faster decode
-12.2% size 1.09x faster decode
-8.2% size 1.06x slower decode
-0.1% size 3.37x faster decode
150000
200000
250000
300000
350000
400000
450000
1,000,000 bits down to ... bits (arrow points to ours)

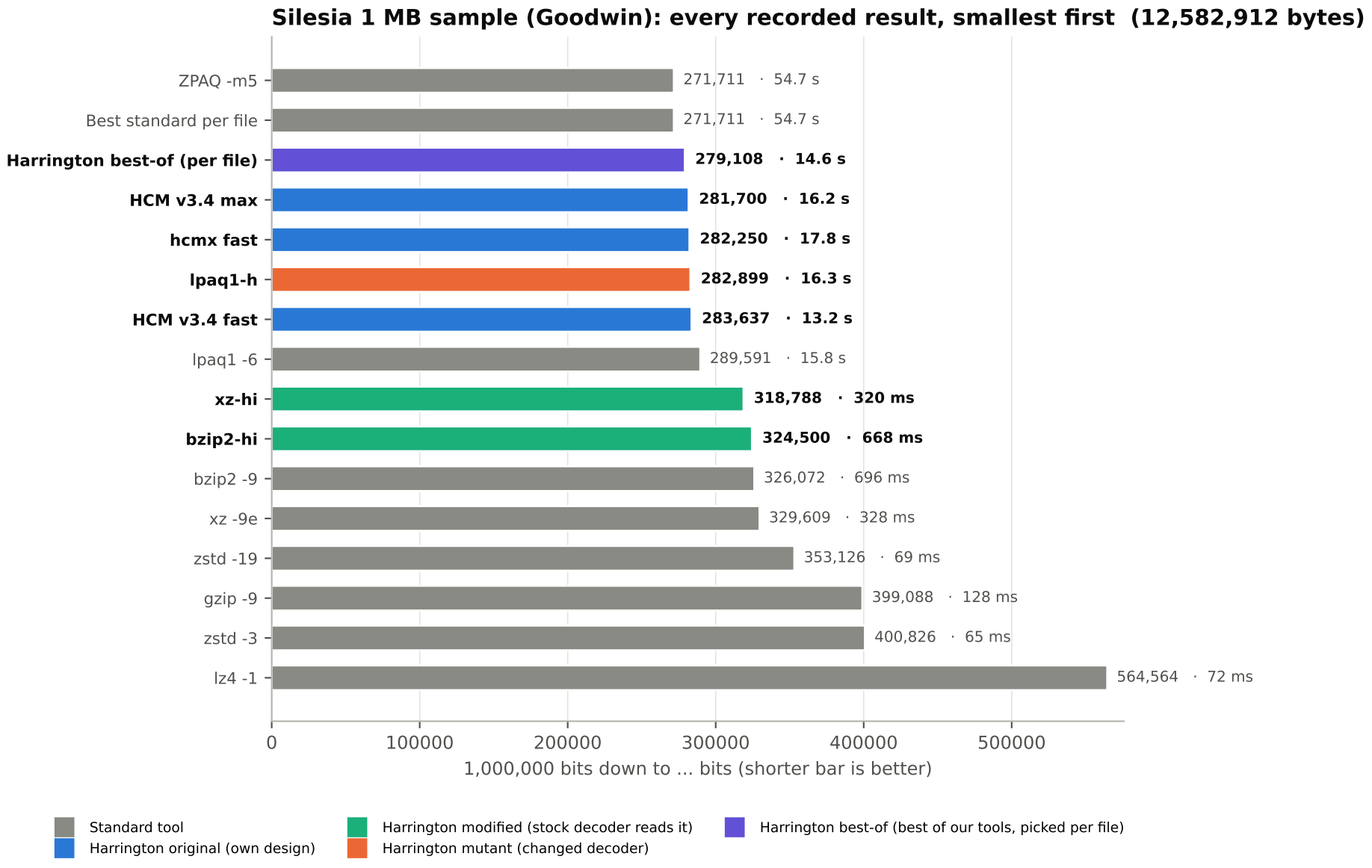

Silesia corpus (full): every recorded result, smallest first (211,938,580 bytes)
Best standard per file
ZPAQ -m5
Harrington best-of (per file)
hcmx fast (best mode per file)
HCM v3.3 max
ZPAQ -m4
HCM v3.3 fast
zip-hi (7-Zip zip/PPMd, per-file)
BSC (max)
bzip3
7z-hi
xz-hi
BSC
xz -9e + record delta
lzma-hi
xz -9e
LZMA -9 (SDK)
7-Zip -mx9
RAR + detector delta
zstd -19 + record delta
zstd -19
RAR 7.30 -m5
gzip -9 + record delta
gzip -9
lz4
184,006 · 435.2 s
184,547 · 554.2 s
197,371 · 258.9 s
198,424 · 275.3 s
199,186 · 216.9 s
201,133 · 160.8 s
202,540 · 186.0 s
207,119 · 13.6 s
219,577 · 12.5 s
221,432 · 23.1 s
223,335 · 3.3 s
223,703 · 3.3 s
223,783 · 8.3 s
226,835 · decode not recorded
227,385 · 2.8 s
228,632 · 3.3 s
230,033 · 2.9 s
230,055 · 3.5 s
246,163 · 1.0 s
248,034 · decode not recorded
249,563 · decode not recorded
250,685 · 1.1 s
313,600 · decode not recorded
319,744 · decode not recorded
473,461 · 300 ms
0
100000
200000
300000
400000
1,000,000 bits down to ... bits (shorter bar is better)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)

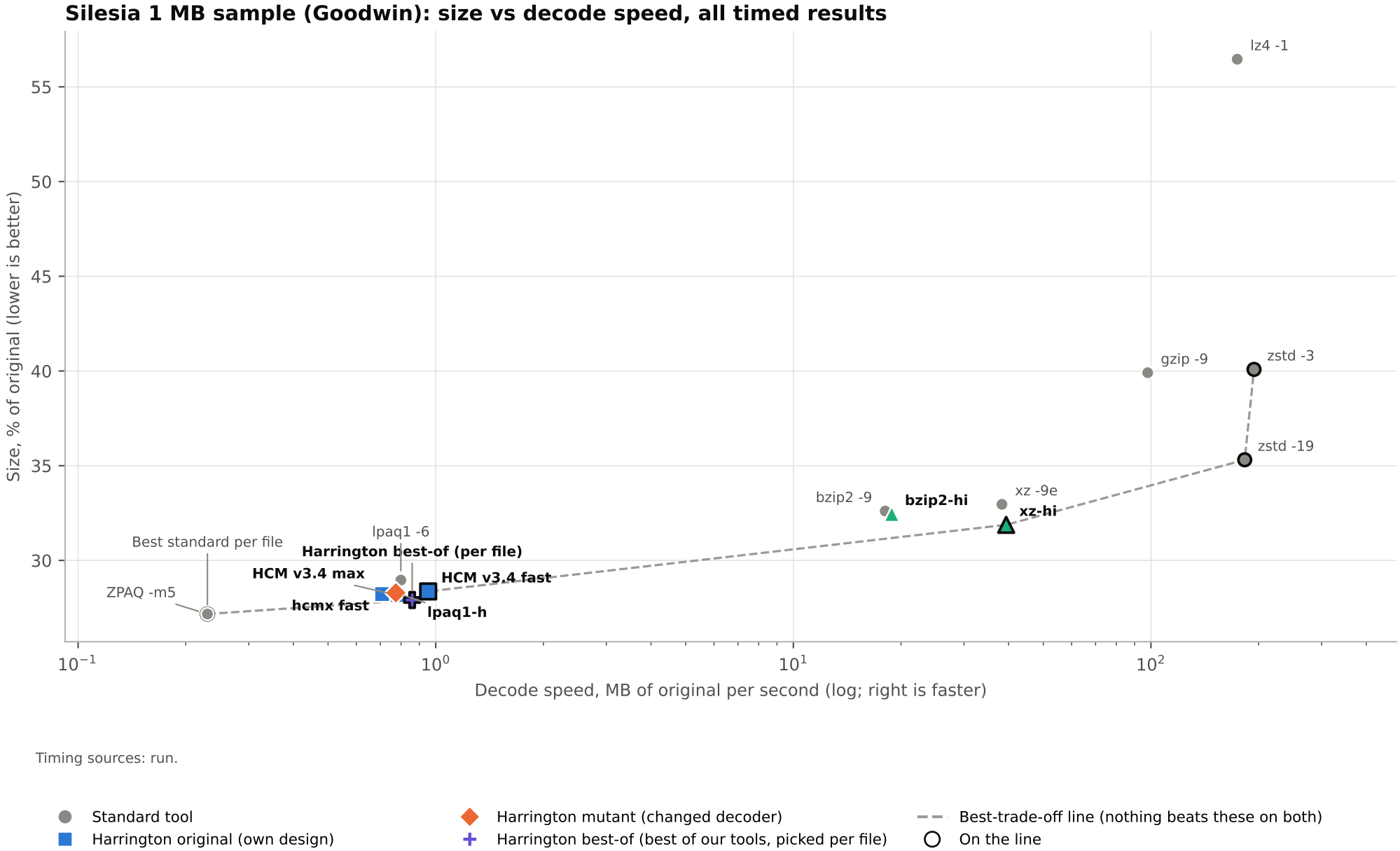


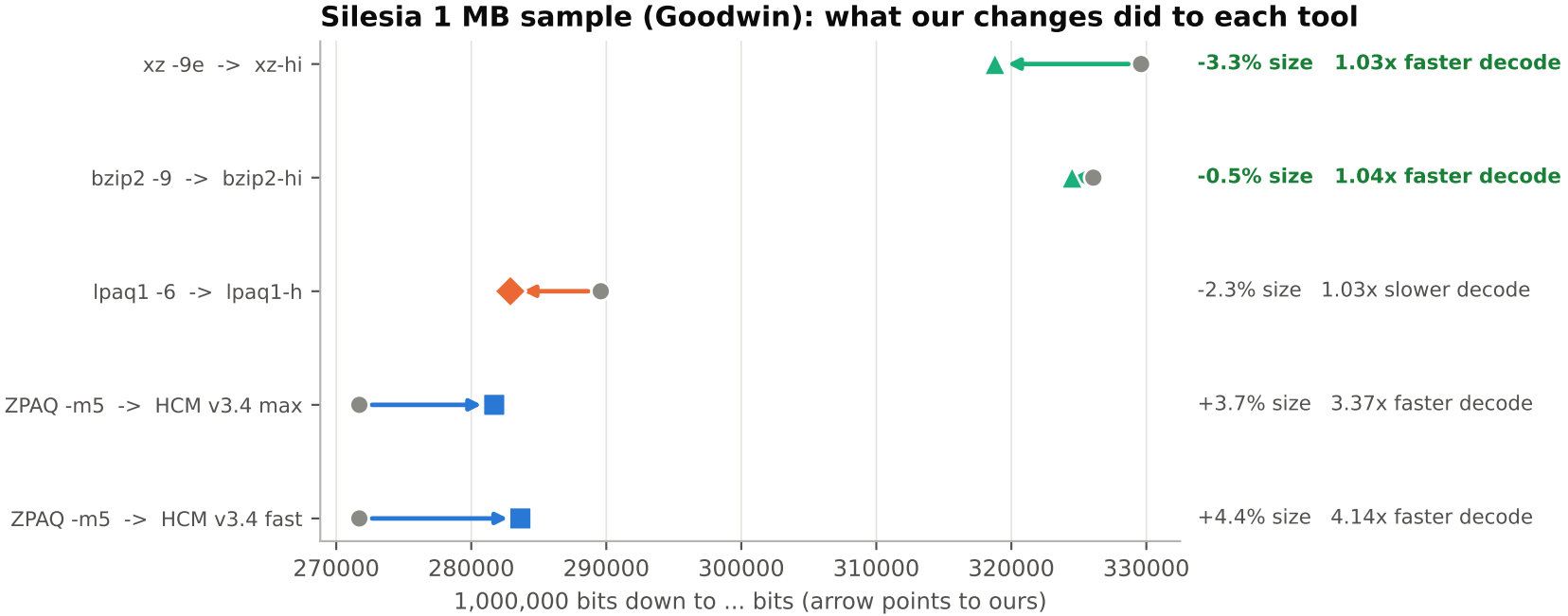
Timing sources: A.1/A.2, ch13, ch5, ch5/ch13/A, composite. Times from different book chapters were measured in different runs; each stock-vs-ours pair comes from the same run.

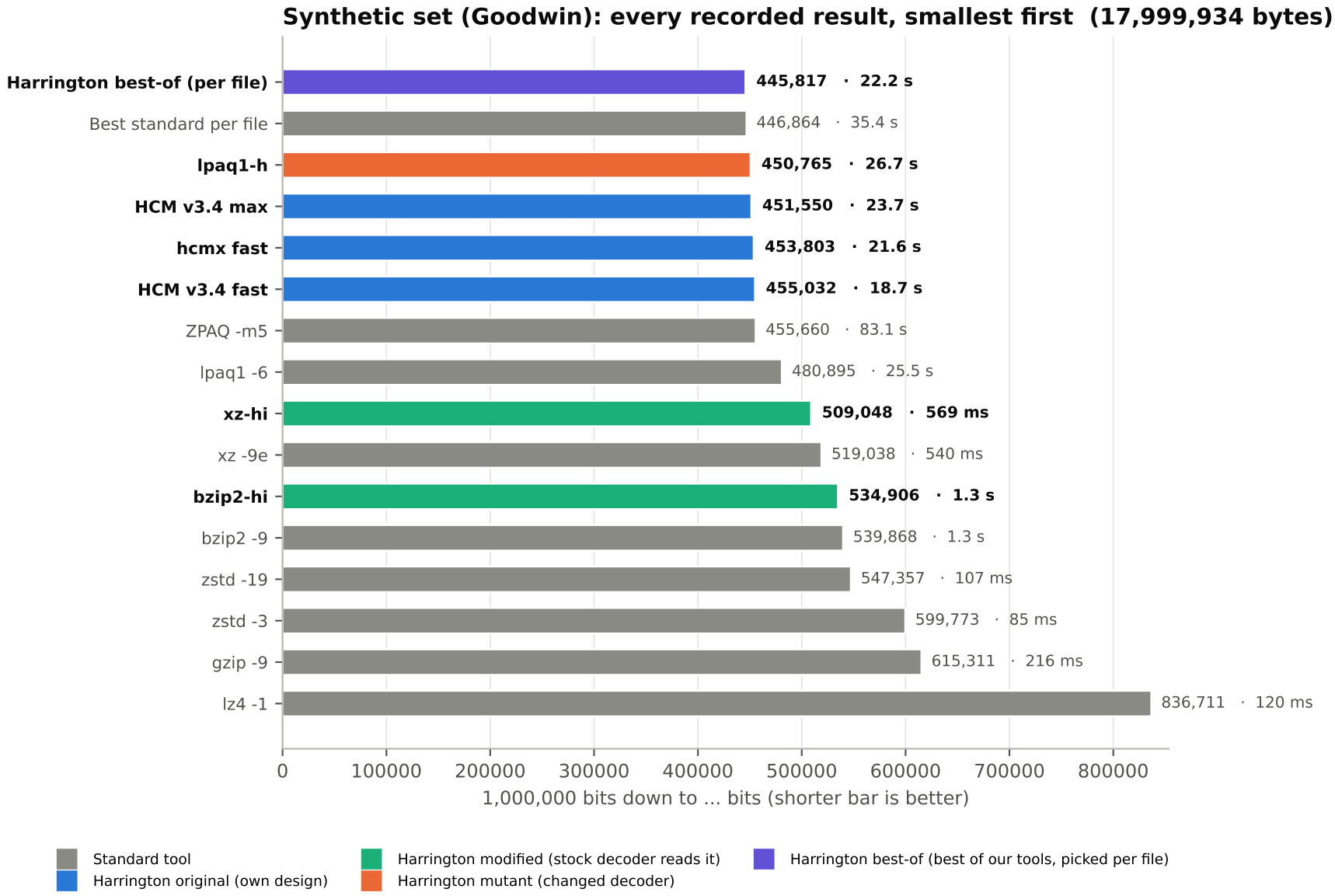
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)
Best-trade-off line (nothing beats these on both)
On the line

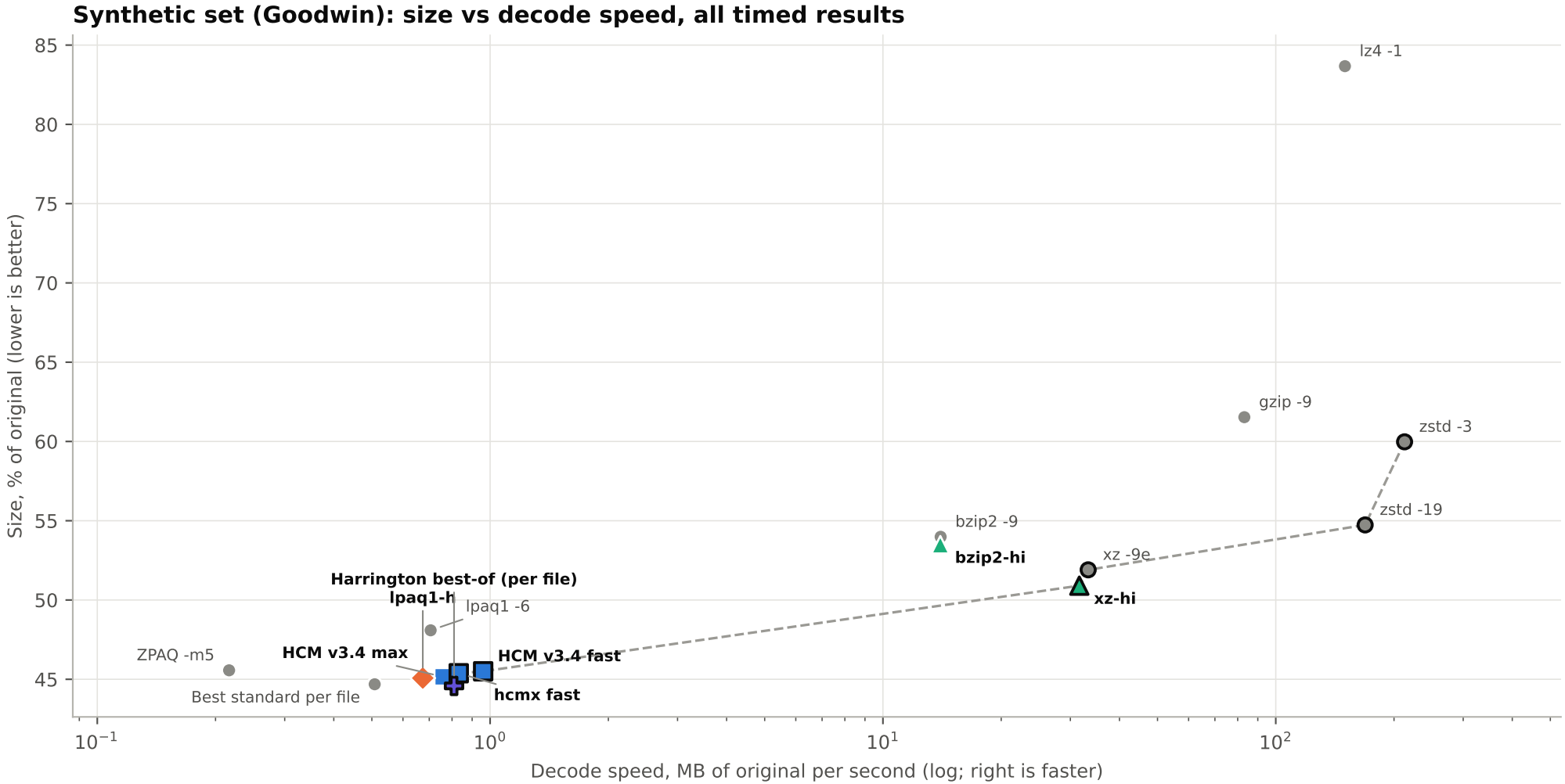

Silesia corpus (full): what our changes did to each tool
xz -9e -> xz-hi
LZMA -9 (SDK) -> lzma-hi
7-Zip -mx9 -> 7z-hi
RAR 7.30 -m5 -> RAR + detector delta
gzip -9 -> gzip -9 + record delta
zstd -19 -> zstd -19 + record delta
xz -9e -> xz -9e + record delta
ZPAQ -m4 -> HCM v3.3 max
-2.2% size 1.02x faster decode
-1.2% size 1.04x faster decode
-2.9% size 1.06x faster decode
-1.8% size 1.04x faster decode
-1.9% size
-0.6% size
-0.8% size
-1.0% size 1.35x slower decode
200000
220000
240000
260000
280000
300000
320000
1,000,000 bits down to ... bits (arrow points to ours)


Silesia 1 MB sample (Goodwin): every recorded result, smallest first (12,582,912 bytes)
ZPAQ -m5
271,711 · 54.7 s
Best standard per file
271,711 · 54.7 s
Harrington best-of (per file)
279,108 · 14.6 s
HCM v3.4 max
281,700 · 16.2 s
hcmx fast
282,250 · 17.8 s
lpaq1-h
282,899 · 16.3 s
HCM v3.4 fast
283,637 · 13.2 s
lpaq1 -6
289,591 · 15.8 s
xz-hi
318,788 · 320 ms
bzip2-hi
324,500 · 668 ms
bzip2 -9
326,072 · 696 ms
xz -9e
329,609 · 328 ms
zstd -19
353,126 · 69 ms
gzip -9
399,088 · 128 ms
zstd -3
400,826 · 65 ms
lz4 -1
564,564 · 72 ms
0
100000
200000
300000
400000
500000
1,000,000 bits down to ... bits (shorter bar is better)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)


Silesia 1 MB sample (Goodwin): size vs decode speed, all timed results
Size, % of original (lower is better)
55
50
45
40
35
30
10^{-1}
10^{0}
10^{1}
10^{2}
Decode speed, MB of original per second (log; right is faster)
lz4 -1
gzip -9
zstd -3
zstd -19
bzip2 -9
bzip2-hi
xz -9e
xz-hi
lpaq1 -6
Best standard per file
Harrington best-of (per file)
HCM v3.4 max
HCM v3.4 fast
ZPAQ -m5
hcmx fast
lpaq1-h
Timing sources: run.
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)
Best-trade-off line (nothing beats these on both)
On the line

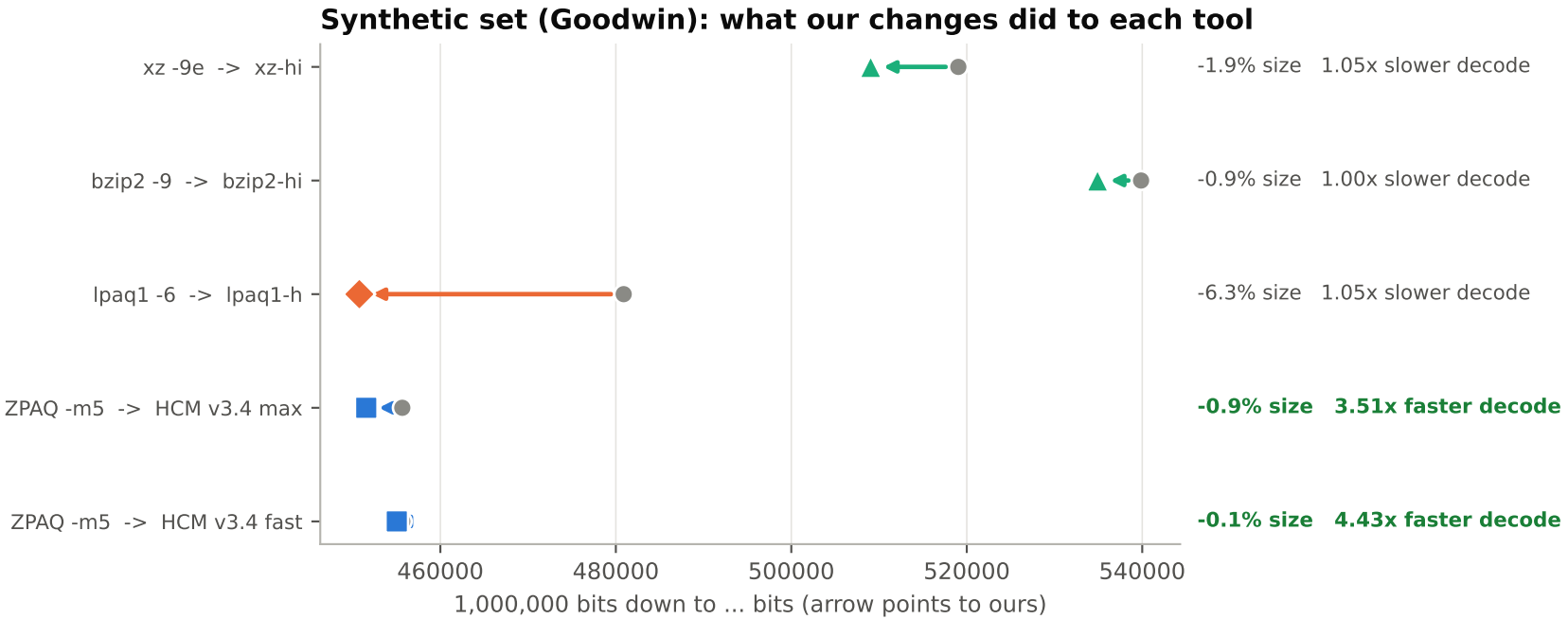

Silesia 1 MB sample (Goodwin): what our changes did to each tool
xz -9e -> xz-hi
bzip2 -9 -> bzip2-hi
lpaq1 -6 -> lpaq1-h
ZPAQ -m5 -> HCM v3.4 max
ZPAQ -m5 -> HCM v3.4 fast
-3.3% size 1.03x faster decode
-0.5% size 1.04x faster decode
-2.3% size 1.03x slower decode
+3.7% size 3.37x faster decode
+4.4% size 4.14x faster decode
270000
280000
290000
300000
310000
320000
330000
1,000,000 bits down to ... bits (arrow points to ours)

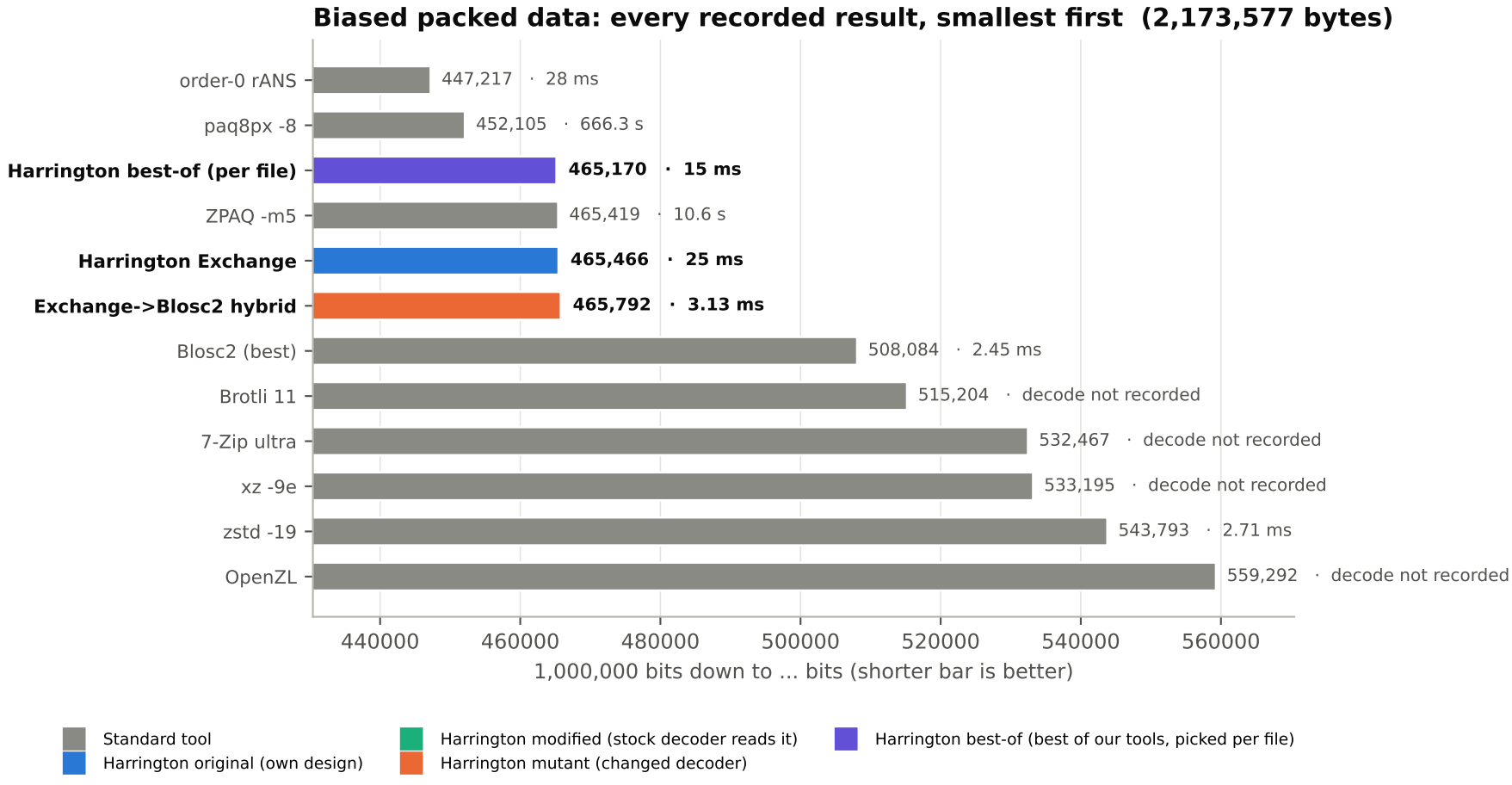

Synthetic set (Goodwin): every recorded result, smallest first (17,999,934 bytes)
Harrington best-of (per file)
445,817 · 22.2 s
Best standard per file
446,864 · 35.4 s
lpaq1-h
450,765 · 26.7 s
HCM v3.4 max
451,550 · 23.7 s
hcmx fast
453,803 · 21.6 s
HCM v3.4 fast
455,032 · 18.7 s
ZPAQ -m5
455,660 · 83.1 s
lpaq1 -6
480,895 · 25.5 s
xz-hi
509,048 · 569 ms
xz -9e
519,038 · 540 ms
bzip2-hi
534,906 · 1.3 s
bzip2 -9
539,868 · 1.3 s
zstd -19
547,357 · 107 ms
zstd -3
599,773 · 85 ms
gzip -9
615,311 · 216 ms
lz4 -1
836,711 · 120 ms
0
100000
200000
300000
400000
500000
600000
700000
800000
1,000,000 bits down to ... bits (shorter bar is better)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)

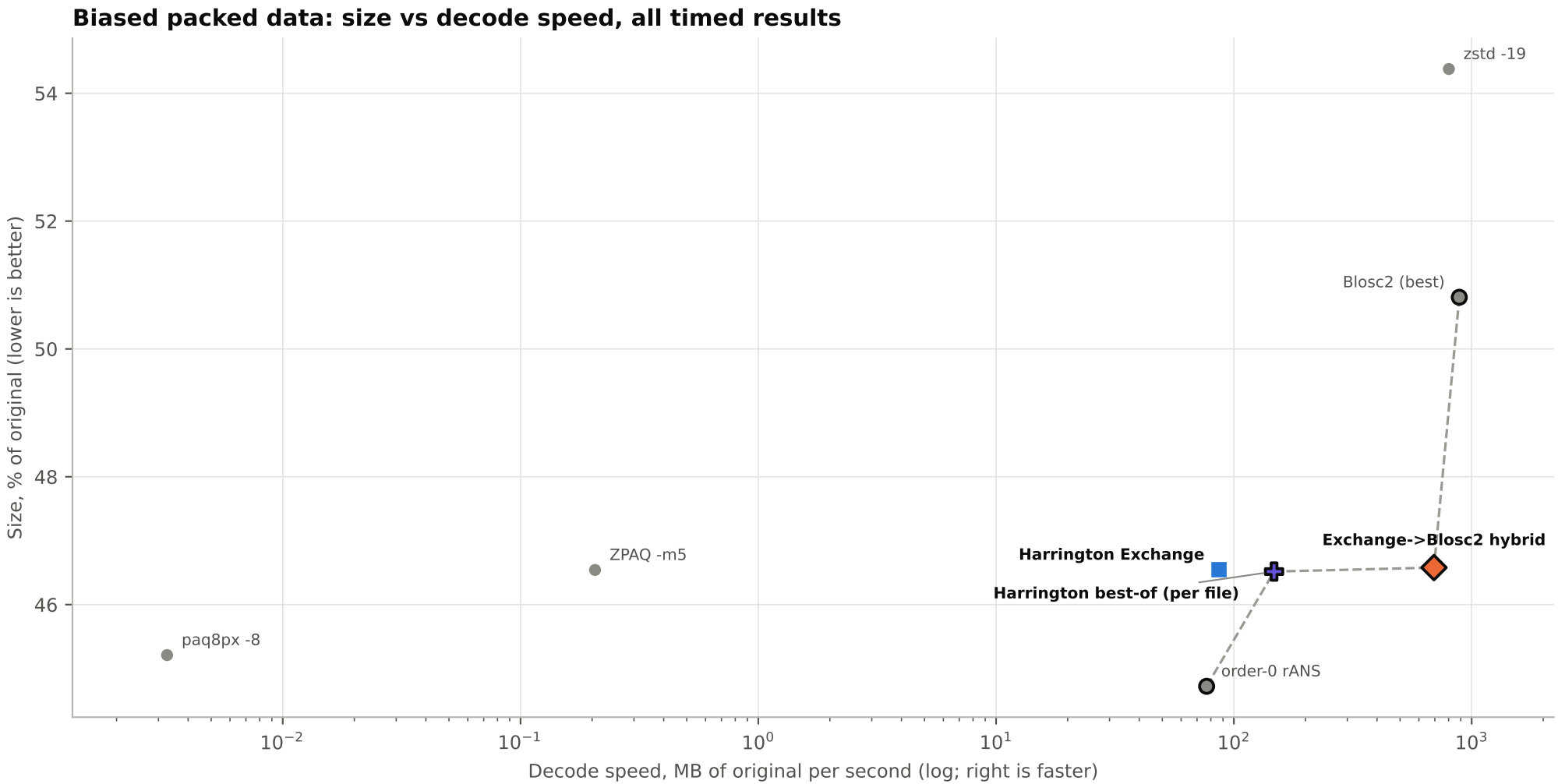


Timing sources: run.

Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)
Best-trade-off line (nothing beats these on both)
On the line

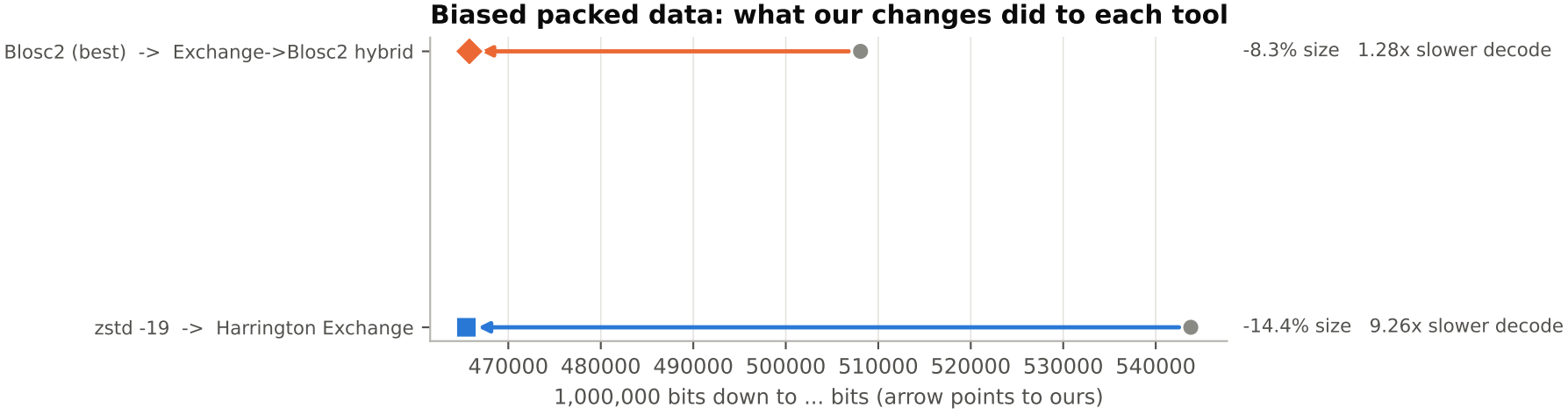

Synthetic set (Goodwin): what our changes did to each tool
xz -9e -> xz-hi
bzip2 -9 -> bzip2-hi
lpaq1 -6 -> lpaq1-h
ZPAQ -m5 -> HCM v3.4 max
ZPAQ -m5 -> HCM v3.4 fast
-1.9% size 1.05x slower decode
-0.9% size 1.00x slower decode
-6.3% size 1.05x slower decode
-0.9% size 3.51x faster decode
-0.1% size 4.43x faster decode
460000
480000
500000
520000
540000
1,000,000 bits down to ... bits (arrow points to ours)

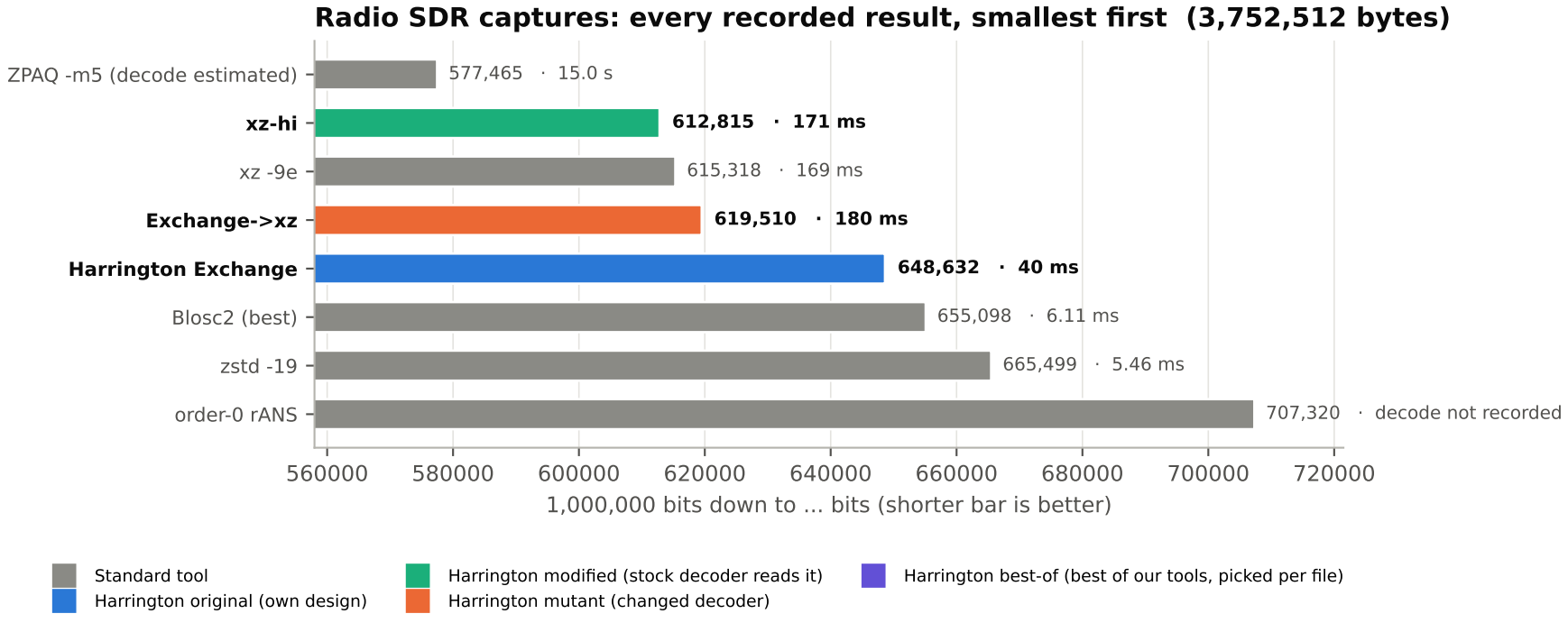

Biased packed data: every recorded result, smallest first (2,173,577 bytes)
order-0 rANS
447,217 · 28 ms
paq8px -8
452,105 · 666.3 s
Harrington best-of (per file)
465,170 · 15 ms
ZPAQ -m5
465,419 · 10.6 s
Harrington Exchange
465,466 · 25 ms
Exchange->Blosc2 hybrid
465,792 · 3.13 ms
Blosc2 (best)
508,084 · 2.45 ms
Brotli 11
515,204 · decode not recorded
7-Zip ultra
532,467 · decode not recorded
xz -9e
533,195 · decode not recorded
zstd -19
543,793 · 2.71 ms
OpenZL
559,292 · decode not recorded
440000
460000
480000
500000
520000
540000
560000
1,000,000 bits down to ... bits (shorter bar is better)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)

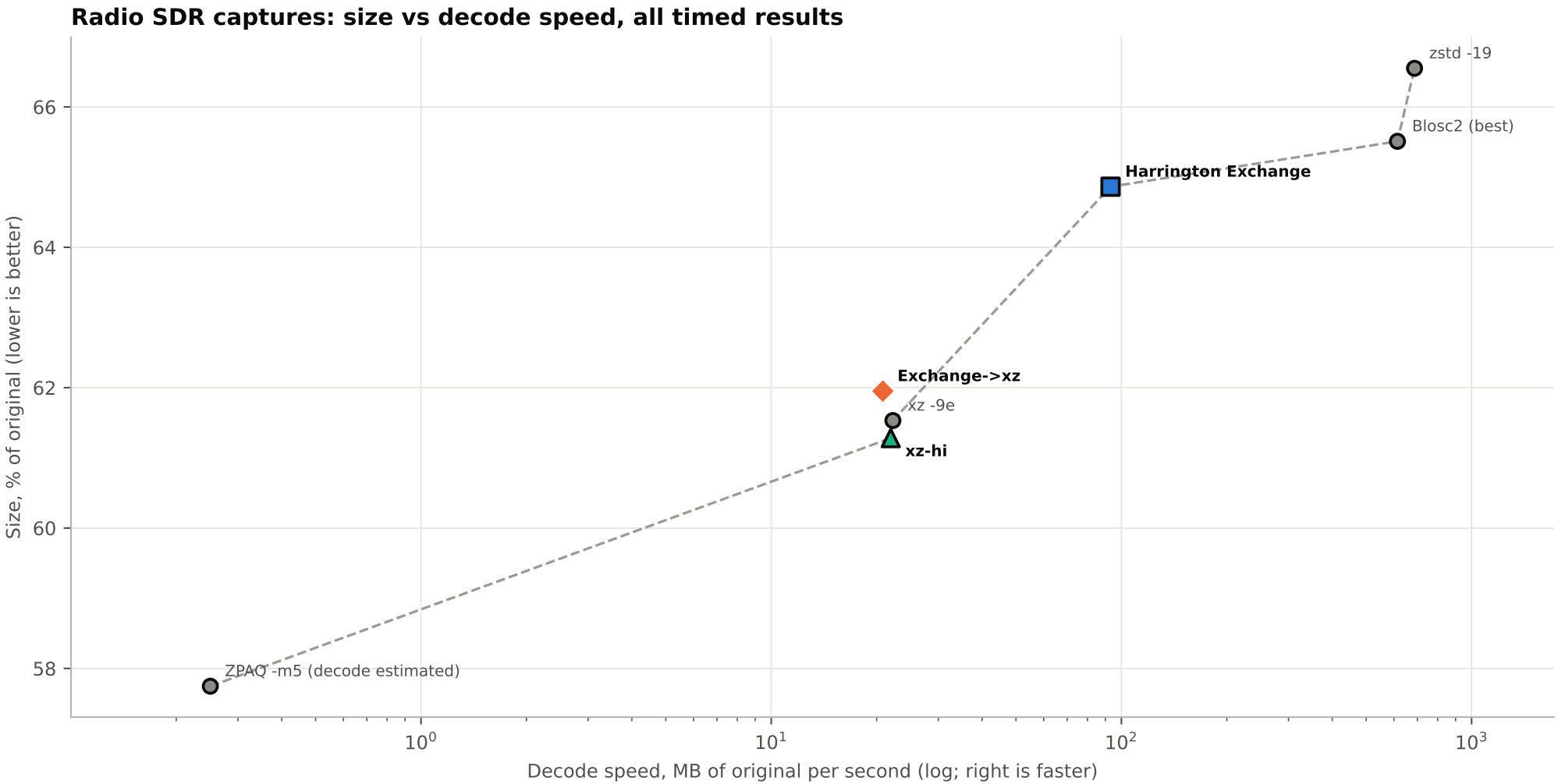


Timing sources: ch15, composite. Times from different book chapters were measured in different runs; each stock-vs-ours pair comes from the same run.

- Standard tool
- Harrington original (own design)
- Harrington modified (stock decoder reads it)
- Harrington mutant (changed decoder)
- Harrington best-of (best of our tools, picked per file)
- Best-trade-off line (nothing beats these on both)
- On the line

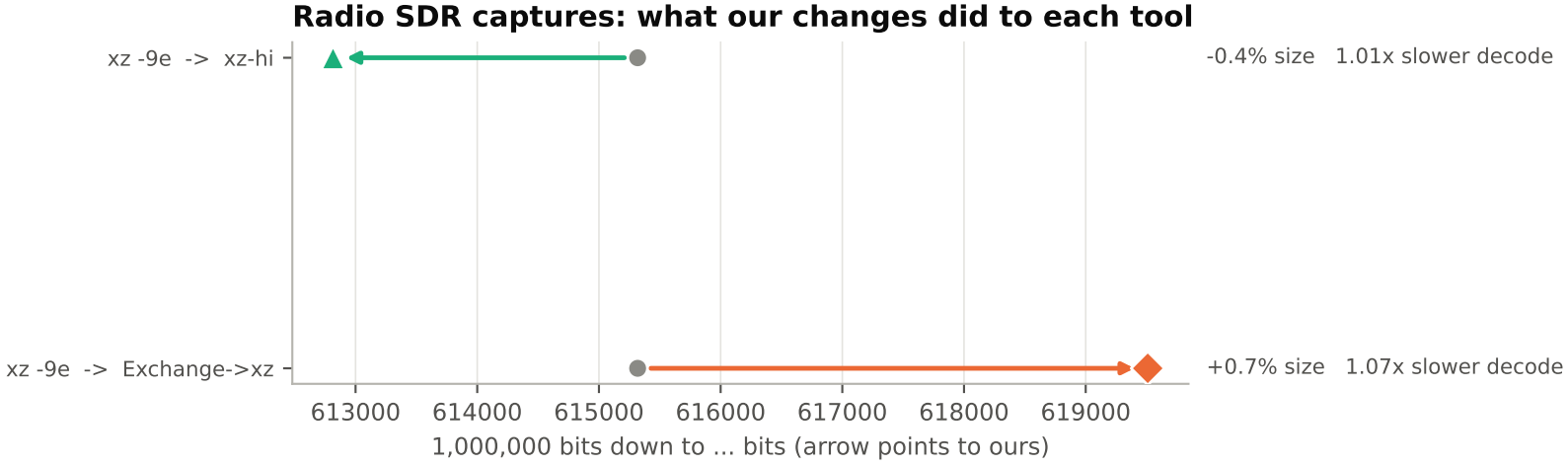

Biased packed data: what our changes did to each tool
Blosc2 (best) -> Exchange->Blosc2 hybrid
-8.3% size 1.28x slower decode
zstd -19 -> Harrington Exchange
-14.4% size 9.26x slower decode
470000 480000 490000 500000 510000 520000 530000 540000
1,000,000 bits down to ... bits (arrow points to ours)

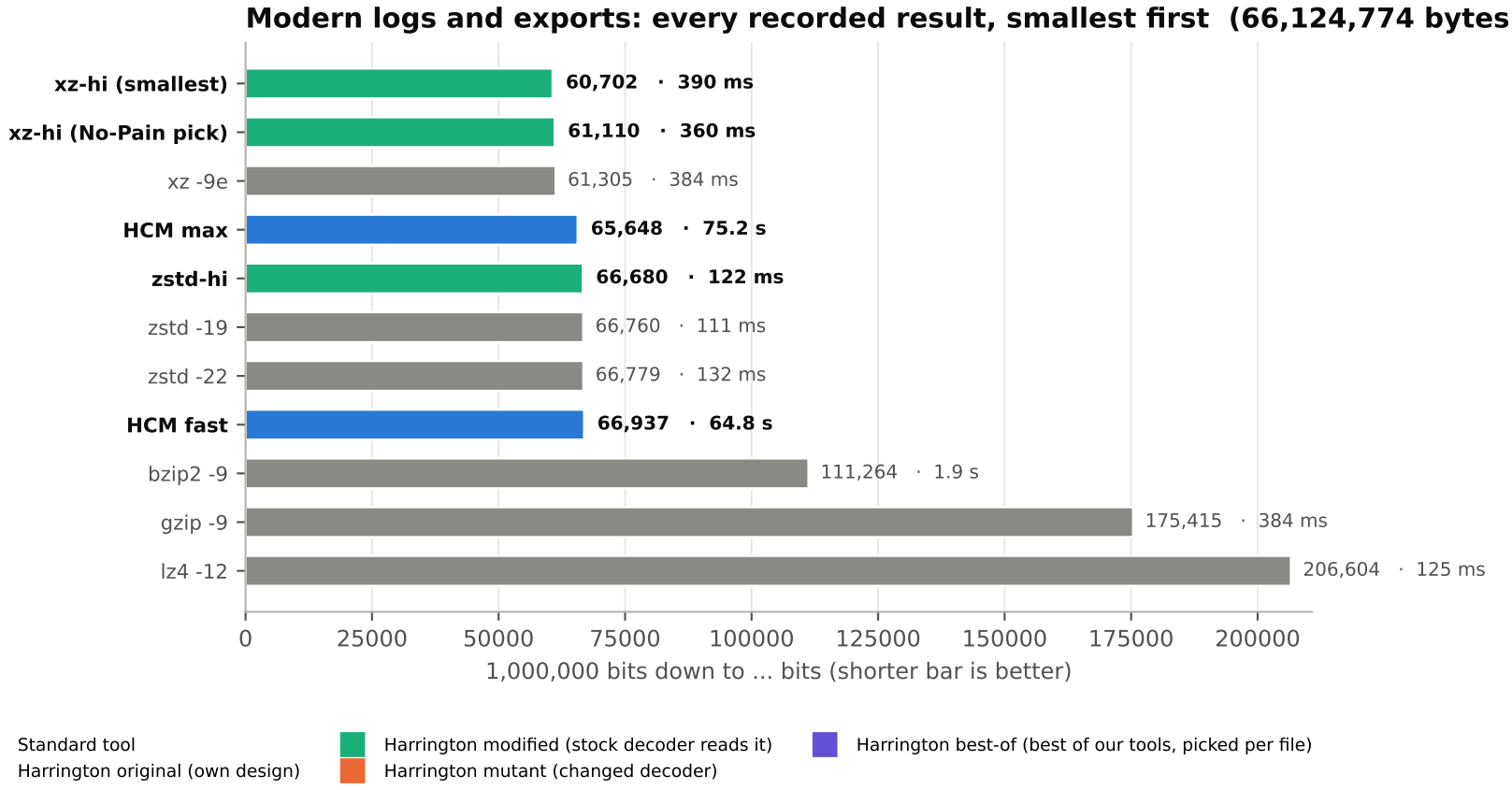

Radio SDR captures: every recorded result, smallest first (3,752,512 bytes)
ZPAQ -m5 (decode estimated)
577,465 · 15.0 s
xz-hi
612,815 · 171 ms
xz -9e
615,318 · 169 ms
Exchange->xz
619,510 · 180 ms
Harrington Exchange
648,632 · 40 ms
Blosc2 (best)
655,098 · 6.11 ms
zstd -19
665,499 · 5.46 ms
order-0 rANS
707,320 · decode not recorded
560000
580000
600000
620000
640000
660000
680000
700000
720000
1,000,000 bits down to ... bits (shorter bar is better)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)



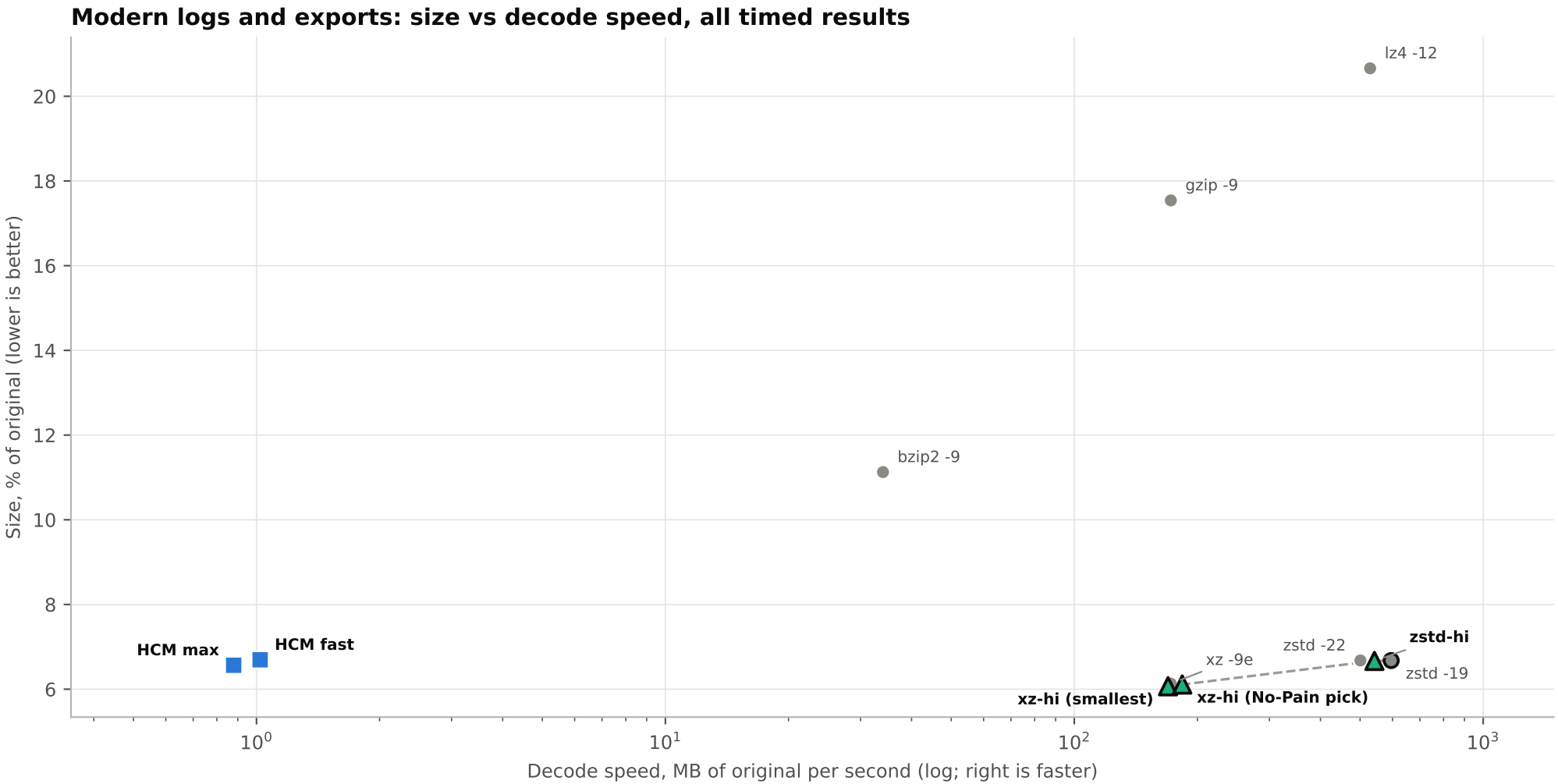
Timing sources: ch15.

Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)
Best-trade-off line (nothing beats these on both)
On the line


Radio SDR captures: what our changes did to each tool
xz -9e -> xz-hi
-0.4% size 1.01x slower decode
xz -9e -> Exchange->xz
+0.7% size 1.07x slower decode
613000
614000
615000
616000
617000
618000
619000
1,000,000 bits down to ... bits (arrow points to ours)

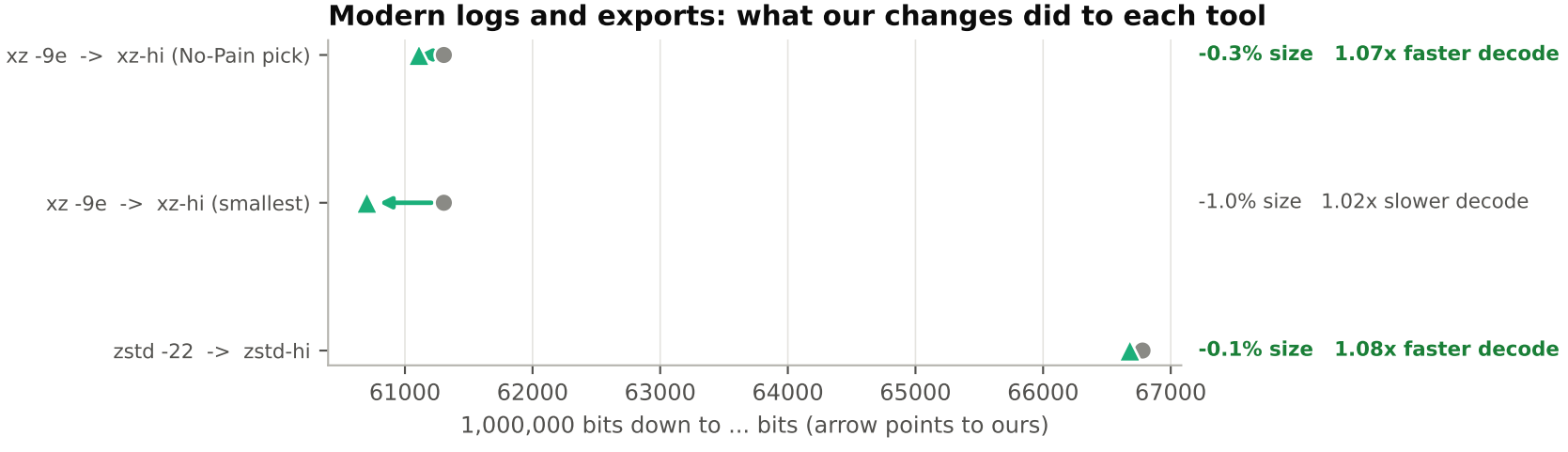


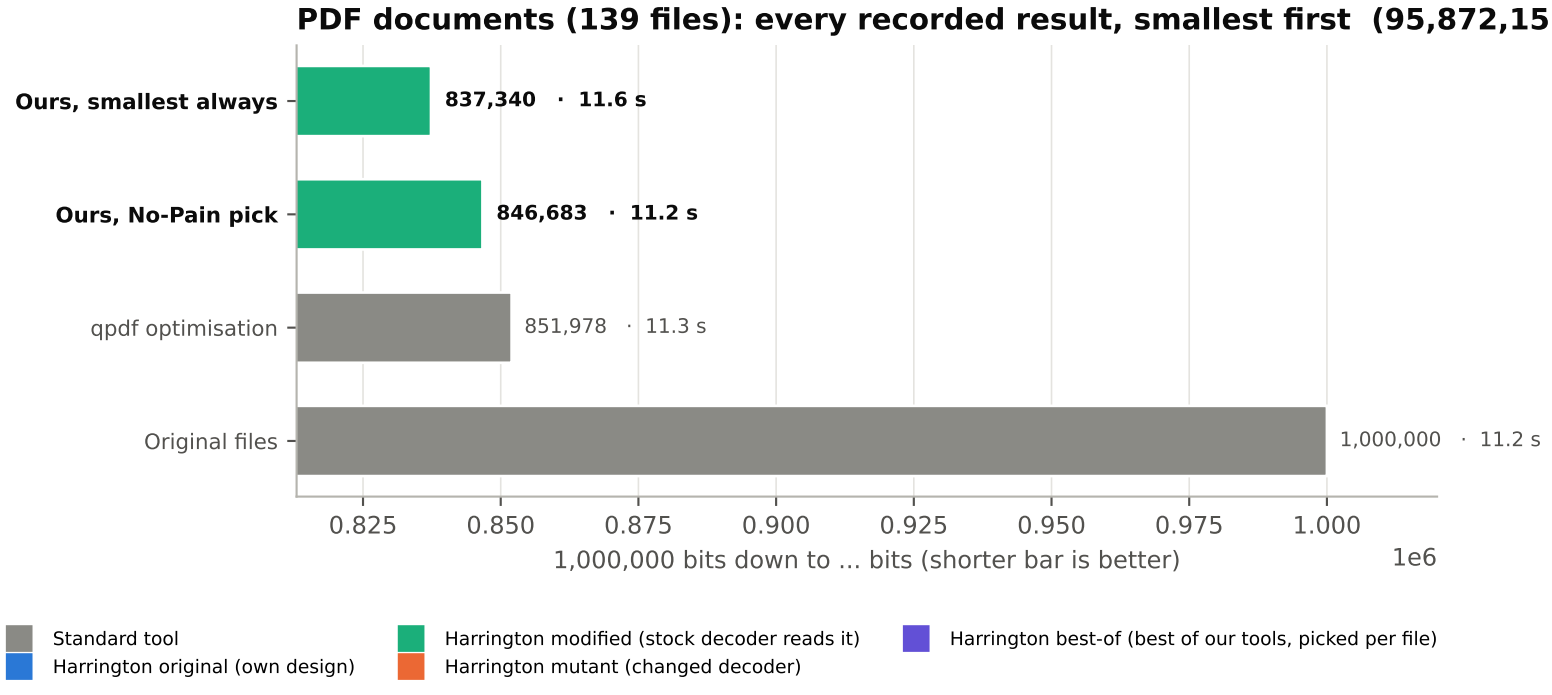
Note: HCM max is the smallest on the four log files (hdfs 42,132 -> 31,068 bytes vs the best stock tool, openstack 38,656 -> 30,192, apache 118,072 -> 80,254, nginx.json 228,132 -> 135,706) but loses on the large exports.



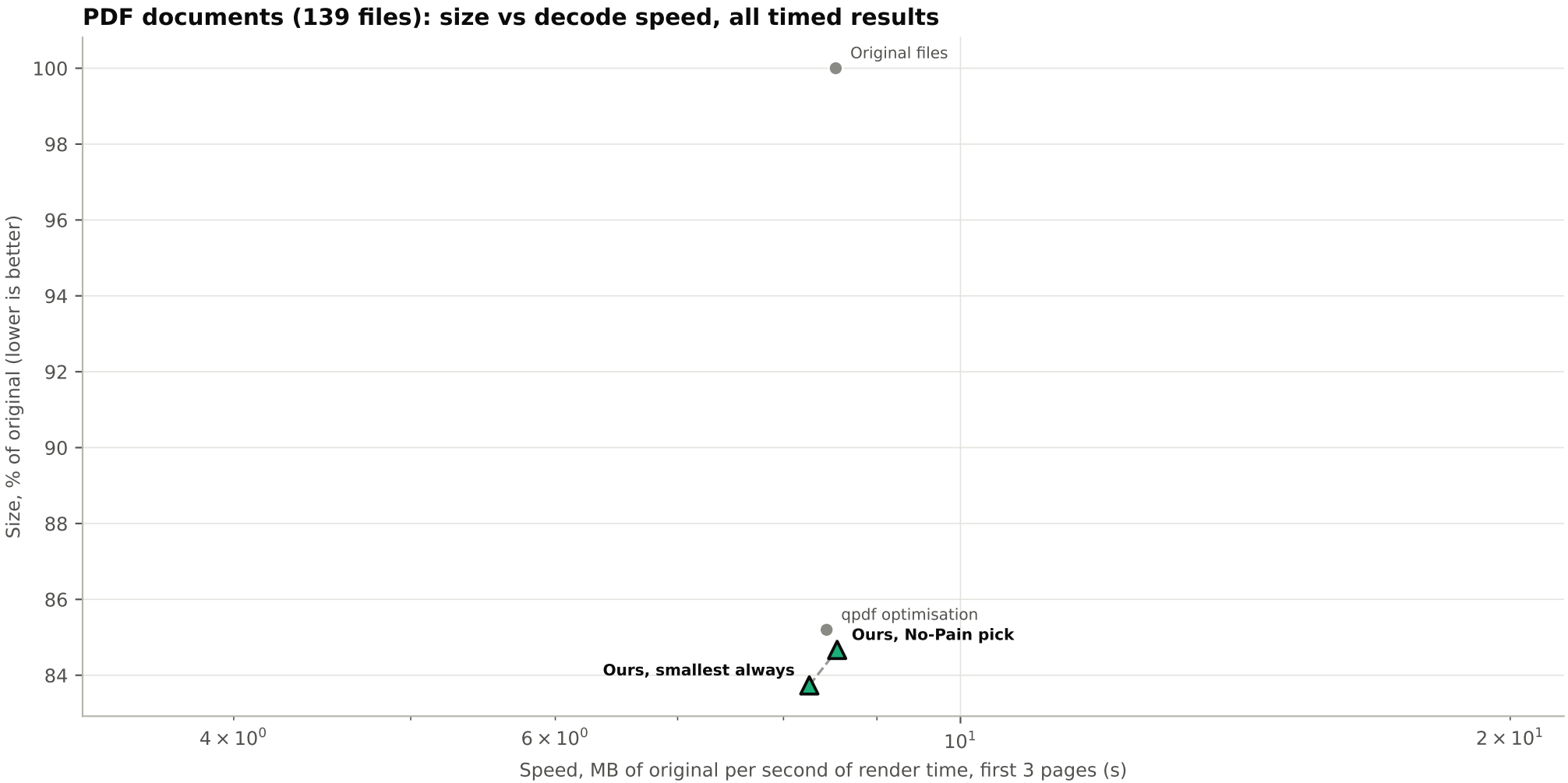
Timing sources: ch18.

Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)
Best-trade-off line (nothing beats these on both)
On the line


Modern logs and exports: what our changes did to each tool
xz -9e -> xz-hi (No-Pain pick)
xz -9e -> xz-hi (smallest)
zstd -22 -> zstd-hi
-0.3% size 1.07x faster decode
-1.0% size 1.02x slower decode
-0.1% size 1.08x faster decode
61000
62000
63000
64000
65000
66000
67000
1,000,000 bits down to ... bits (arrow points to ours)

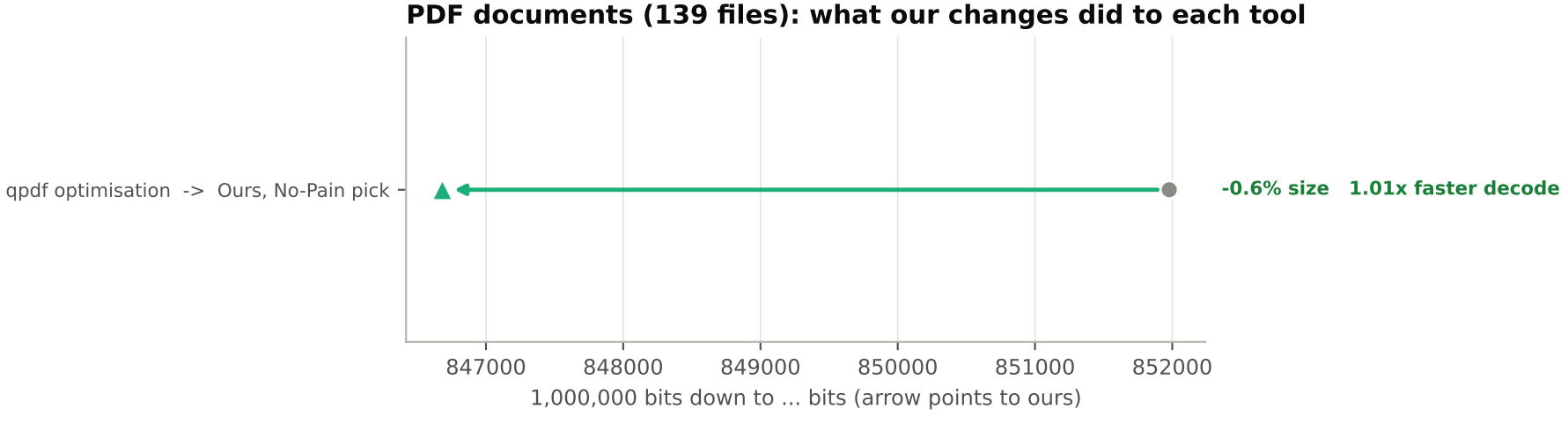


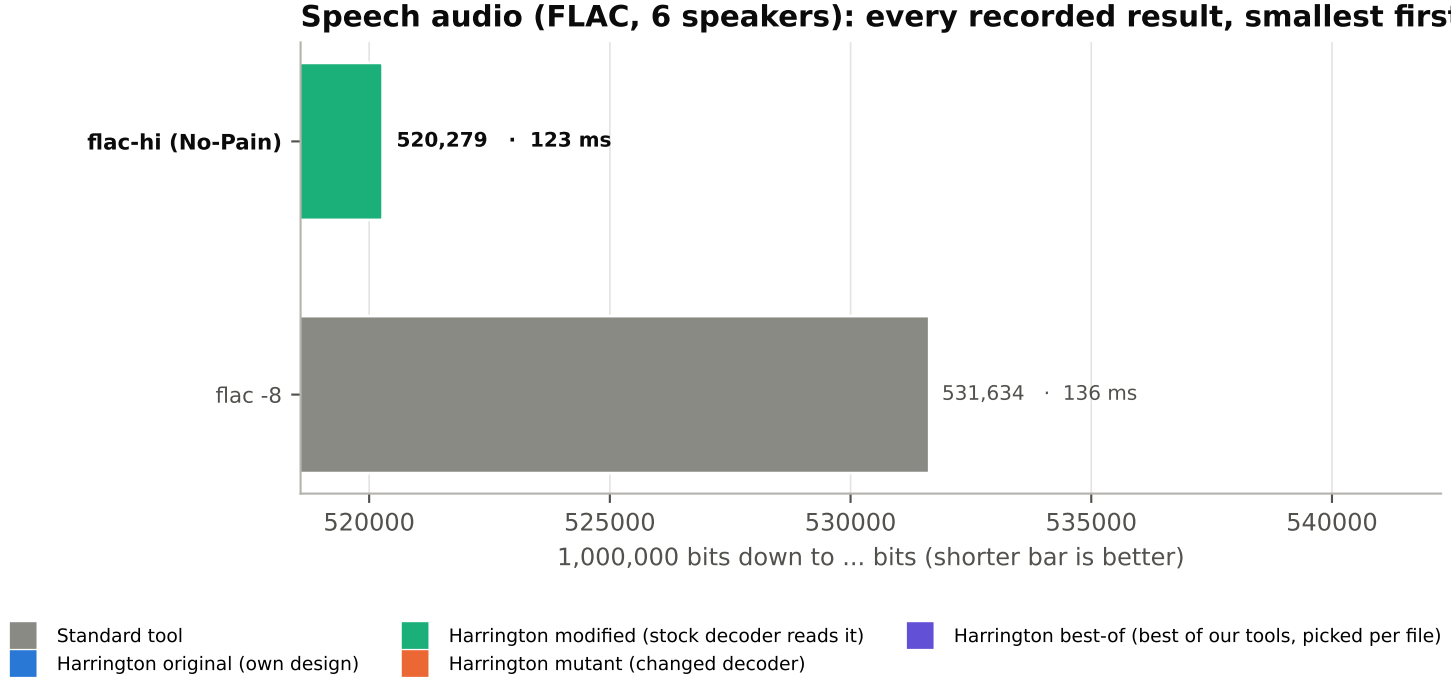
Time shown is render time, first 3 pages (s).

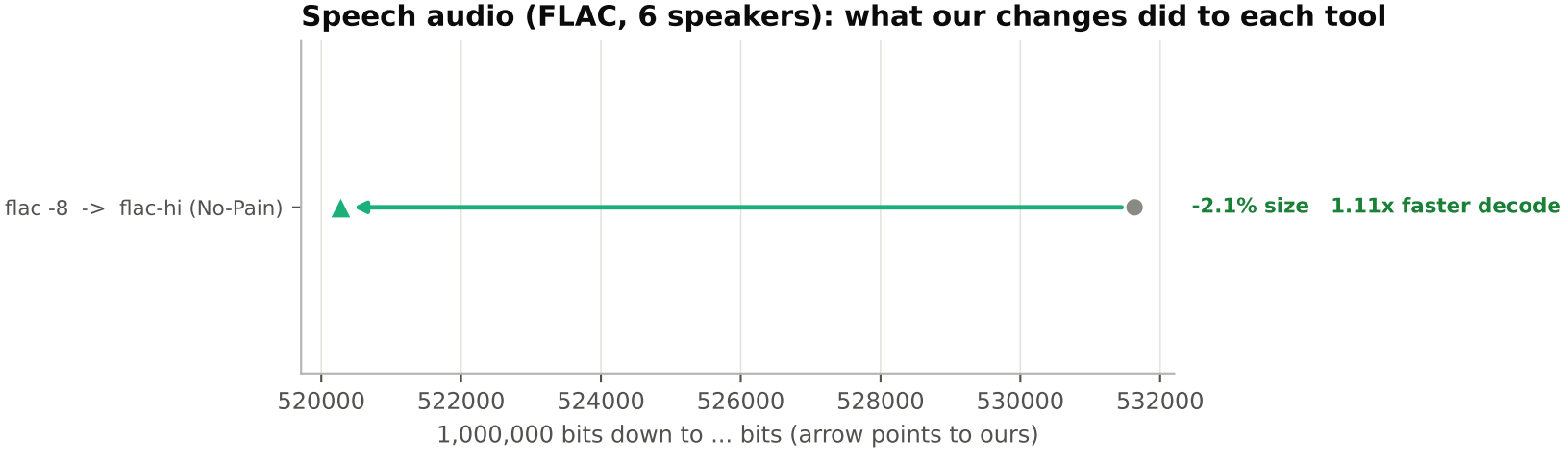


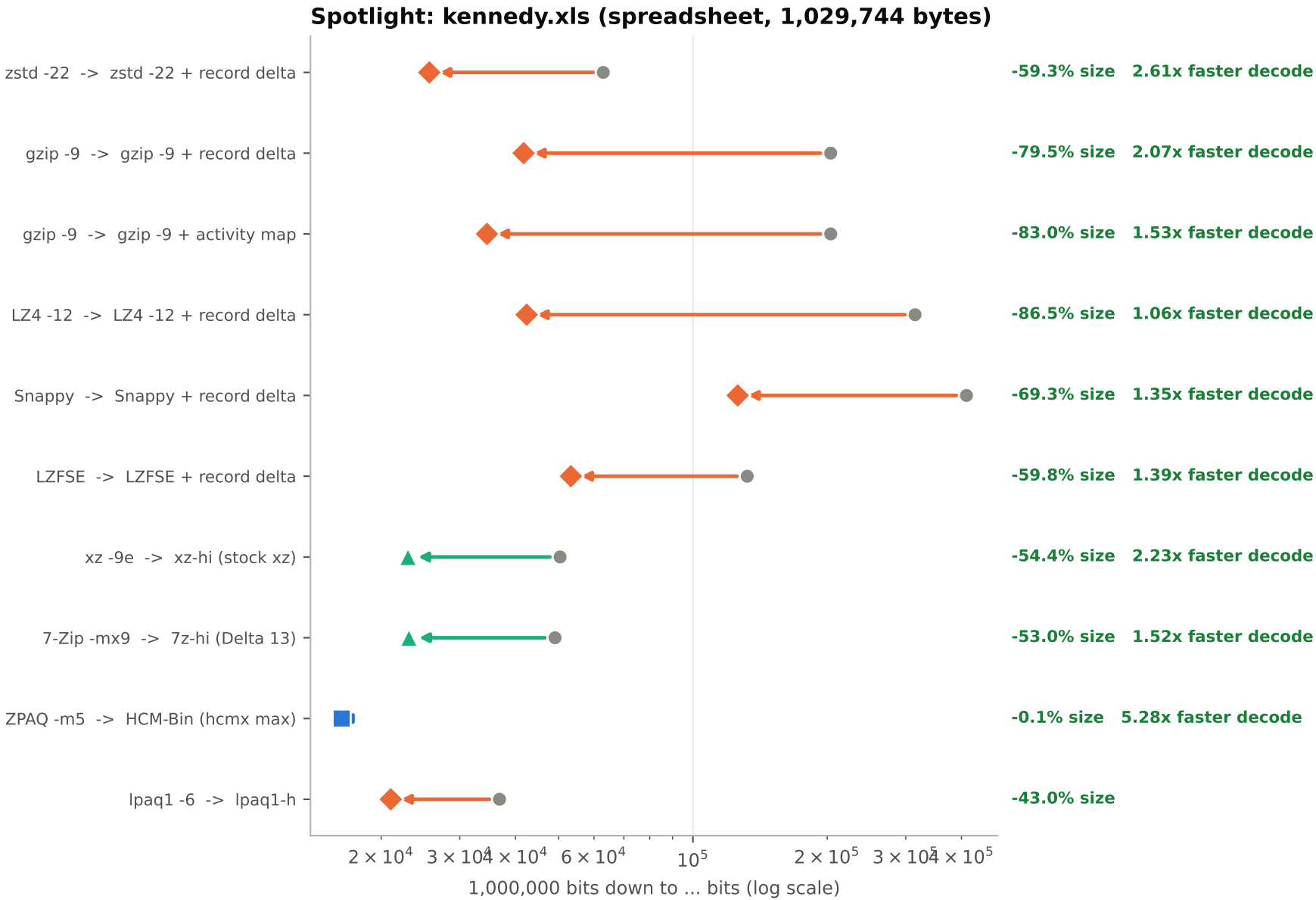
Timing sources: ch6.

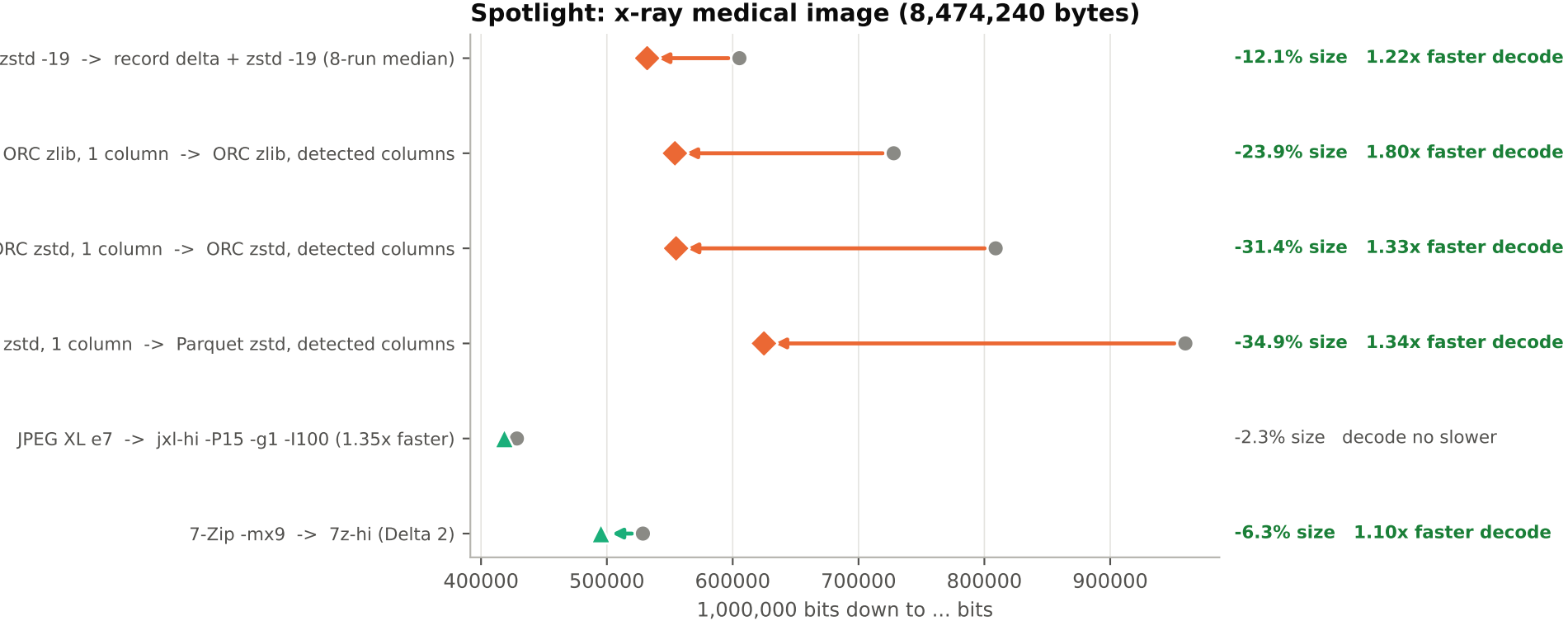
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)
Best-trade-off line (nothing beats these on both)
On the line

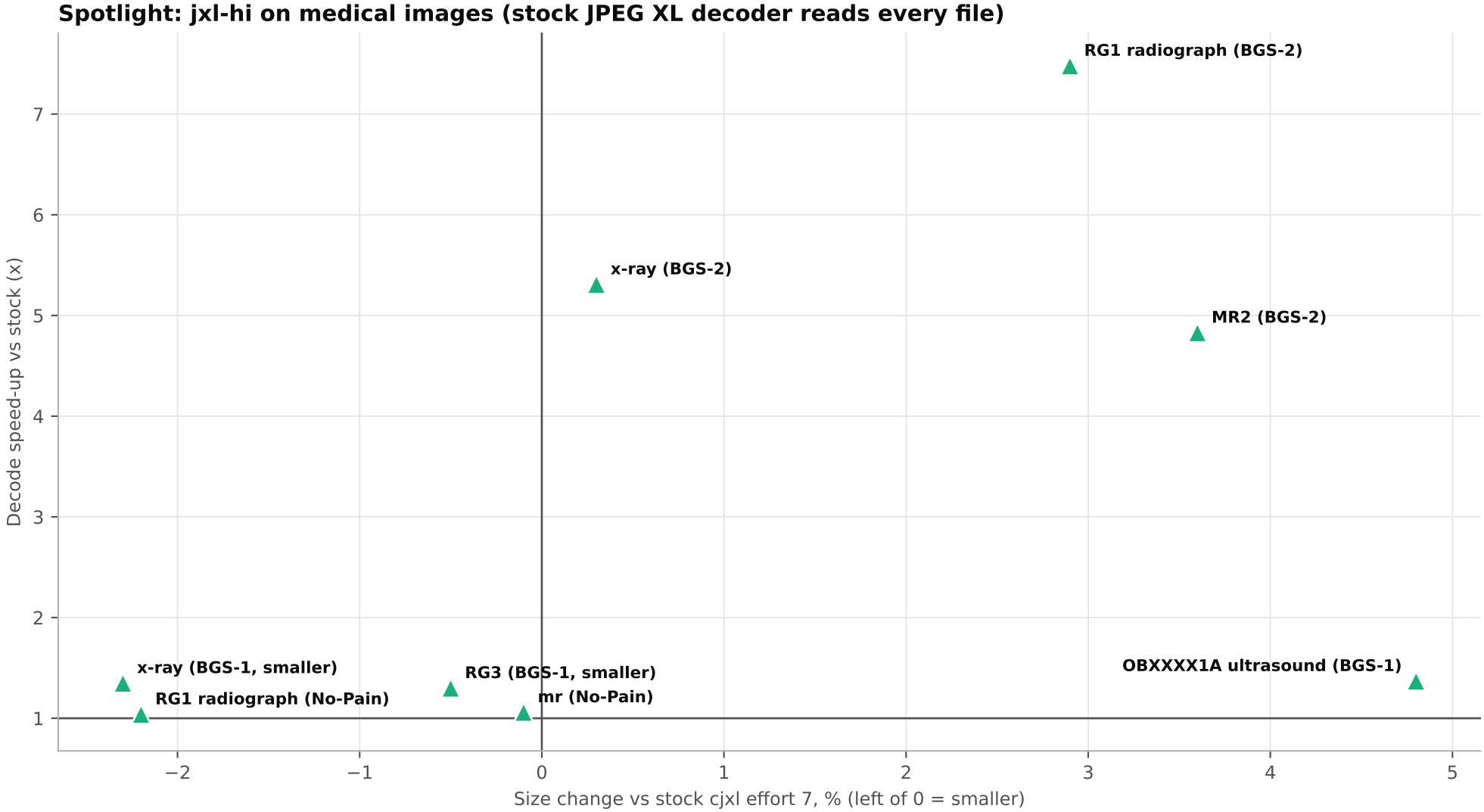

PDF documents (139 files): what our changes did to each tool
qpdf optimisation -> Ours, No-Pain pick
-0.6% size
1.01x faster decode
847000
848000
849000
850000
851000
852000
1,000,000 bits down to ... bits (arrow points to ours)


Speech audio (FLAC, 6 speakers): every recorded result, smallest first (20,997,112 bytes
flac-hi (No-Pain)
520,279 · 123 ms
flac -8
531,634 · 136 ms
520000
525000
530000
535000
540000
1,000,000 bits down to ... bits (shorter bar is better)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Harrington best-of (best of our tools, picked per file)


Speech audio (FLAC, 6 speakers): what our changes did to each tool
flac -8 -> flac-hi (No-Pain)
-2.1% size
1.11x faster decode
520000
522000
524000
526000
528000
530000
532000
1,000,000 bits down to ... bits (arrow points to ours)

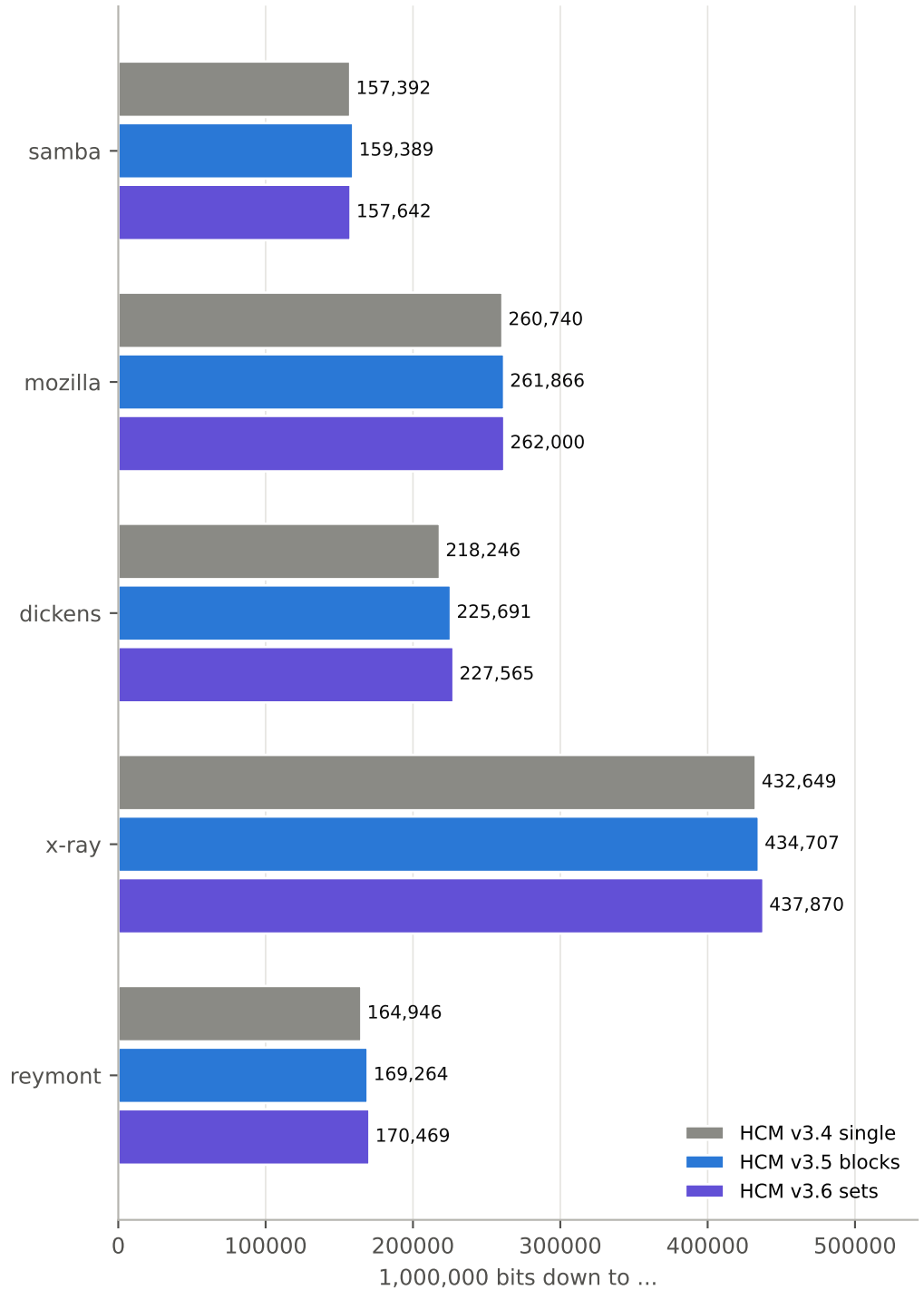

Spotlight: kennedy.xls (spreadsheet, 1,029,744 bytes)
zstd -22 -> zstd -22 + record delta
gzip -9 -> gzip -9 + record delta
gzip -9 -> gzip -9 + activity map
LZ4 -12 -> LZ4 -12 + record delta
Snappy -> Snappy + record delta
LZFSE -> LZFSE + record delta
xz -9e -> xz-hi (stock xz)
7-Zip -mx9 -> 7z-hi (Delta 13)
ZPAQ -m5 -> HCM-Bin (hcmx max)
lpaq1 -6 -> lpaq1-h
-59.3% size 2.61x faster decode
-79.5% size 2.07x faster decode
-83.0% size 1.53x faster decode
-86.5% size 1.06x faster decode
-69.3% size 1.35x faster decode
-59.8% size 1.39x faster decode
-54.4% size 2.23x faster decode
-53.0% size 1.52x faster decode
-0.1% size 5.28x faster decode
-43.0% size
2×10^4
3×10^4
4×10^4
6×10^4
10^5
2×10^5
3×10^5
4×10^5
1,000,000 bits down to ... bits (log scale)

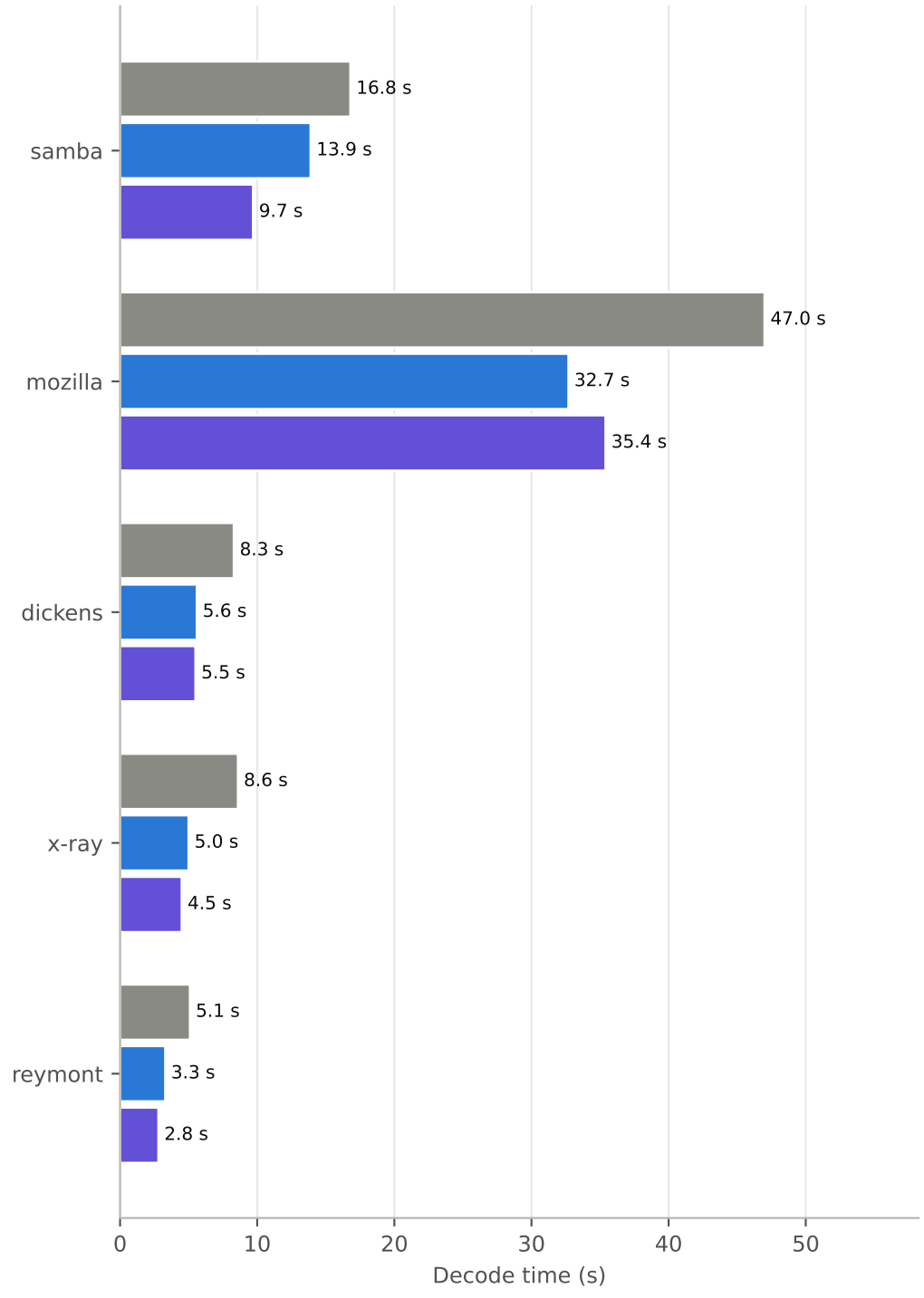

Spotlight: x-ray medical image (8,474,240 bytes)
zstd -19 -> record delta + zstd -19 (8-run median)
ORC zlib, 1 column -> ORC zlib, detected columns
ORC zstd, 1 column -> ORC zstd, detected columns
zstd, 1 column -> Parquet zstd, detected columns
JPEG XL e7 -> jxl-hi -P15 -g1 -I100 (1.35x faster)
7-Zip -mx9 -> 7z-hi (Delta 2)
-12.1% size 1.22x faster decode
-23.9% size 1.80x faster decode
-31.4% size 1.33x faster decode
-34.9% size 1.34x faster decode
-2.3% size decode no slower
-6.3% size 1.10x faster decode
400000
500000
600000
700000
800000
900000
1,000,000 bits down to ... bits


Spotlight: jxl-hi on medical images (stock JPEG XL decoder reads every file)
RG1 radiograph (BGS-2)
x-ray (BGS-2)
MR2 (BGS-2)
x-ray (BGS-1, smaller)
RG1 radiograph (No-Pain)
RG3 (BGS-1, smaller)
mr (No-Pain)
OBXXXX1A ultrasound (BGS-1)
Decode speed-up vs stock (x)
Size change vs stock cjxl effort 7, % (left of 0 = smaller)
7
6
5
4
3
2
1
−2
−1
0
1
2
3
4
5

HCM: one core vs multi-core

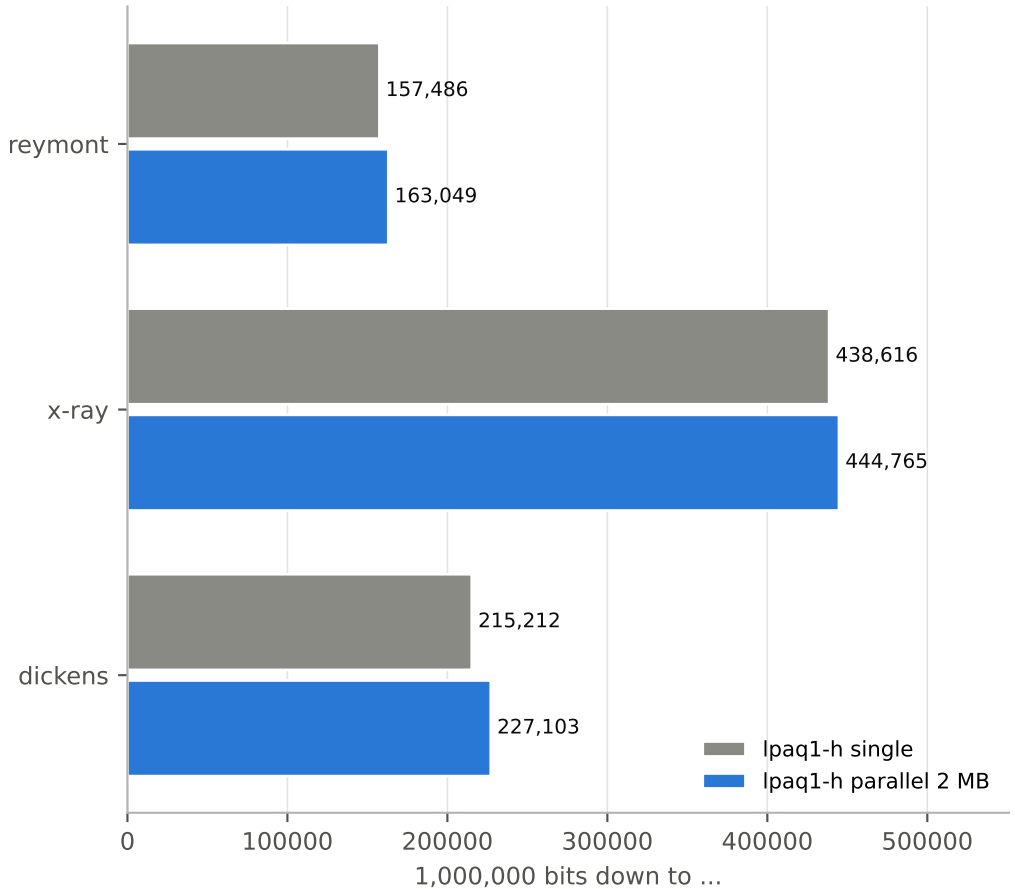
Left: 1,000,000 bits down to X (shorter is smaller). Right: decode time (shorter is faster). Measured on a 2-core machine; more cores give more speed.
Every file decoded back exactly.

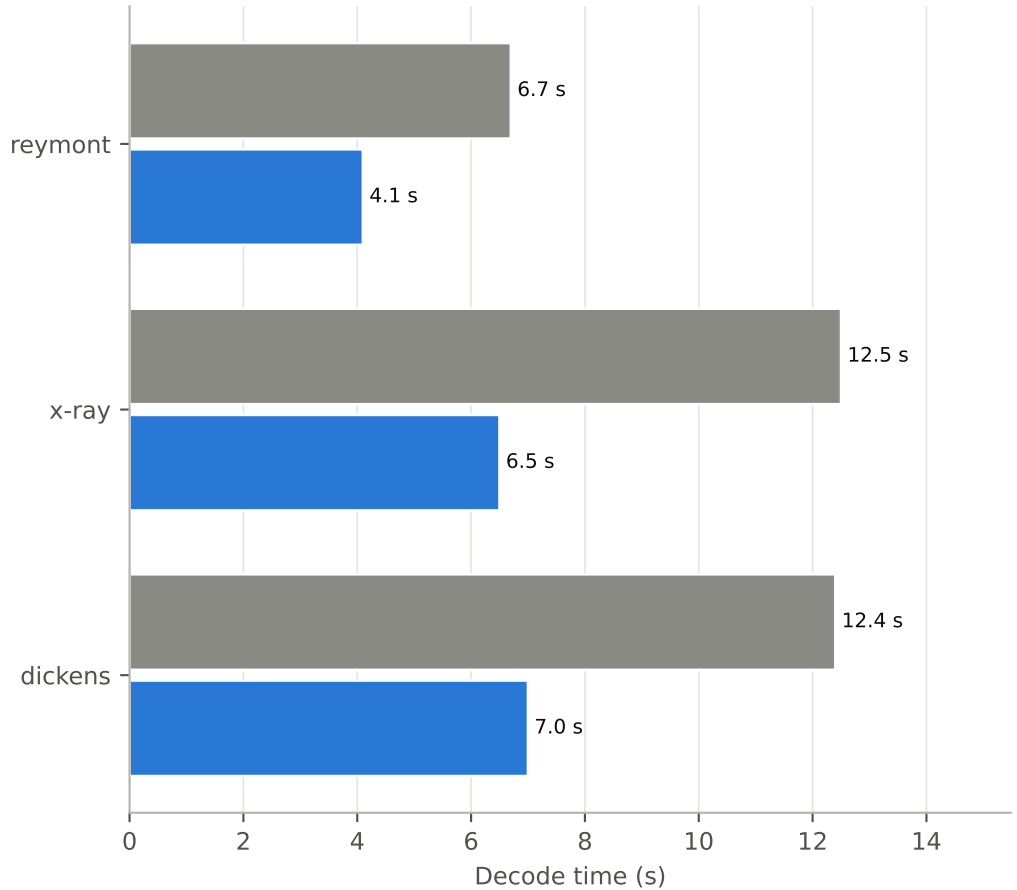




lpaq1-h: one core vs multi-core

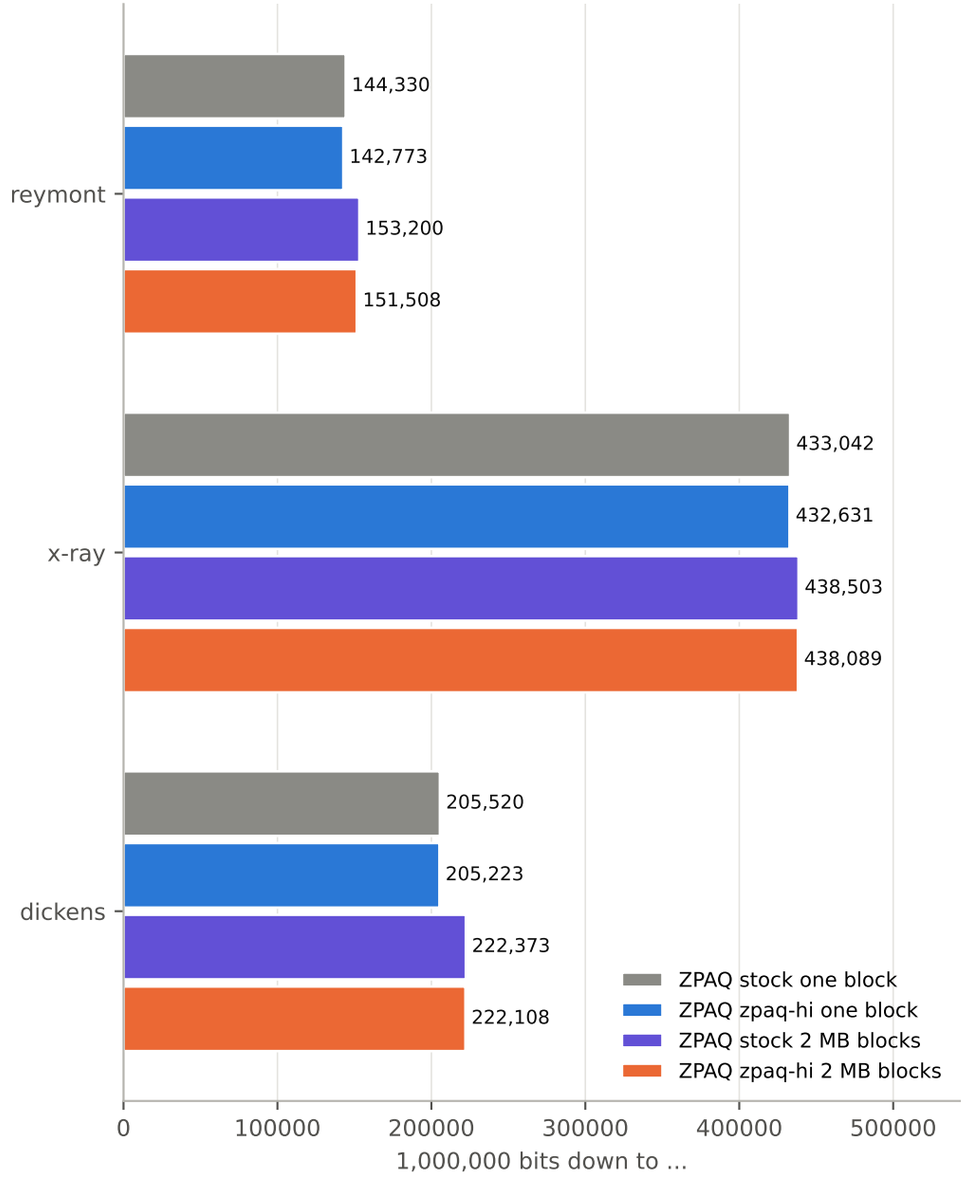
Left: 1,000,000 bits down to X (shorter is smaller). Right: decode time (shorter is faster). Measured on a 2-core machine; more cores give more speed.
Every file decoded back exactly.

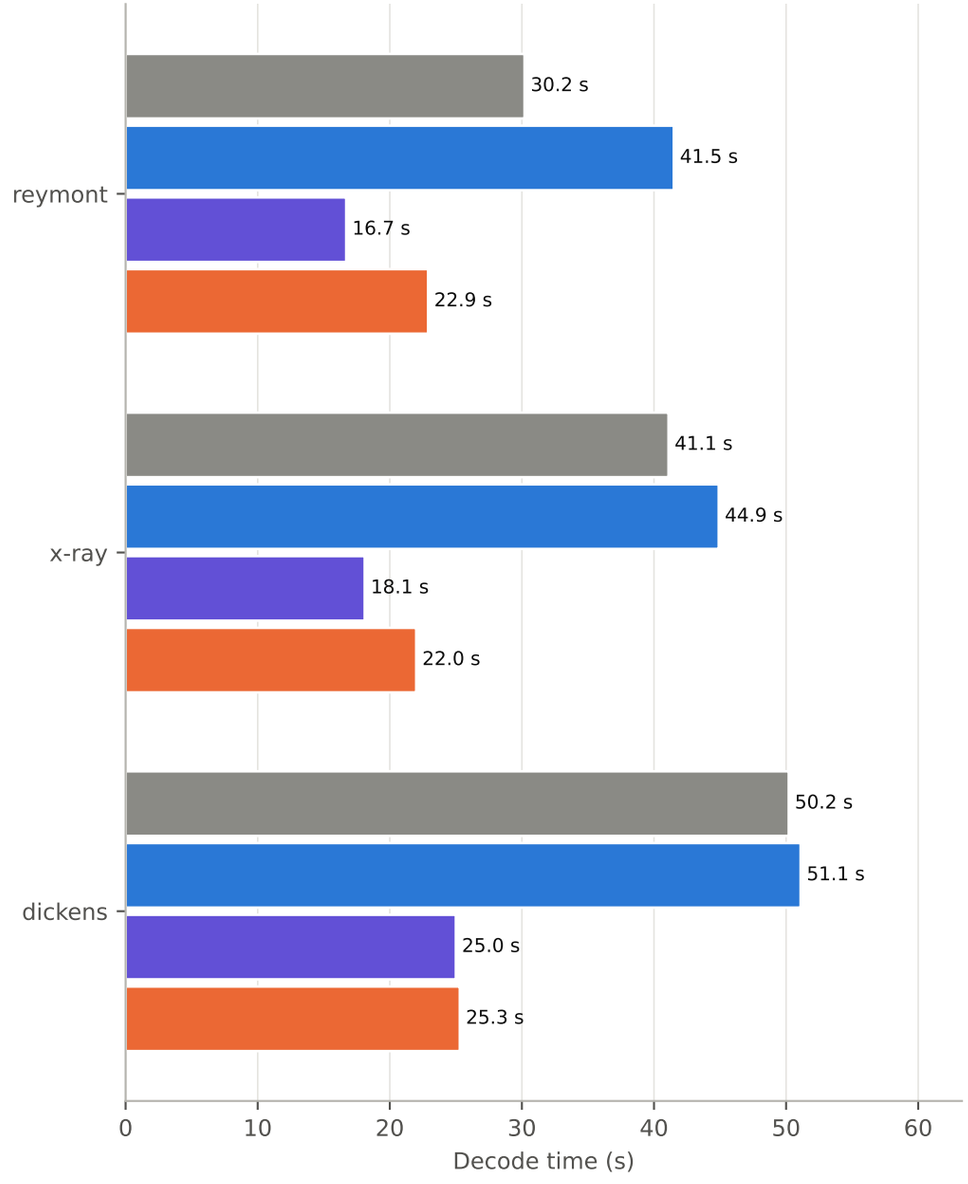




ZPAQ: one core vs multi-core

Left: 1,000,000 bits down to X (shorter is smaller). Right: decode time (shorter is faster). Measured on a 2-core machine; more cores give more speed.
Every file decoded back exactly.





Benchmark Charts

Harrington's Insights: benchmark charts, standard compressors against the author's originals, modified tools and mutants. Assembled 29 October 2026 from recorded results; one run per tool per file, and every result was decompressed and checked byte for byte when it was recorded.

Classes. Standard: the stock tool. Original: the author's own designs (HCM, hcmx, Harrington Exchange). Modified: a tuned encoder whose output the stock decoder reads unchanged (bzip2-hi, xz-hi). Mutant: an existing compressor whose decoder was changed (lpaq1-h, the Exchange->Blosc2 hybrid, Exchange->xz).

Sets follow Andrew Goodwin's review: the Canterbury corpus, the first 1 MB of each Silesia file, and his synthetic set; plus the author's biased packed-data set and six radio captures from earlier runs.

How to read the dot charts. Each dot is one tool's total over a set. Up and down is size (lower is smaller); left to right is speed in MB of original data per second on a log scale (further right is faster). The dashed line joins the tools that nothing else beats on both size and speed; dots with a dark ring sit on it. Our tools are drawn larger with bold labels. The line charts show each file separately.

Data sources. All three of Goodwin's sets were run fresh, one run of all 14 tools per file on the same machine (29 Oct 2026), so speeds compare directly. Peak memory is the highest of compression and decode across the set's files.

Timing notes. Times are wall clock including program start-up, so on small files the fast tools all look about the same; sizes are exact. lpaq1-h's compression time is the sum of its six tries (it keeps the best). ZPAQ sizes have its 1,015-byte per-file archive header removed, as in Goodwin's review. On the biased and radio sets decode times are earlier in-process measurements and compression times were mostly not recorded, so those two sets have decode charts only; ZPAQ's radio decode time is an estimate.

RESULTS

Biased packed data (4 files) (2,173,577 bytes). Smallest overall: order-0 rANS. Our best: Harrington Exchange at 1,011,727 bytes (decode 25.1 ms).

Radio SDR captures (6 files) (3,752,512 bytes). Smallest overall: ZPAQ -m5 (decode estimated). Our best: xz-hi at 2,299,596 bytes (decode 171.0 ms).

Biased file 12bit_noise (524,288 bytes): our best Harrington Exchange 324,507; best fast standard tool (decode under 0.1 s) order-0 rANS 323,335; smallest of all paq8px -8 323,153. The hybrid beats Blosc2 on both size (324,765 vs 360,045) and decode (0.69 vs 0.76 ms).

Biased file biased_bits (125,000 bytes): our best Harrington Exchange 19,722; best fast standard tool (decode under 0.1 s) order-0 rANS 17,925; smallest of all order-0 rANS 17,925.

Biased file biased_bytes (1,000,000 bytes): our best Harrington Exchange 321,128; best fast standard tool (decode under 0.1 s) order-0 rANS 290,572; smallest of all paq8px -8 290,460.

Biased file 5bit_skewed (524,289 bytes): our best Exchange->Blosc2 hybrid 345,725; best fast standard tool (decode under 0.1 s) order-0 rANS 340,229; smallest of all order-0 rANS 340,229.

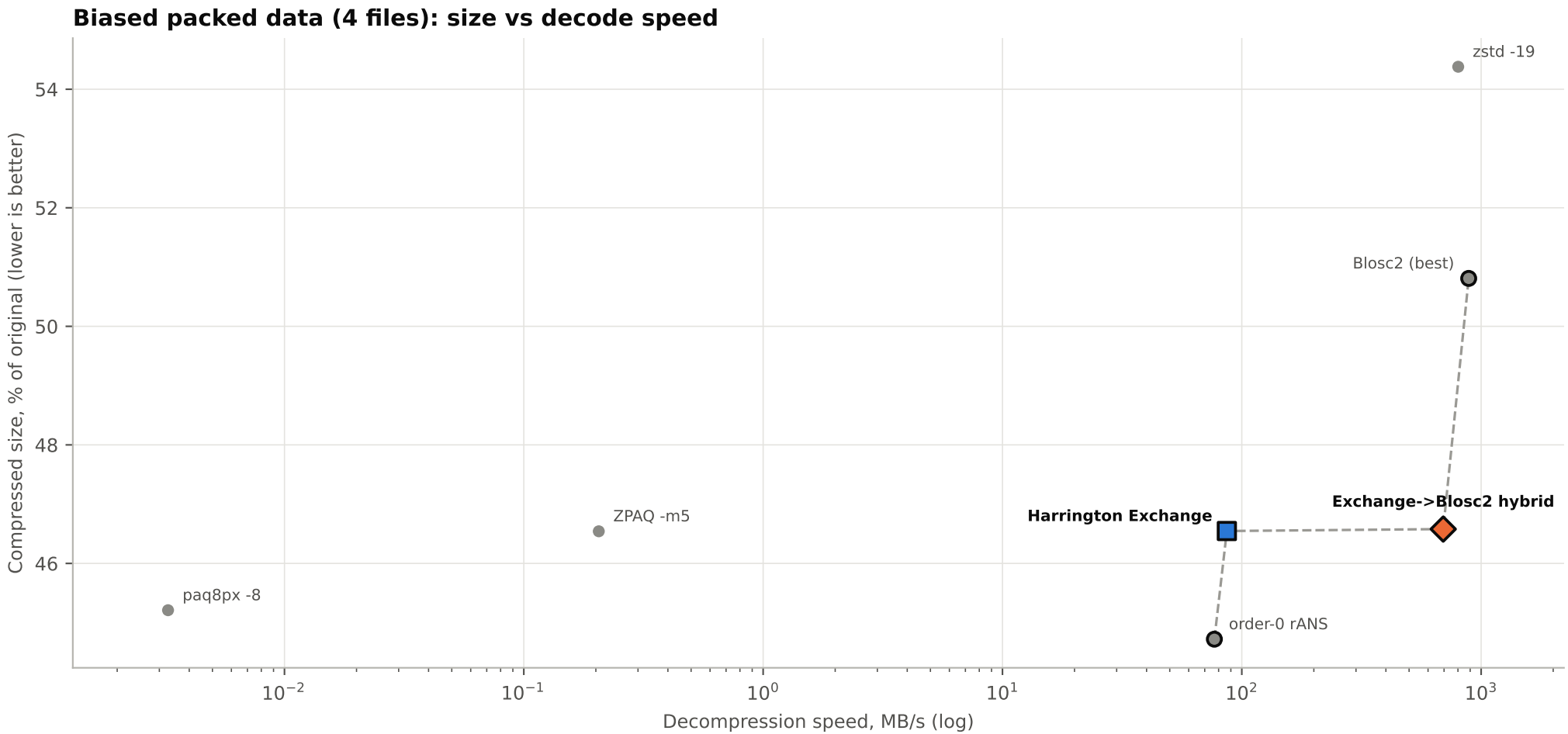
Benchmark Charts (continued)

Biased set note: the order-0 rANS coder was smaller than the Exchange on all four files (972,061 vs 1,011,727 bytes in total) but decodes about 9x slower than the hybrid; paq8px is smallest on three files but takes minutes to decode. On the 5-bit file the Exchange and the hybrid beat paq8px outright.

Canterbury (2,810,784 bytes, 14 tools). Smallest overall: HCM v3.4 max at 350,326 bytes. Best standard: ZPAQ -m5 at 350,542. Our best: HCM v3.4 max at 350,326 bytes (-0.06% vs best standard). Ours on the decode-speed trade-off line: HCM v3.4 max, hcmx fast, HCM v3.4 fast, xz-hi. Smaller AND faster to decode: xz-hi beats bzip2 -9, xz -9e; HCM v3.4 fast beats lpaq1 -6; HCM v3.4 max beats lpaq1 -6, ZPAQ -m5; hcmx fast beats lpaq1 -6.

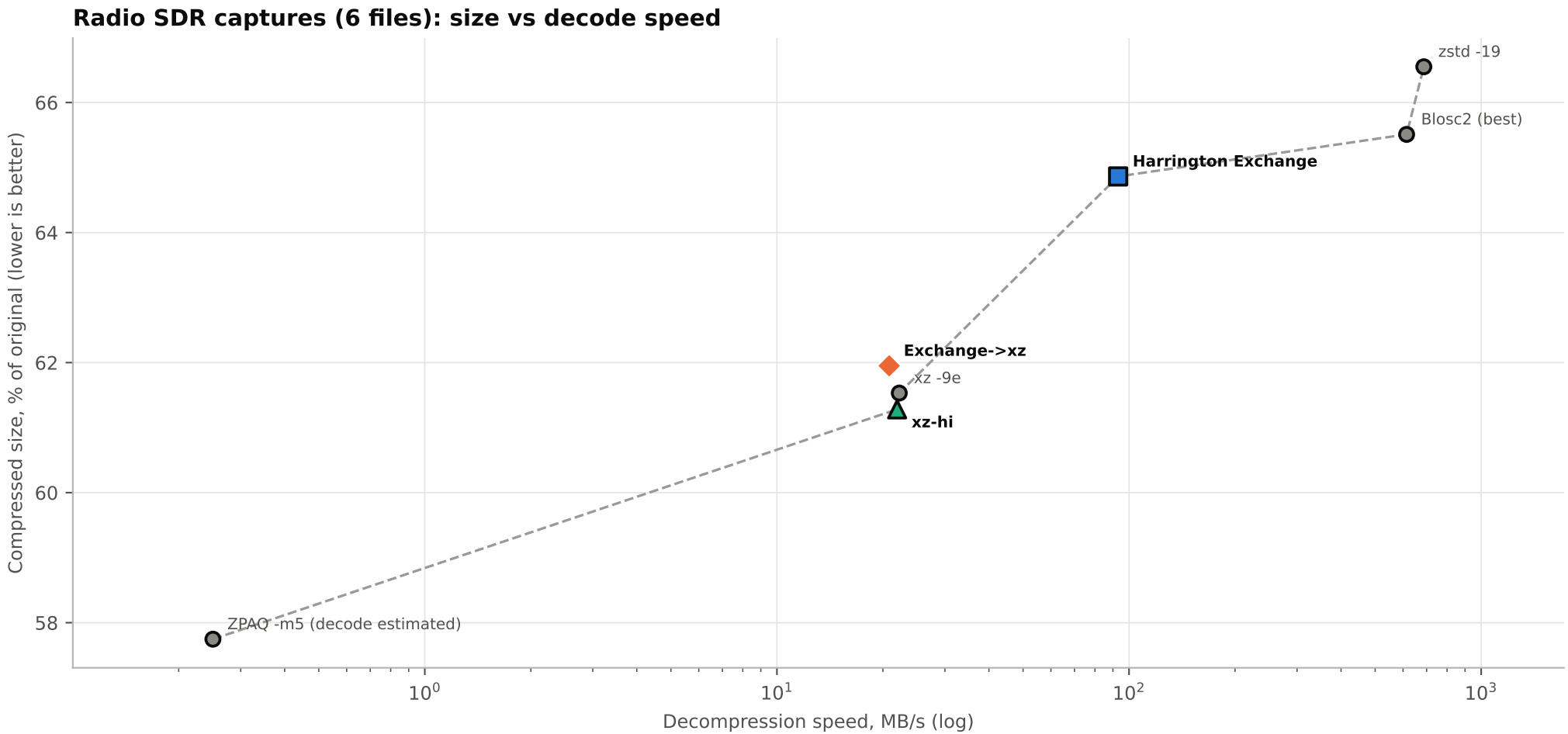
Silesia 1 MB sample (12,582,912 bytes, 14 tools). Smallest overall: ZPAQ -m5 at 3,418,919 bytes. Best standard: ZPAQ -m5 at 3,418,919. Our best: HCM v3.4 max at 3,544,604 bytes (+3.68% vs best standard). Ours on the decode-speed trade-off line: HCM v3.4 max, HCM v3.4 fast, xz-hi. Smaller AND faster to decode: bzip2-hi beats bzip2 -9; xz-hi beats bzip2 -9, xz -9e; HCM v3.4 fast beats lpaq1 -6.

Synthetic set (17,999,934 bytes, 14 tools). Smallest overall: lpaq1-h at 8,113,745 bytes. Best standard: ZPAQ -m5 at 8,201,855. Our best: lpaq1-h at 8,113,745 bytes (-1.07% vs best standard). Ours on the decode-speed trade-off line: lpaq1-h, HCM v3.4 max, hcmx fast, HCM v3.4 fast, xz-hi. Smaller AND faster to decode: xz-hi beats bzip2 -9; HCM v3.4 fast beats lpaq1 -6, ZPAQ -m5; HCM v3.4 max beats lpaq1 -6, ZPAQ -m5; hcmx fast beats lpaq1 -6, ZPAQ -m5; lpaq1-h beats ZPAQ -m5.



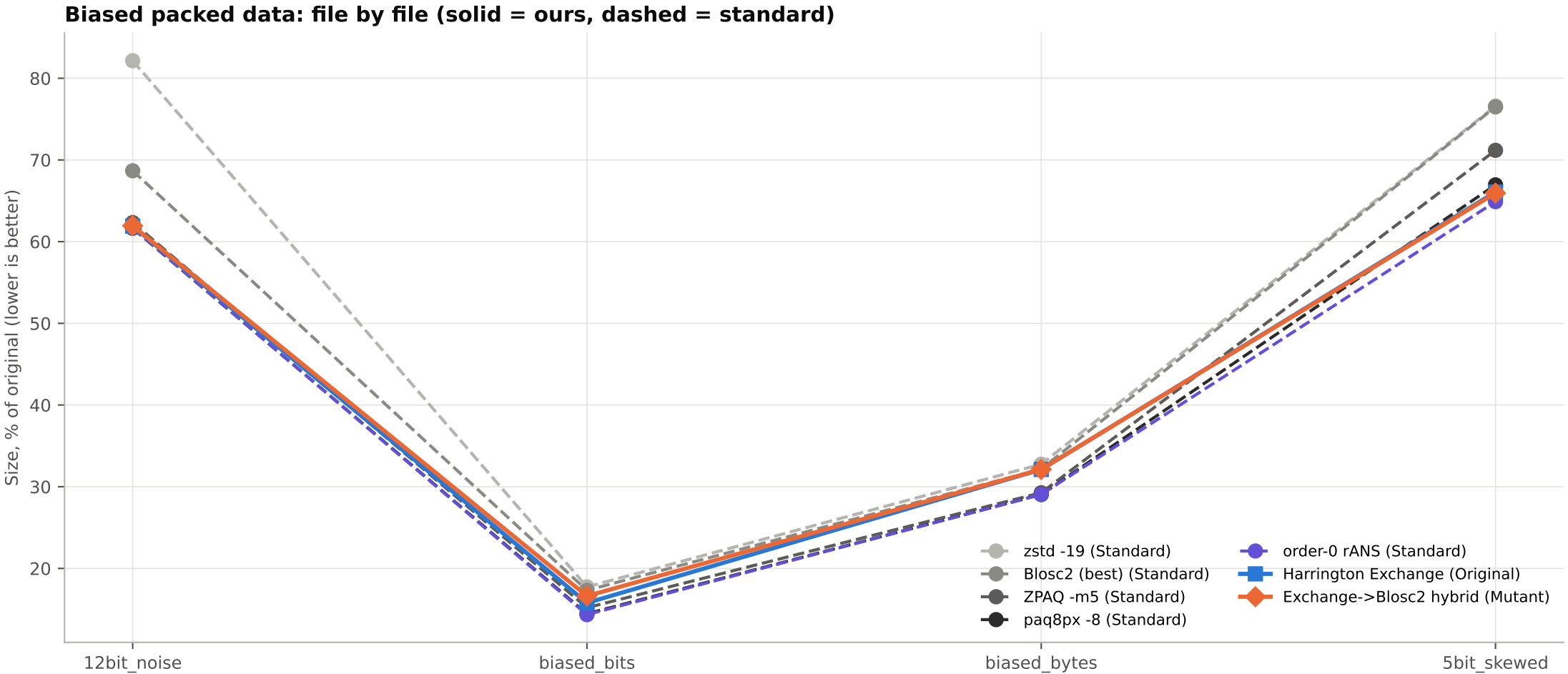
Compression times were not recorded for this set; decode times are earlier in-process measurements. Size only (no decode time recorded): Brotli 11 51.5%; 7-Zip ultra 53.2%; xz -9e 53.3%; OpenZL 55.9%.

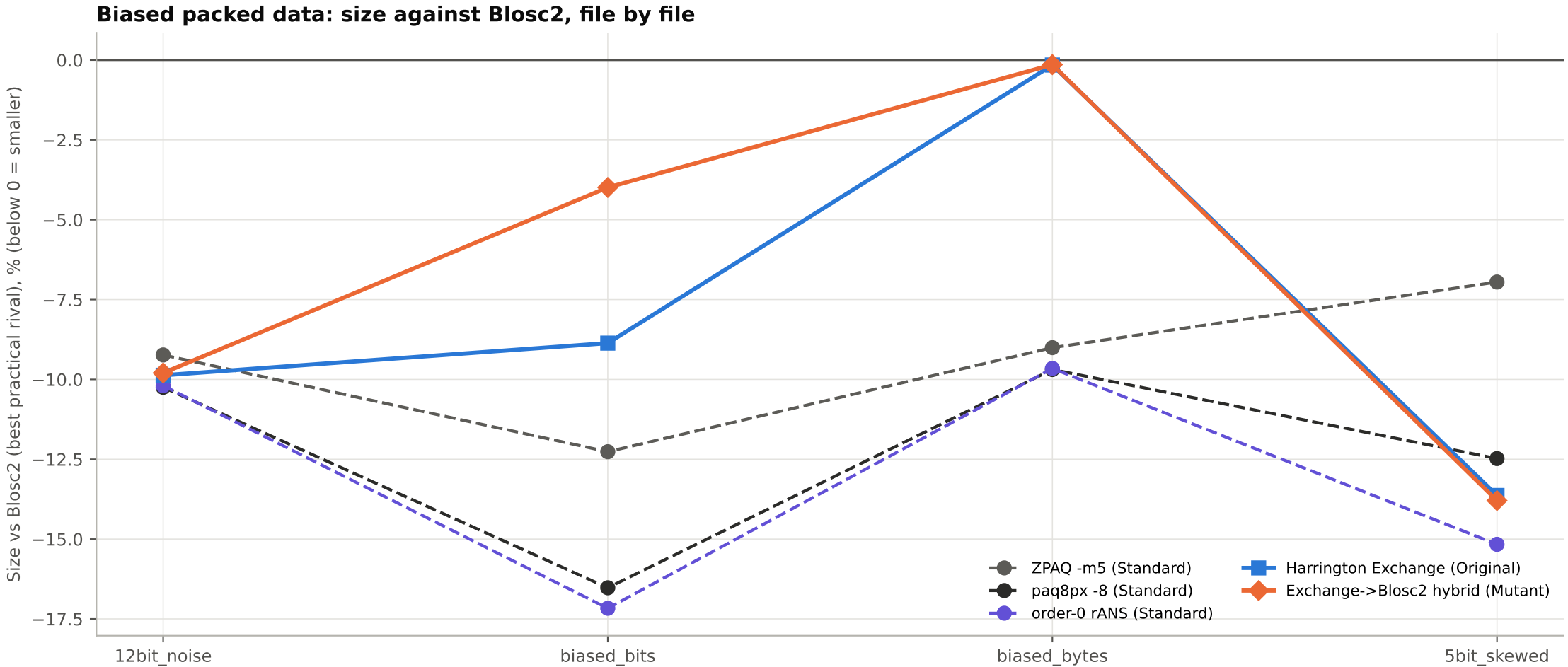
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

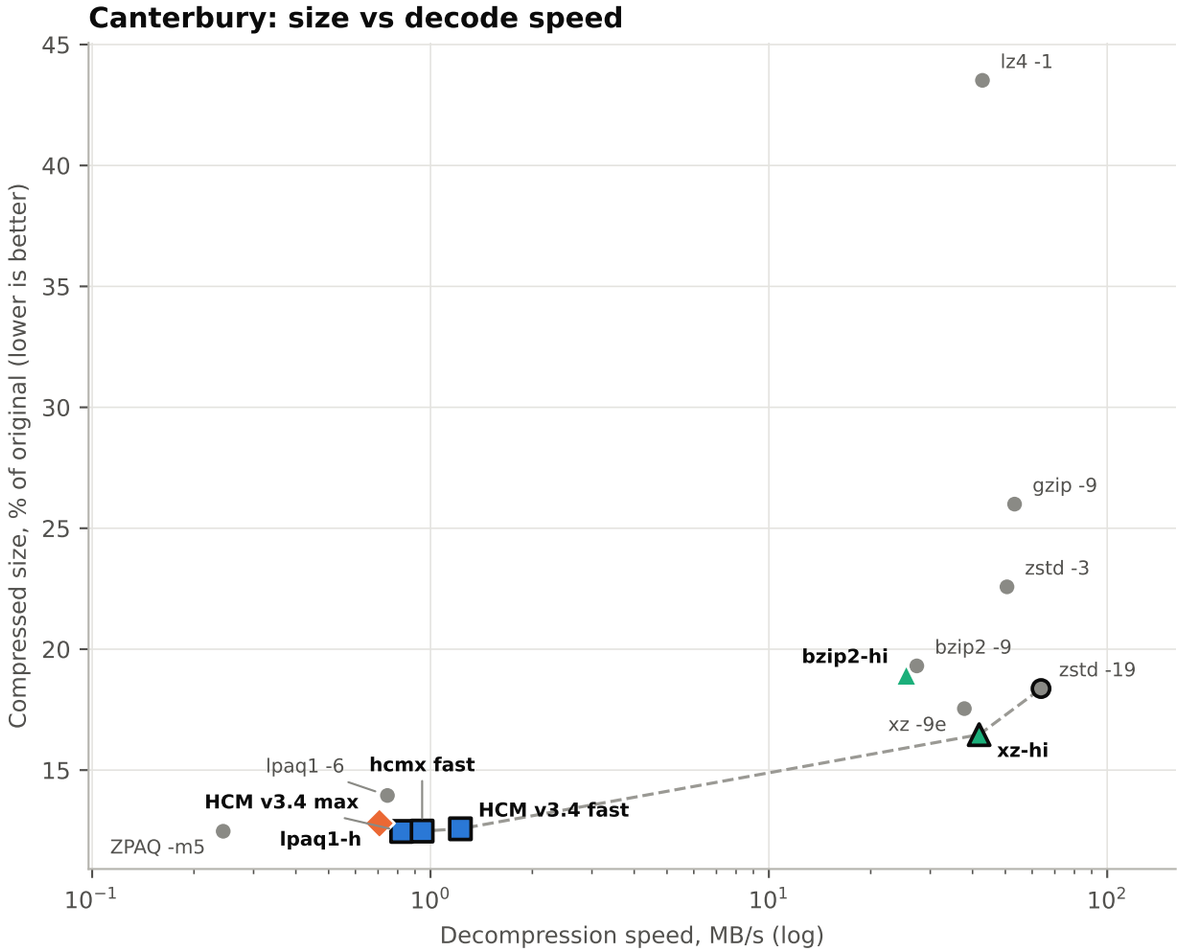


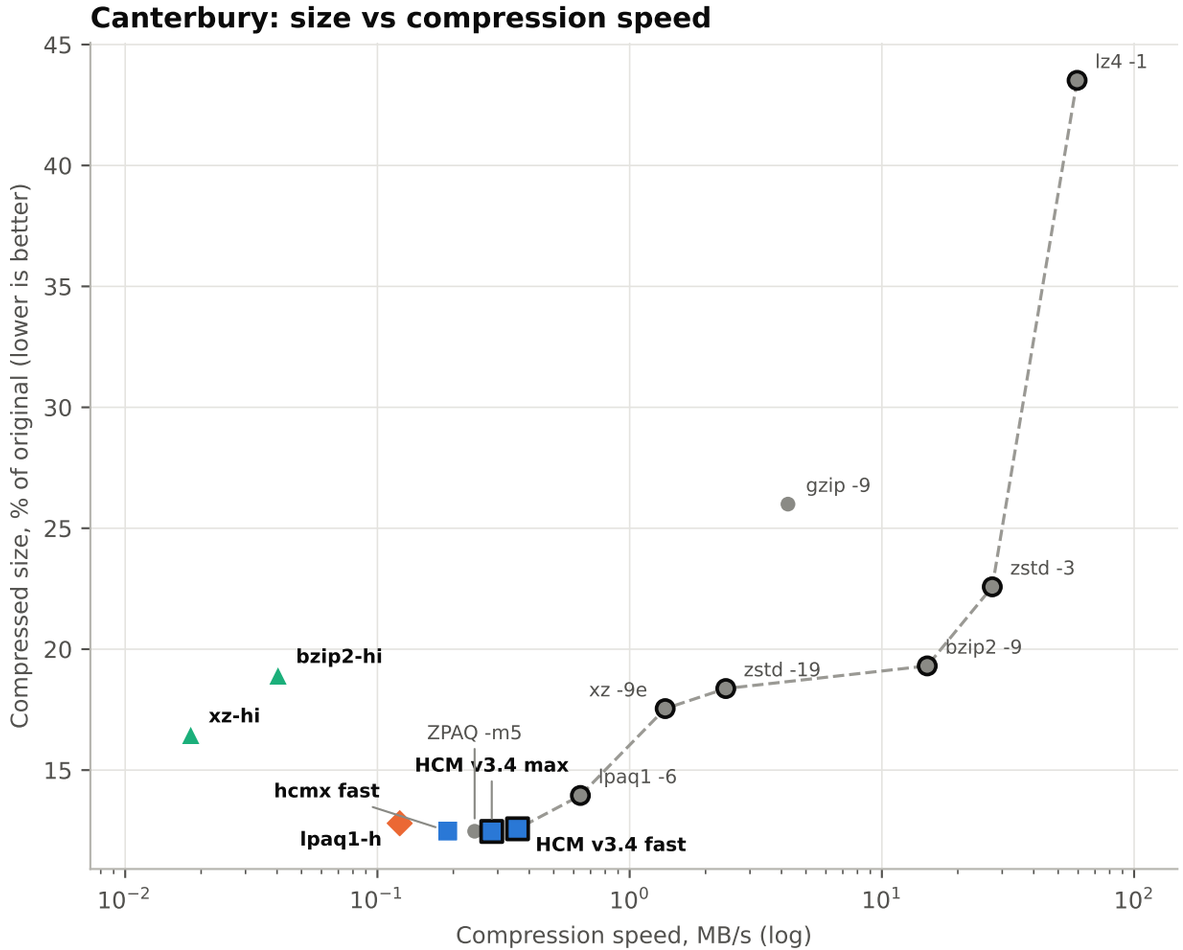
Compression times were not recorded for this set; decode times are earlier in-process measurements. Size only (no decode time recorded): order-0 rANS 70.7%.

Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

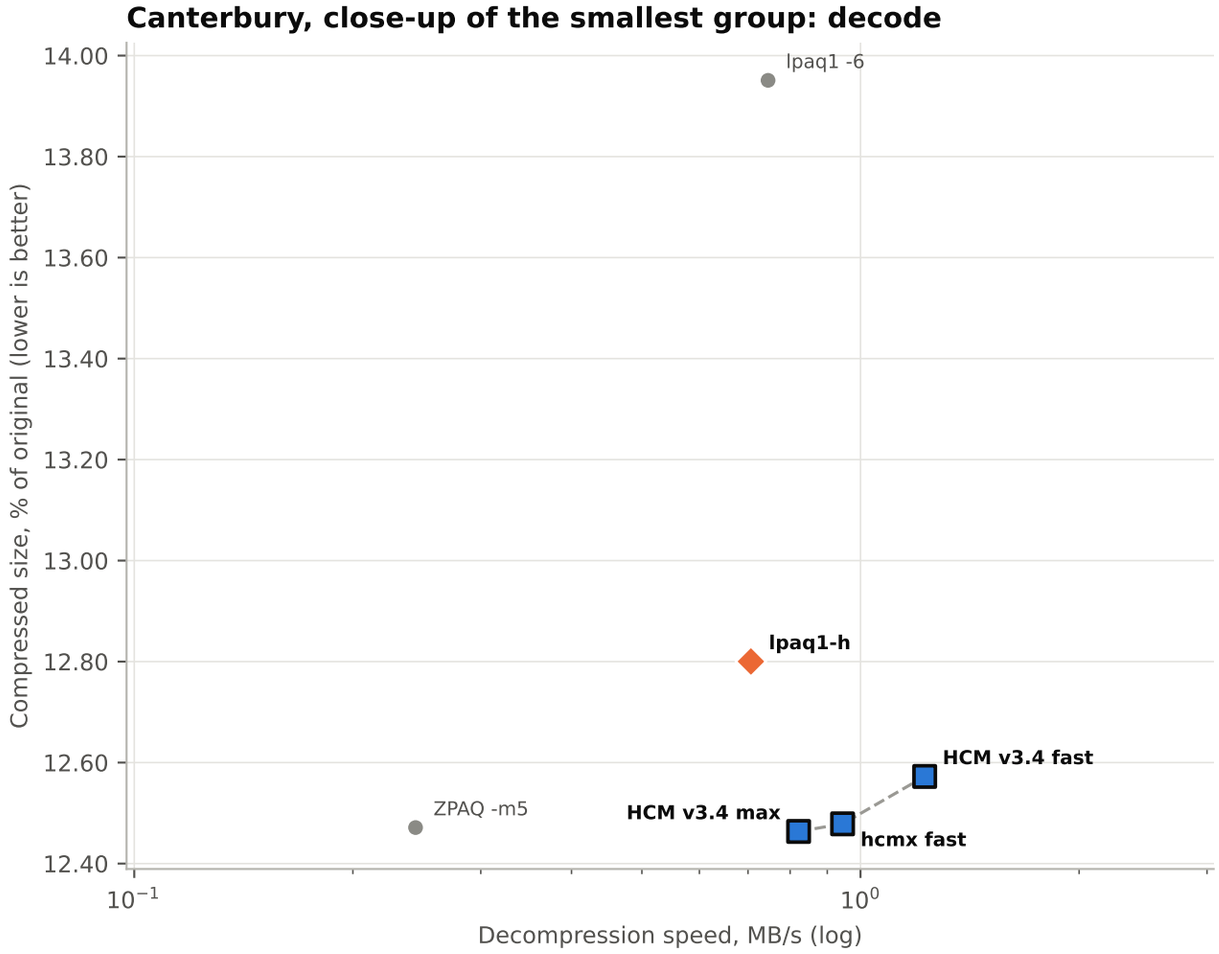

Biased packed data: file by file (solid = ours, dashed = standard)
Size, % of original (lower is better)
80
70
60
50
40
30
20
12bit_noise
biased_bits
biased_bytes
5bit_skewed
zstd -19 (Standard)
Blosc2 (best) (Standard)
ZPAQ -m5 (Standard)
paq8px -8 (Standard)
order-0 rANS (Standard)
Harrington Exchange (Original)
Exchange->Blosc2 hybrid (Mutant)

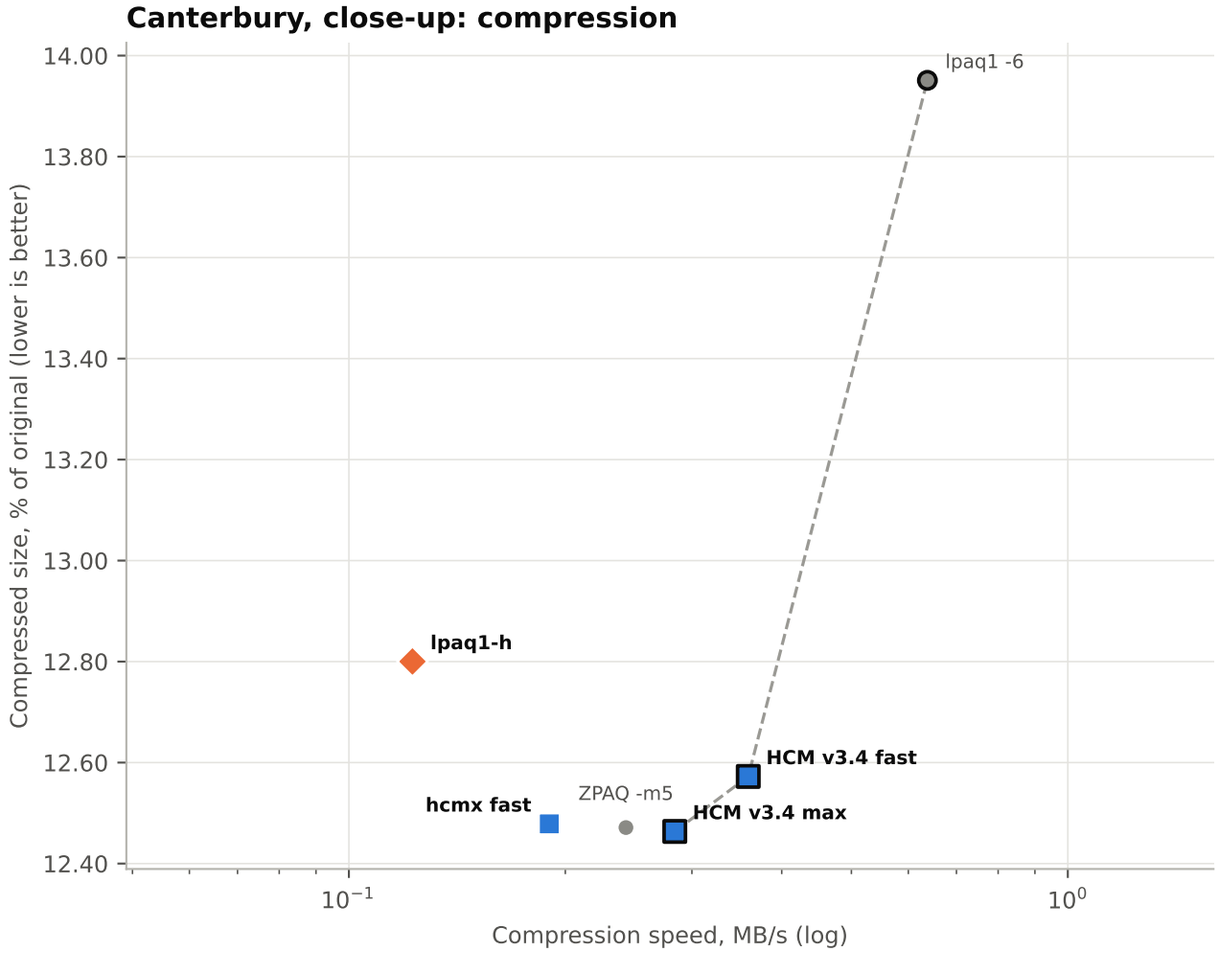

Biased packed data: size against Blosc2, file by file
Size vs Blosc2 (best practical rival), % (below 0 = smaller)
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−17.5
12bit_noise
biased_bits
biased_bytes
5bit_skewed
ZPAQ -m5 (Standard)
paq8px -8 (Standard)
order-0 rANS (Standard)
Harrington Exchange (Original)
Exchange->Blosc2 hybrid (Mutant)

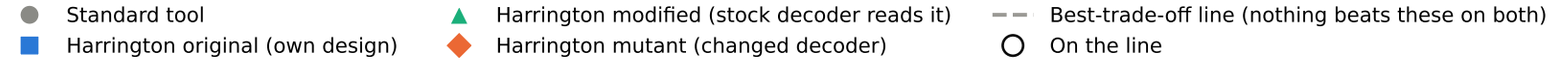


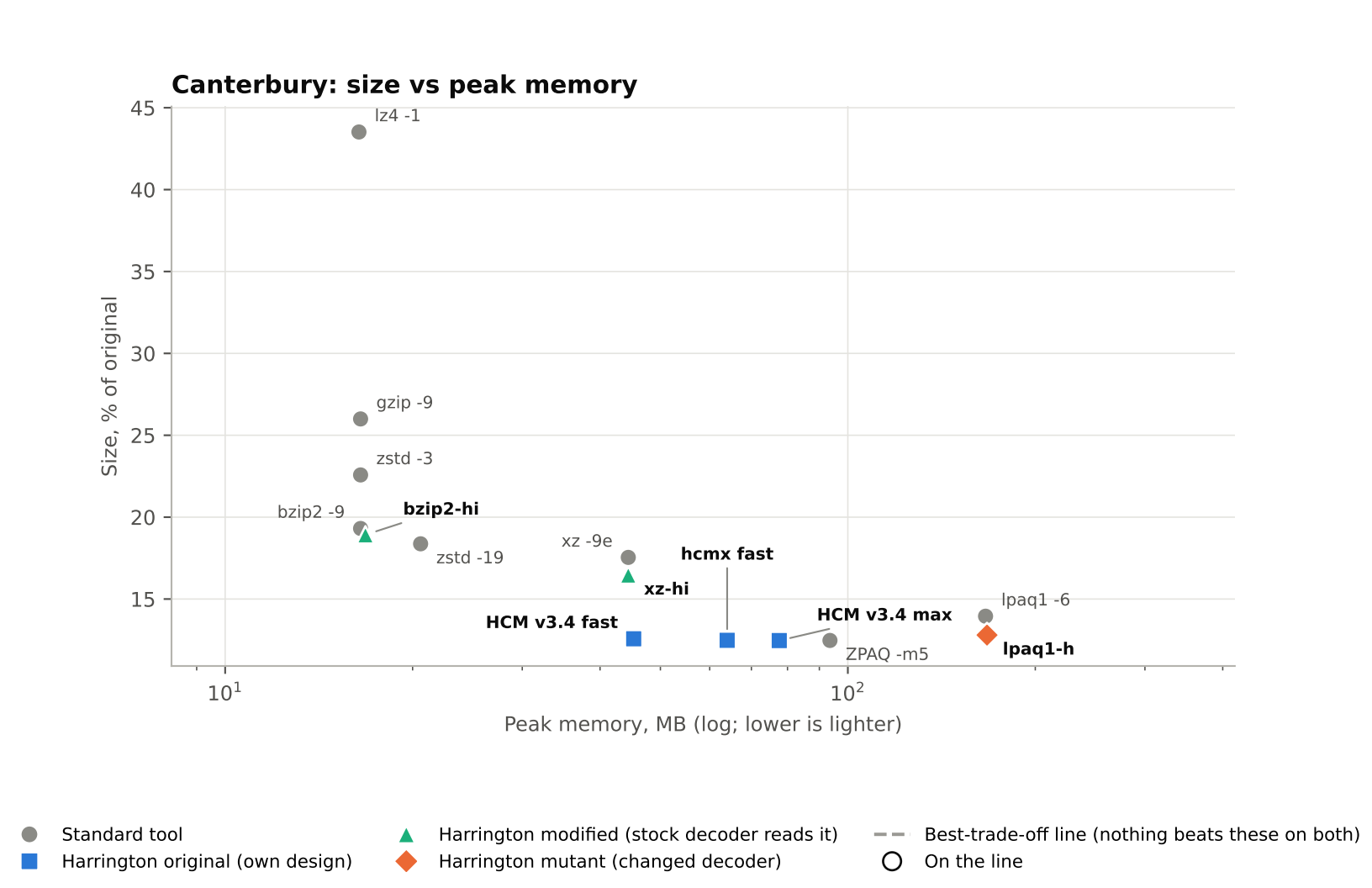


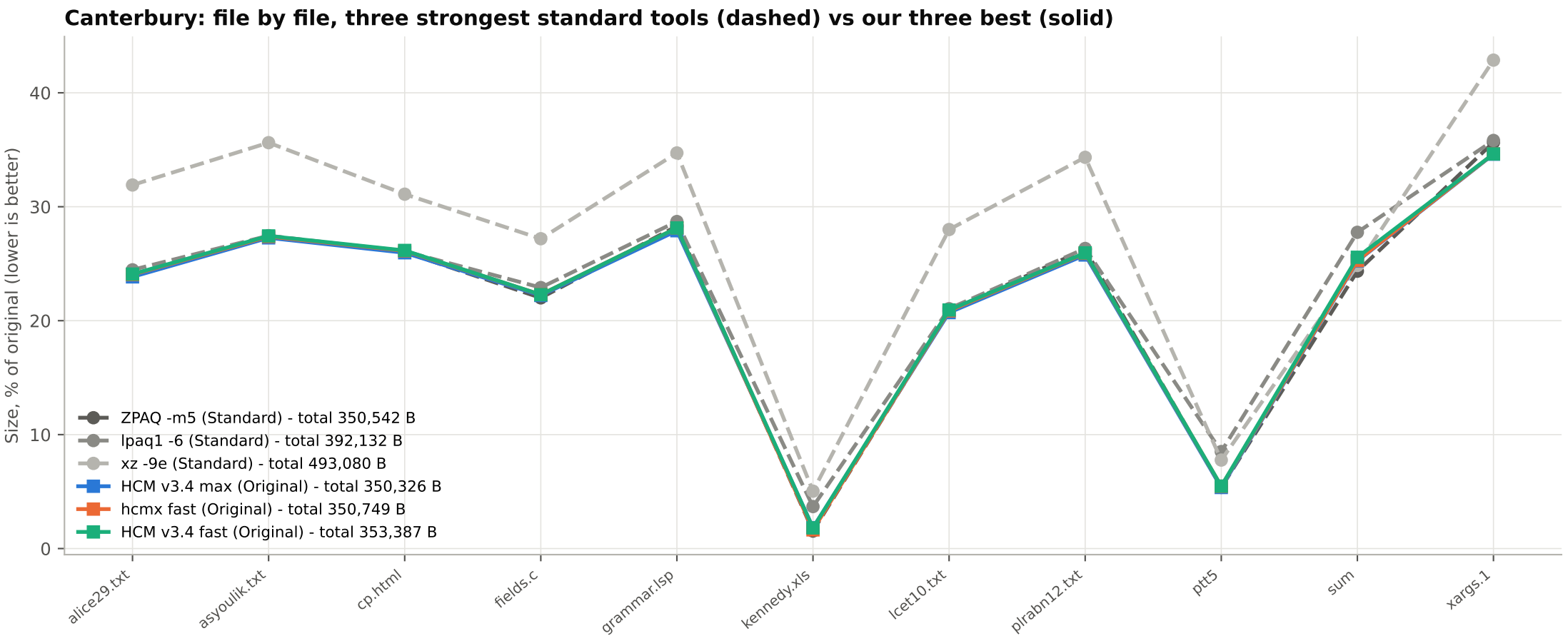
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

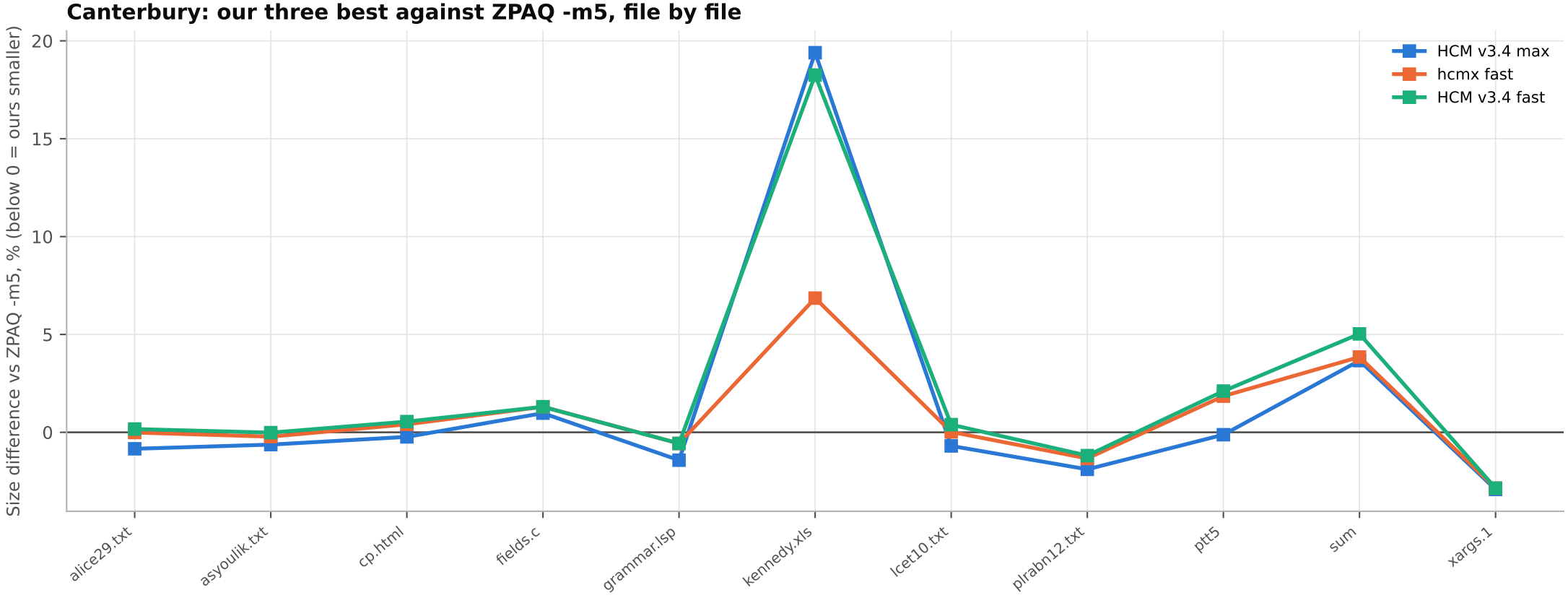

Canterbury, close-up of the smallest group: decode
Compressed size, % of original (lower is better)
Decompression speed, MB/s (log)
lpaq1 -6
lpaq1-h
HCM v3.4 fast
ZPAQ -m5
HCM v3.4 max
hcmx fast

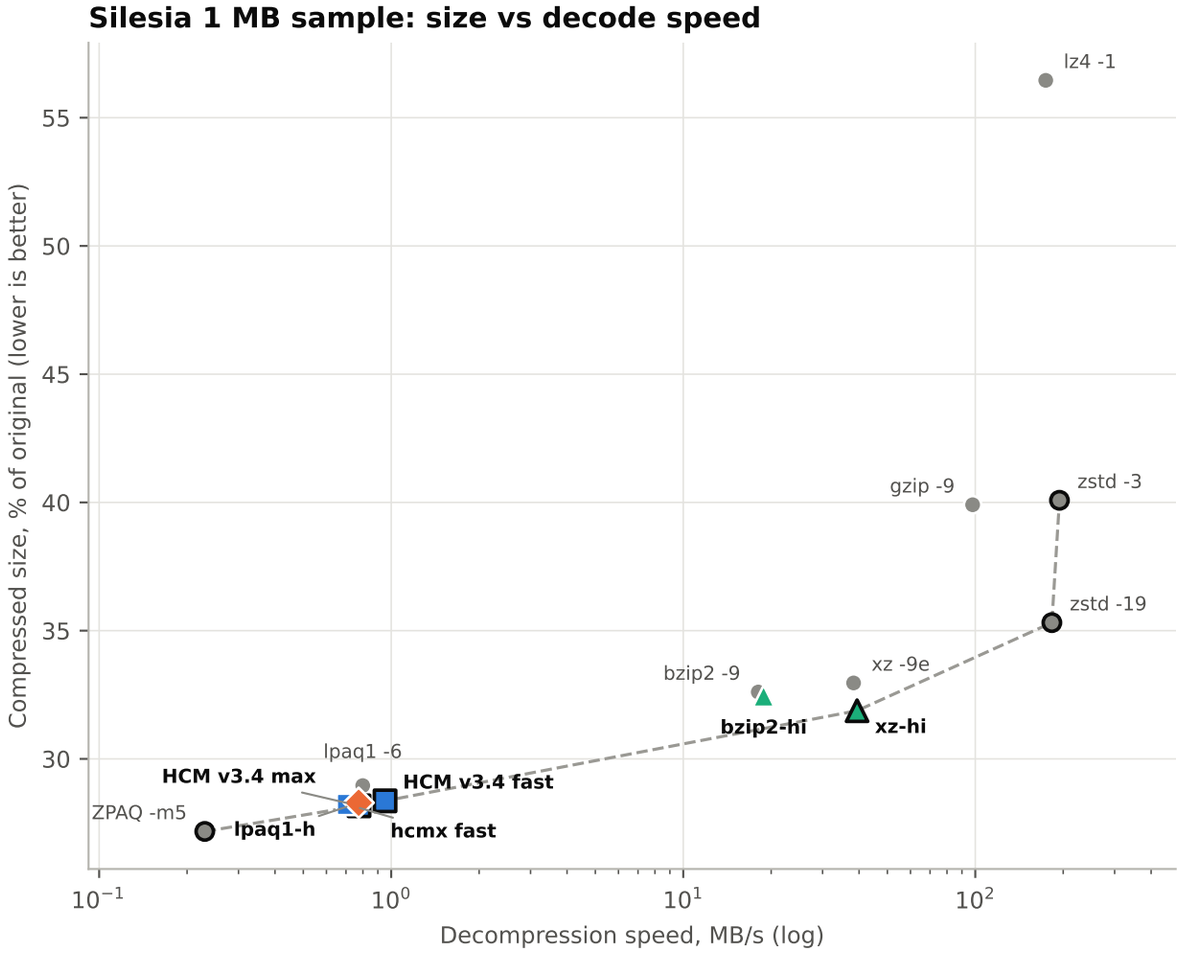

Canterbury, close-up: compression
Compressed size, % of original (lower is better)
Compression speed, MB/s (log)
lpaq1 -6
lpaq1-h
HCM v3.4 fast
hcmx fast
ZPAQ -m5
HCM v3.4 max

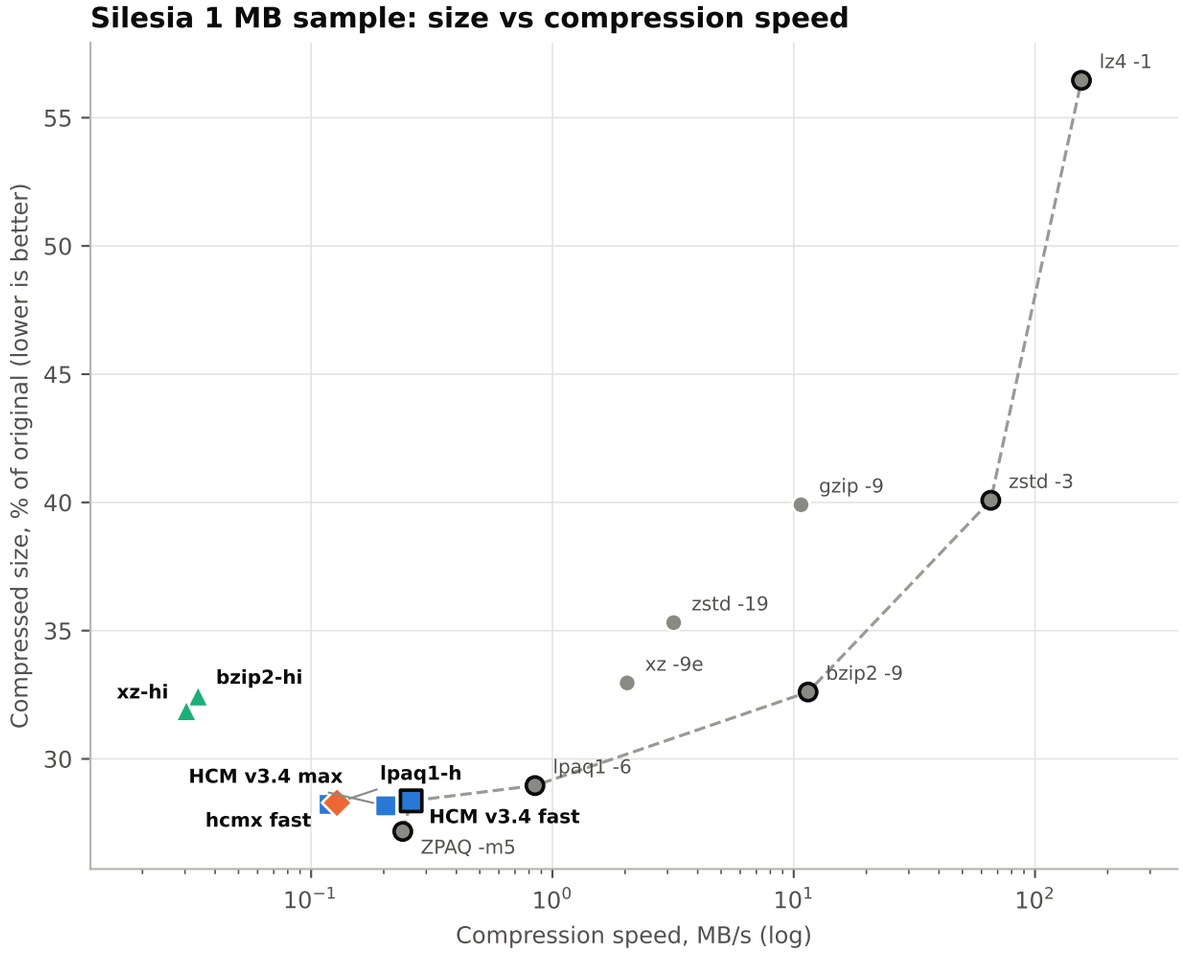

Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line


Canterbury: size vs peak memory
Size, % of original
Peak memory, MB (log; lower is lighter)
45
40
35
30
25
20
15
10^1
10^2
lz4 -1
gzip -9
zstd -3
bzip2 -9
bzip2-hi
zstd -19
xz -9e
hcmx fast
xz-hi
HCM v3.4 fast
HCM v3.4 max
ZPAQ -m5
lpaq1 -6
lpaq1-h
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

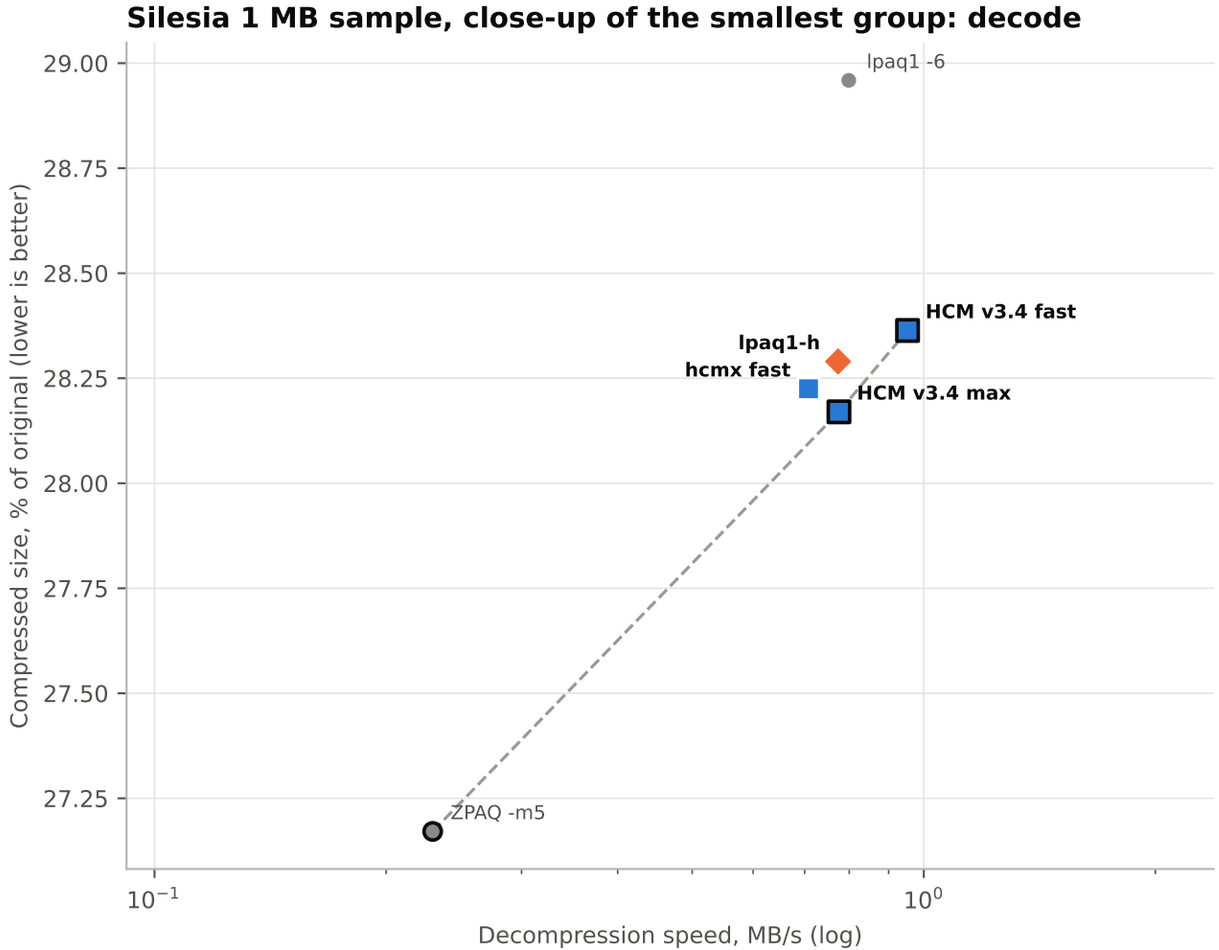

Canterbury: file by file, three strongest standard tools (dashed) vs our three best (solid)
Size, % of original (lower is better)
0
10
20
30
40
ZPAQ -m5 (Standard) - total 350,542 B
lpaq1 -6 (Standard) - total 392,132 B
xz -9e (Standard) - total 493,080 B
HCM v3.4 max (Original) - total 350,326 B
hcmx fast (Original) - total 350,749 B
HCM v3.4 fast (Original) - total 353,387 B
alice29.txt
asyoulik.txt
cp.html
fields.c
grammar.lsp
kennedy.xls
lcet10.txt
plrabn12.txt
ptt5
sum
xargs.1

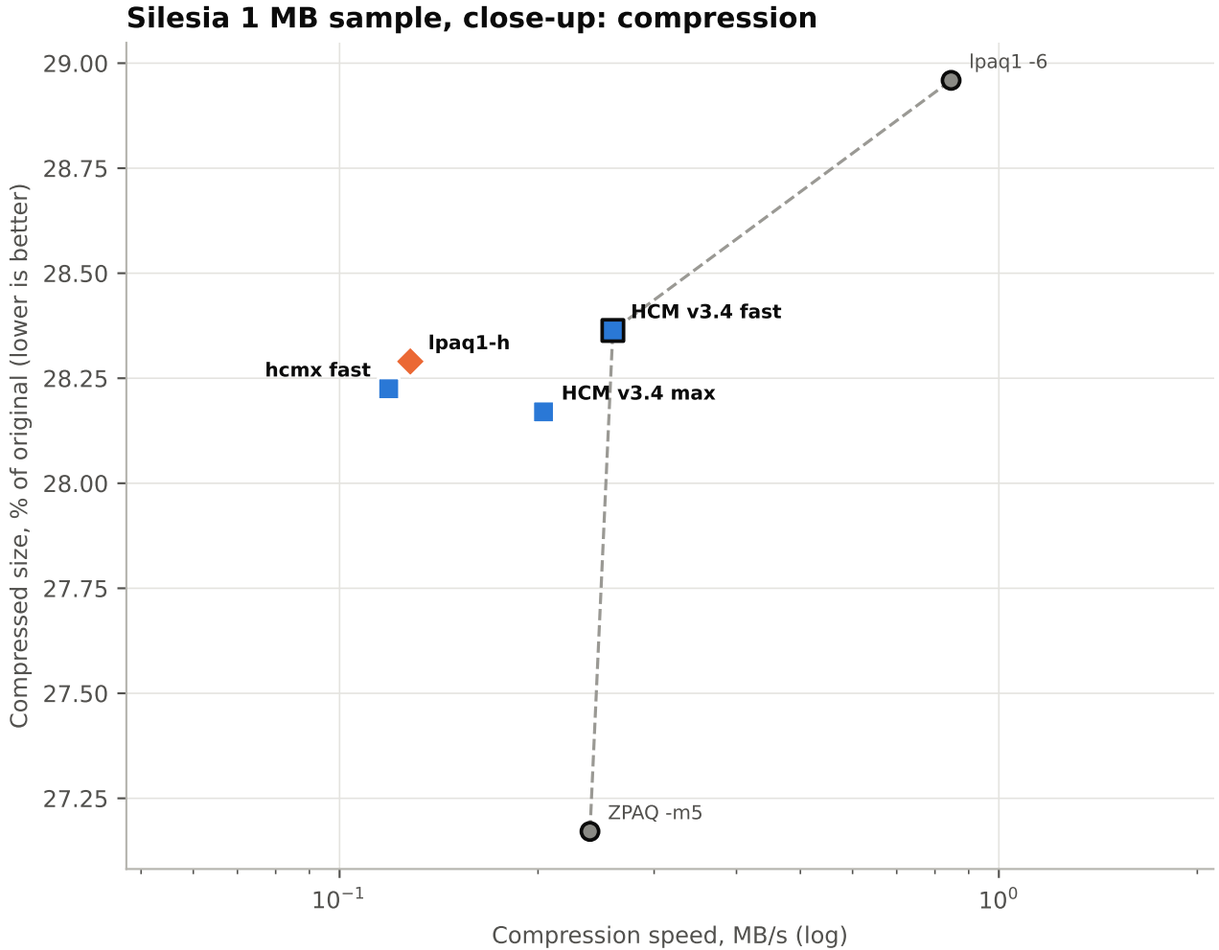

Canterbury: our three best against ZPAQ -m5, file by file
Size difference vs ZPAQ -m5, % (below 0 = ours smaller)
20
15
10
5
0
HCM v3.4 max
hcmx fast
HCM v3.4 fast
alice29.txt
asyoulik.txt
cp.html
fields.c
grammar.lsp
kennedy.xls
lcet10.txt
plrabn12.txt
ptt5
sum
xargs.1



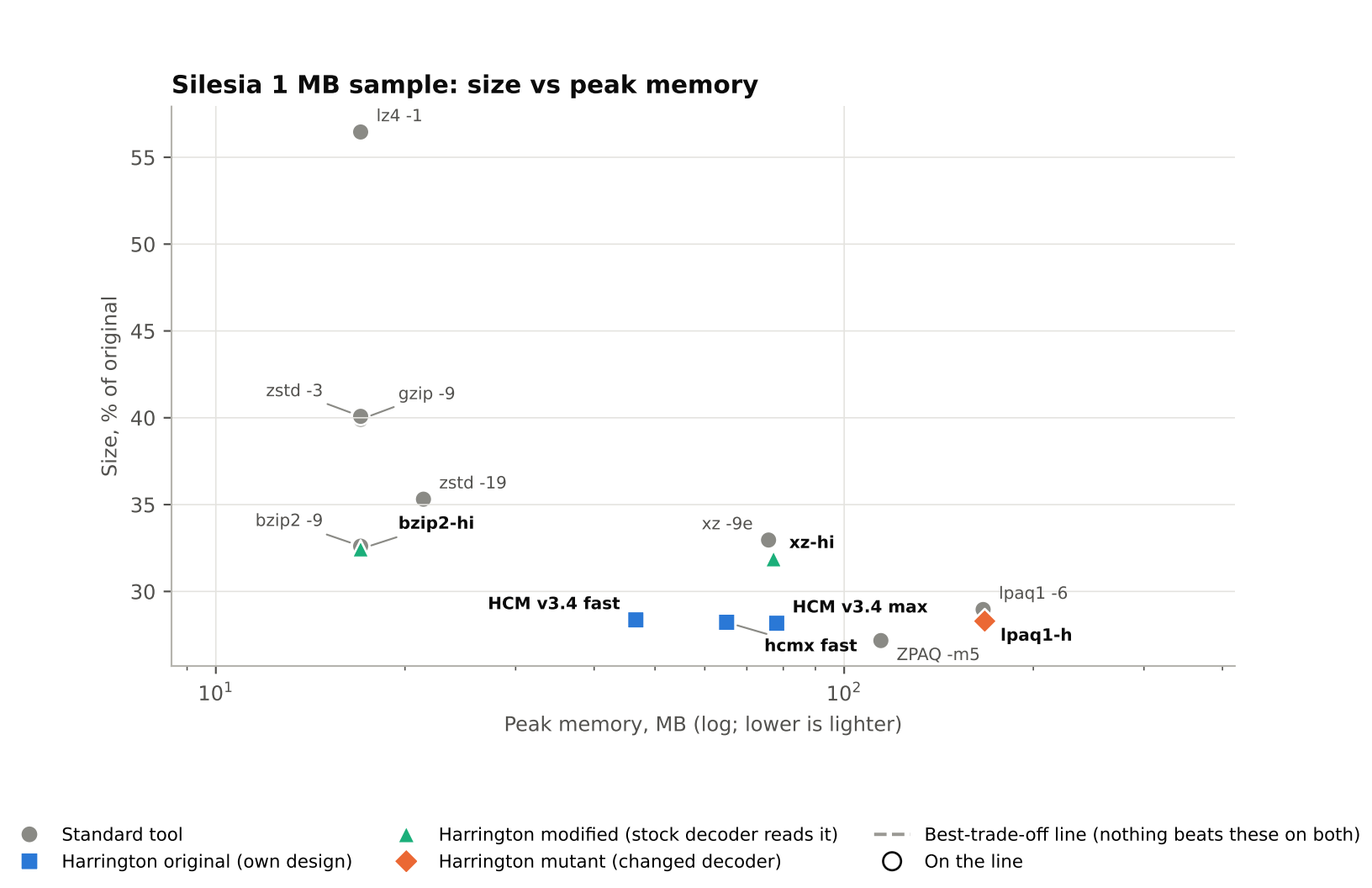


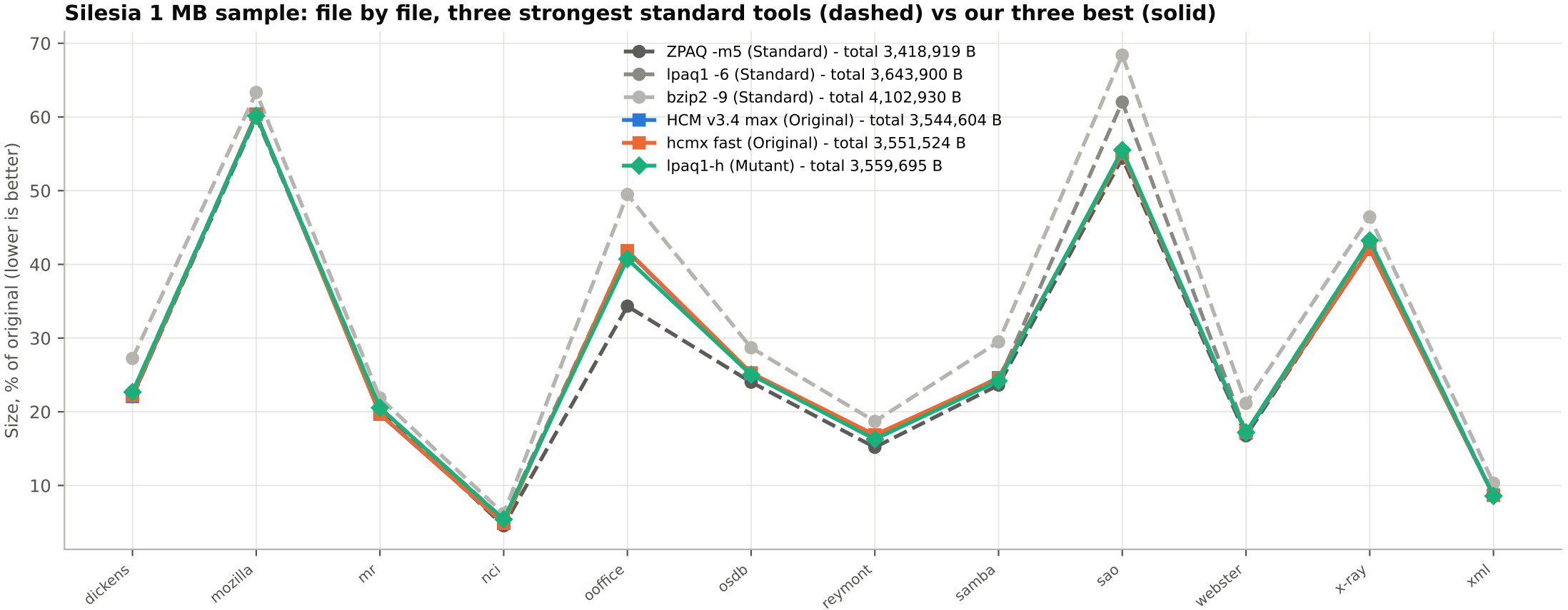
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

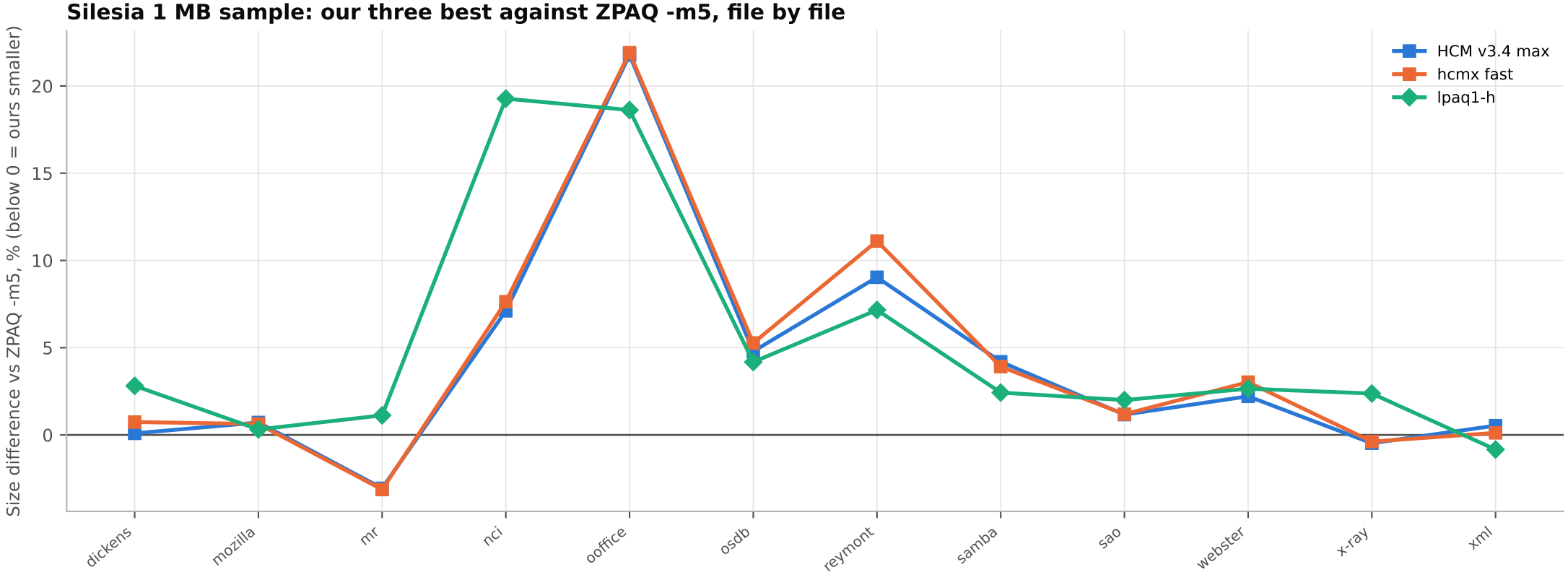


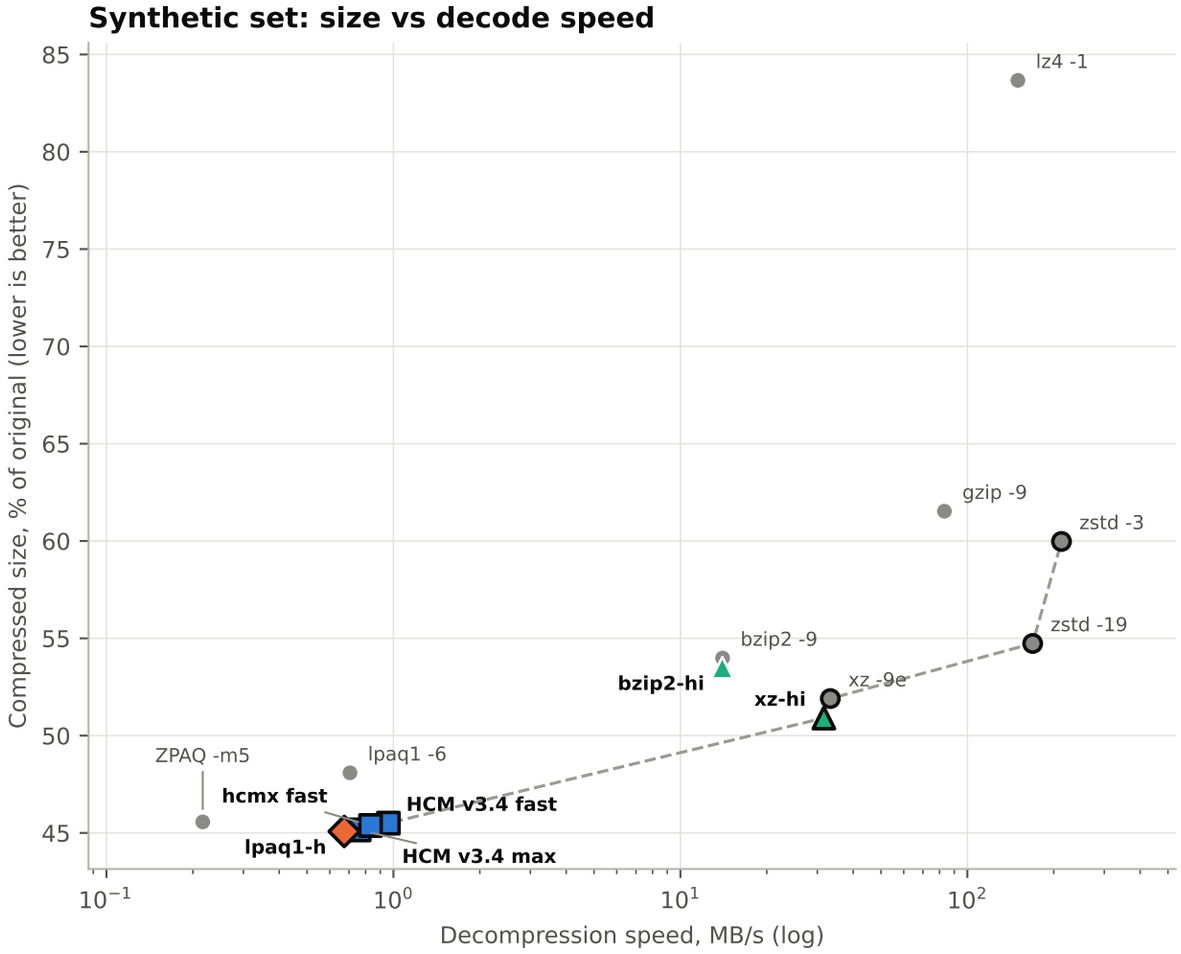


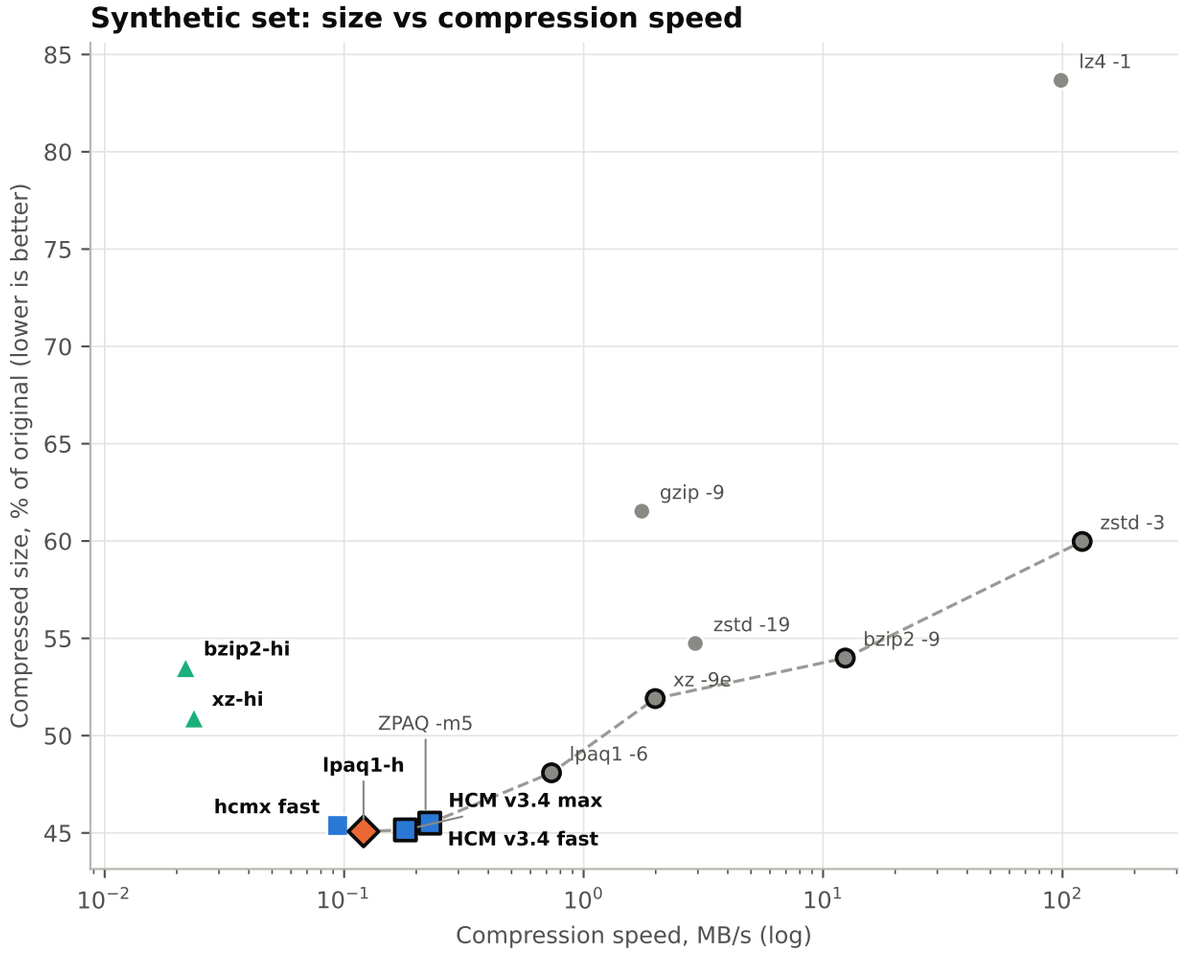
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

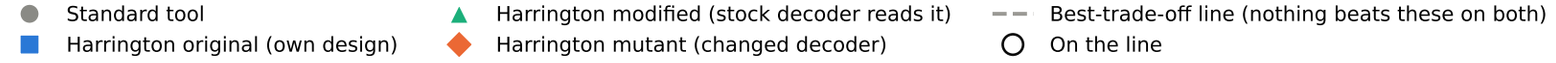

Silesia 1 MB sample: size vs peak memory
lz4 -1
zstd -3
gzip -9
zstd -19
bzip2 -9
bzip2-hi
xz -9e
xz-hi
HCM v3.4 fast
HCM v3.4 max
lpaq1 -6
lpaq1-h
hcmx fast
ZPAQ -m5
55
50
45
40
35
30
Size, % of original
10^1
10^2
Peak memory, MB (log; lower is lighter)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line

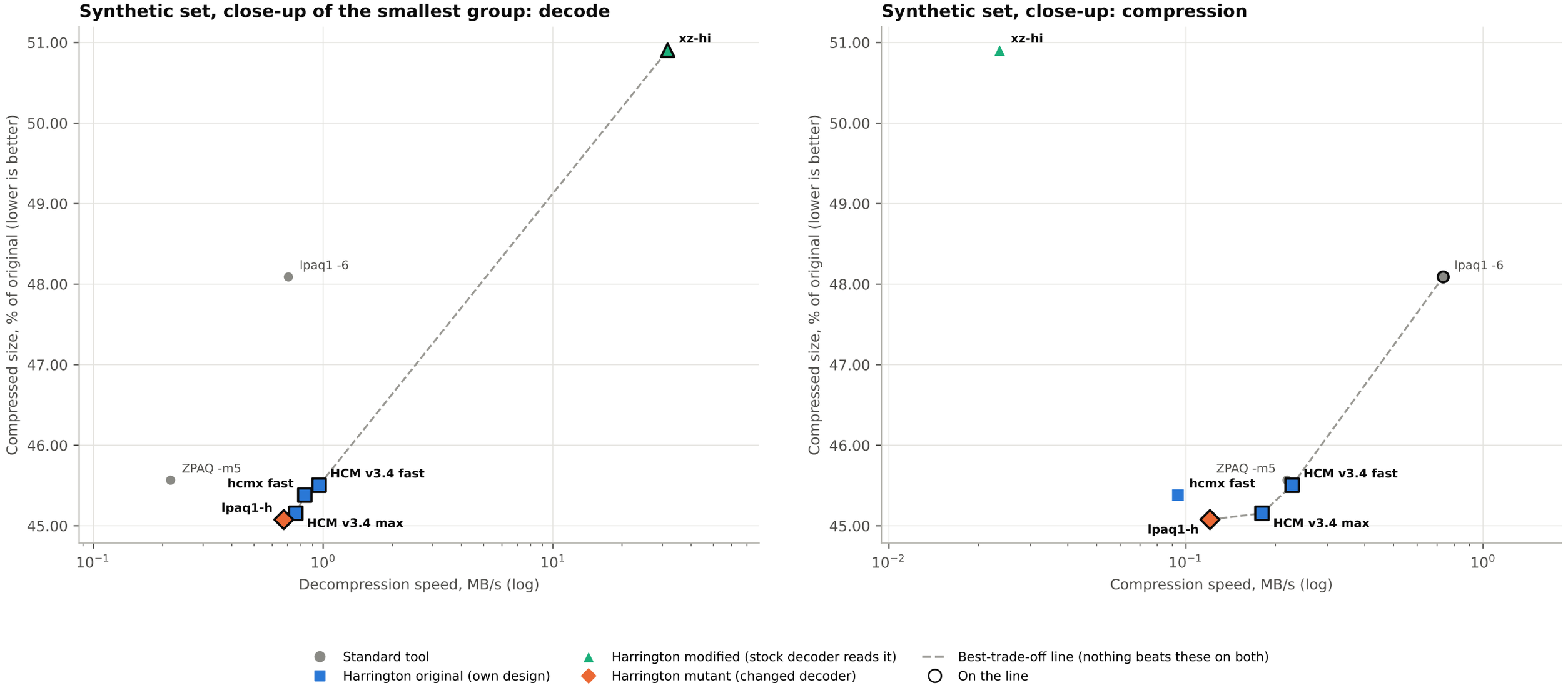

Silesia 1 MB sample: file by file, three strongest standard tools (dashed) vs our three best (solid)
Size, % of original (lower is better)
70
60
50
40
30
20
10
ZPAQ -m5 (Standard) - total 3,418,919 B
lpaq1 -6 (Standard) - total 3,643,900 B
bzip2 -9 (Standard) - total 4,102,930 B
HCM v3.4 max (Original) - total 3,544,604 B
hcmx fast (Original) - total 3,551,524 B
lpaq1-h (Mutant) - total 3,559,695 B
dickens
mozilla
mr
nci
ooffice
osdb
reymont
samba
sao
webster
x-ray
xml

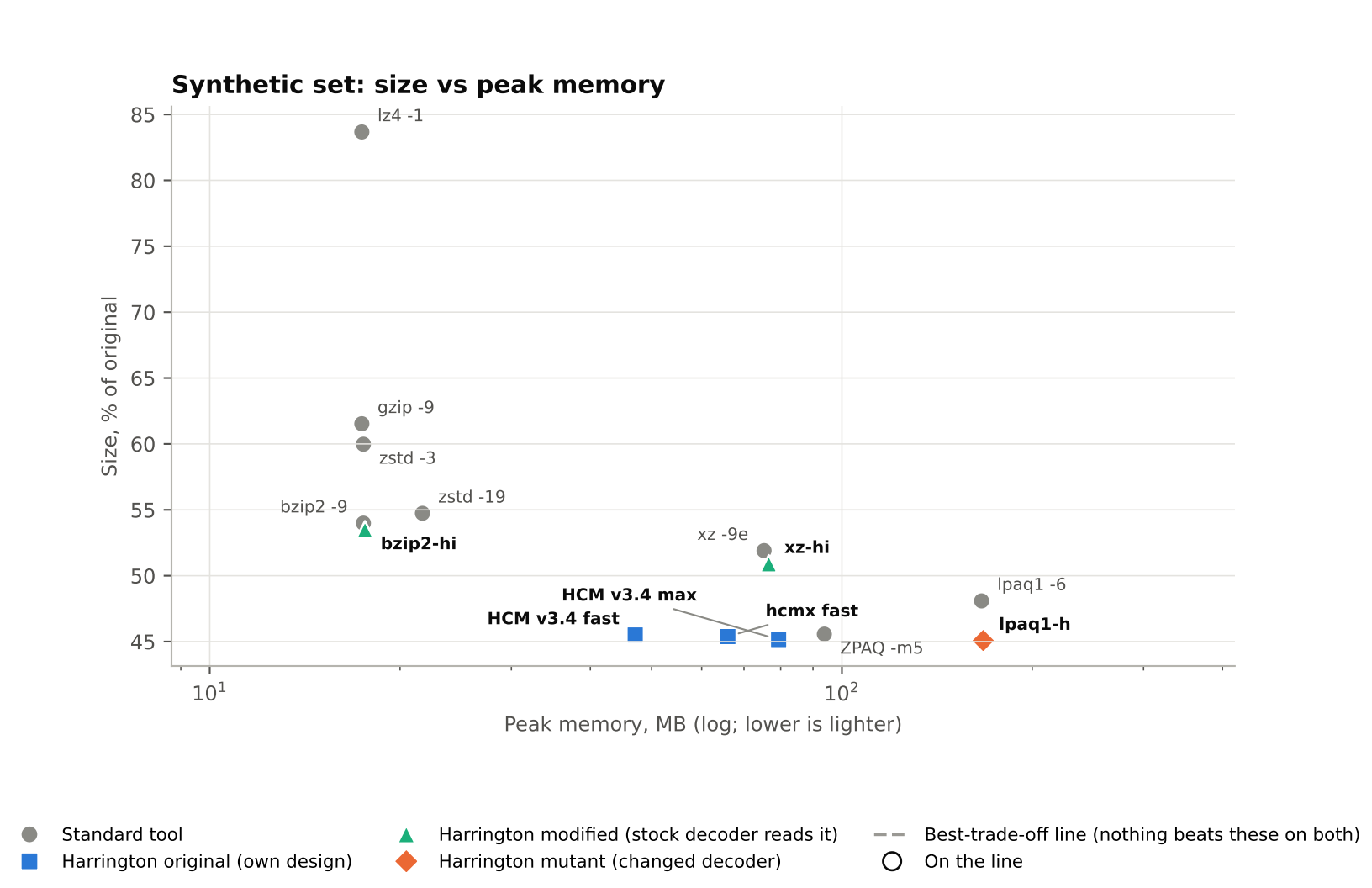

Silesia 1 MB sample: our three best against ZPAQ -m5, file by file
Size difference vs ZPAQ -m5, % (below 0 = ours smaller)
20
15
10
5
0
HCM v3.4 max
hcmx fast
lpaq1-h
dickens
mozilla
mr
nci
ooffice
osdb
reymont
samba
sao
webster
x-ray
xml

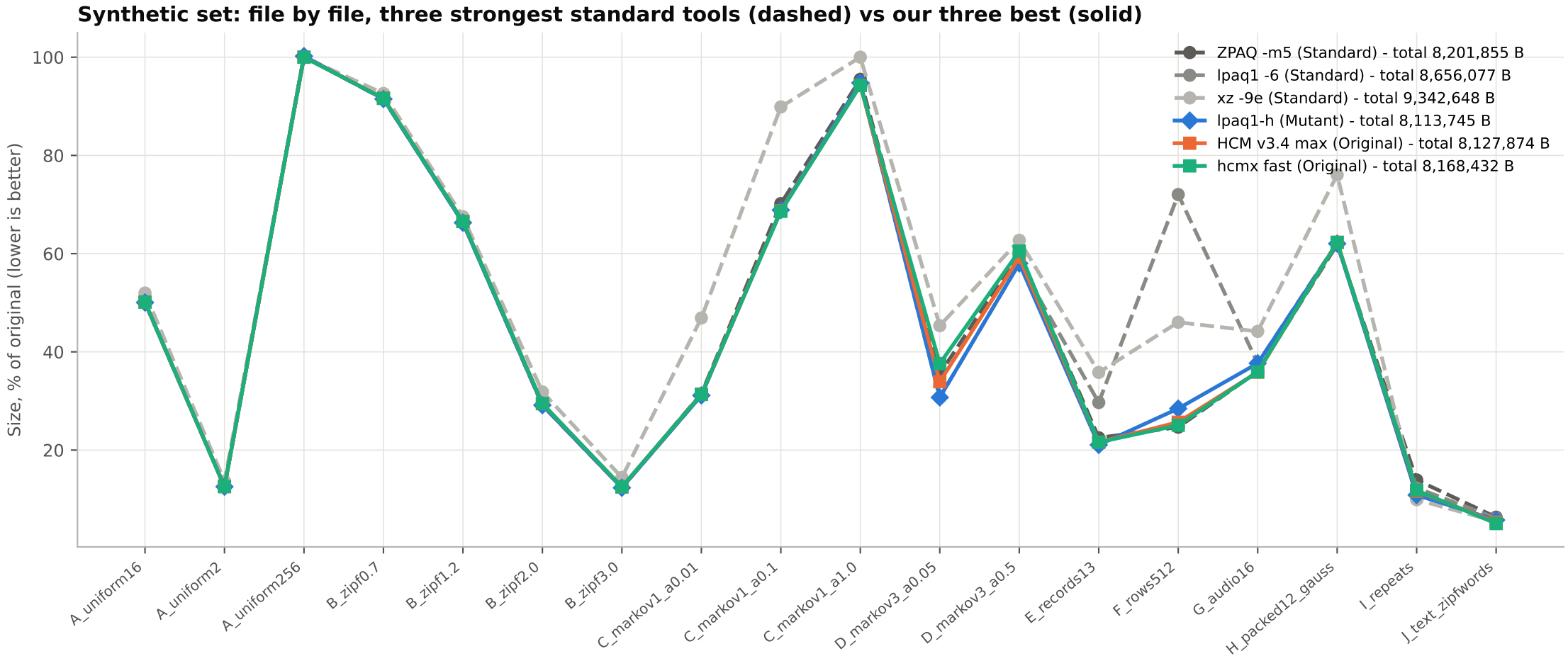

Synthetic set: size vs decode speed
Compressed size, % of original (lower is better)
Decompression speed, MB/s (log)
lz4 -1
gzip -9
zstd -3
zstd -19
bzip2 -9
xz -9e
bzip2-hi
xz-hi
ZPAQ -m5
lpaq1 -6
hcmx fast
HCM v3.4 fast
lpaq1-h
HCM v3.4 max

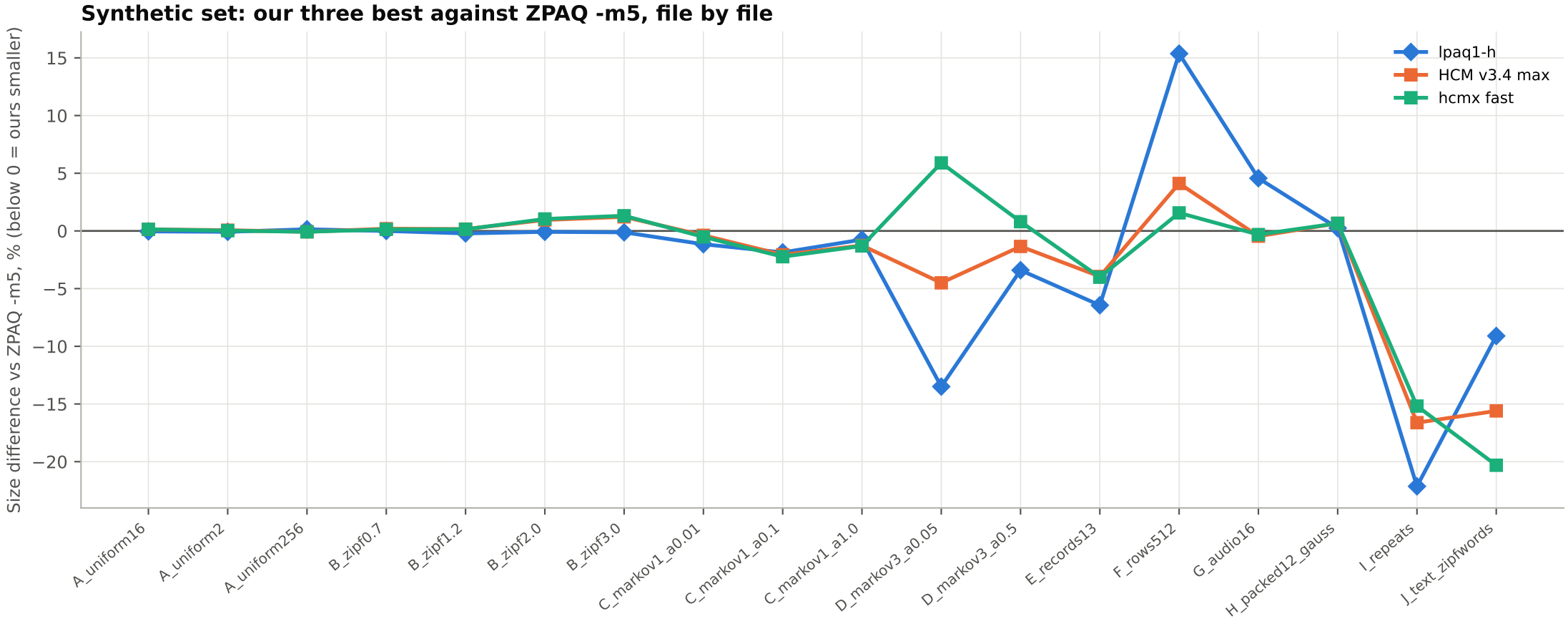

Synthetic set: size vs compression speed
Compressed size, % of original (lower is better)
Compression speed, MB/s (log)
lz4 -1
gzip -9
zstd -3
zstd -19
bzip2 -9
xz -9e
bzip2-hi
xz-hi
ZPAQ -m5
lpaq1-h
lpaq1 -6
hcmx fast
HCM v3.4 max
HCM v3.4 fast


Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line


Synthetic set, close-up of the smallest group: decode
Compressed size, % of original (lower is better)
51.00
50.00
49.00
48.00
47.00
46.00
45.00
xz-hi
lpaq1 -6
ZPAQ -m5
hcmx fast
HCM v3.4 fast
lpaq1-h
HCM v3.4 max
10^{-1}
10^{0}
10^{1}
Decompression speed, MB/s (log)
Synthetic set, close-up: compression
Compressed size, % of original (lower is better)
51.00
50.00
49.00
48.00
47.00
46.00
45.00
xz-hi
lpaq1 -6
ZPAQ -m5
hcmx fast
HCM v3.4 fast
lpaq1-h
HCM v3.4 max
10^{-2}
10^{-1}
10^{0}
Compression speed, MB/s (log)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line


Synthetic set: size vs peak memory
Size, % of original
85
80
75
70
65
60
55
50
45
lz4 -1
gzip -9
zstd -3
bzip2 -9
zstd -19
bzip2-hi
xz -9e
xz-hi
HCM v3.4 max
HCM v3.4 fast
hcmx fast
ZPAQ -m5
lpaq1 -6
lpaq1-h
10^1
10^2
Peak memory, MB (log; lower is lighter)
Standard tool
Harrington original (own design)
Harrington modified (stock decoder reads it)
Harrington mutant (changed decoder)
Best-trade-off line (nothing beats these on both)
On the line


Synthetic set: file by file, three strongest standard tools (dashed) vs our three best (solid)
Size, % of original (lower is better)
100
80
60
40
20
ZPAQ -m5 (Standard) - total 8,201,855 B
lpaq1 -6 (Standard) - total 8,656,077 B
xz -9e (Standard) - total 9,342,648 B
lpaq1-h (Mutant) - total 8,113,745 B
HCM v3.4 max (Original) - total 8,127,874 B
hcmx fast (Original) - total 8,168,432 B
A_uniform16
A_uniform2
A_uniform256
B_zipf0.7
B_zipf1.2
B_zipf2.0
B_zipf3.0
C_markov1_a0.01
C_markov1_a0.1
C_markov1_a1.0
D_markov3_a0.05
D_markov3_a0.5
E_records13
F_rows512
G_audio16
H_packed12_gauss
I_repeats
J_text_zipfwords


Synthetic set: our three best against ZPAQ -m5, file by file
Size difference vs ZPAQ -m5, % (below 0 = ours smaller)
15
10
5
0
−5
−10
−15
−20
lpaq1-h
HCM v3.4 max
hcmx fast
A_uniform16
A_uniform2
A_uniform256
B_zipf0.7
B_zipf1.2
B_zipf2.0
B_zipf3.0
C_markov1_a0.01
C_markov1_a0.1
C_markov1_a1.0
D_markov3_a0.05
D_markov3_a0.5
E_records13
F_rows512
G_audio16
H_packed12_gauss
I_repeats
J_text_zipfwords